MEDICAL TECHNOLOGY LICENSURE EXAM REVIEW – **CLINICAL MICROBIOLOGY**

CLINICAL BACTERIOLOGY

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Notes by: Xiao - The Conqueror of Demons, The Vigilant Yaksha, & Alatus, the Golden-Winged King

IMPORTANT CHARACTERISTICS OF BACTERIA

- Prokaryotic (organisms with no true nucleus)
- Has both RNA and DNA
- Multiplies by Binary fission
- Measured in µm; Average size: 0.4-2 µm
- Smallest living organism: genus *Mycoplasma*
- Largest living organism: genus *Bacillus*
- Produce either exotoxins or endotoxins; one of their virulence factors.
- Toxins: are biologically produced poisons.

EXOTOXINS	ENDOTOXINS
- Released by all major gram positive organisms except: *Listeria* - Gram negative said to produce exotoxin: *Vibrio and E. coli*	- Usually produced by gram negative organisms
- Usually excreted or released by living bacterial cells; does not require cell death for release	- Released only when cells are destroyed (cell death/lysis)
- Composition: protein	- Composition: polysaccharide
- Effect: systemic	- Effect: local
- Toxicity: high	- Toxicity: low
- Stability to heating: unstable	- Stability to heating: stable
- Stimulates antitoxin production once released into the host	- Stimulates antitoxin production once released into the host
- Examples o *Diphtheria* toxin by *C. diphtheria* o Botulinum toxin by *C. botulinum* o TSST-1 by *Staphylococcus aureus* o Enzymes such as Coagulase by *S. aureus* - Types o Cytotoxin: kills host cells o Enterotoxin: damages cells of the GIT (i.e. *E. coli and S. aureus*) - Neurotoxin: interferes with nerve impulses	- Limulus lysate test o Used to detect the presence of endotoxin o Uses aqueous extract of blood cells of horse shoe crabs o (+) result: clumping

IMPORTANT PARTS OF A BACTERIAL CELL WALL

1. CAPSULE

- Slimy area surrounding the cell wall
- Responsible for mucoid colonies (encapsulated bacteria)
- Function: prevents phagocytosis
- Neufeld Quellung Test
  - o A capsular swelling test
  - o Used to determine if an organism is encapsulated
- Capsule (+):
  - o *Bacillus anthracis*
  - o *Klebsiella pneumoniae*
  - o *Haemophilus influenzae*
  - o *Streptococcus pneumoniae*

2. CELL WALL *(MUREIN LAYER, PEPTIDOGLYCAN LAYER)*

- Defines the shape of bacteria; main component is peptidoglycan
- Point of anchorage for flagella
- Site of antibiotic action
- Basis of gram staining; determines gram positivity/negativity of an organism
  - o Gram (+) with rigid peptidoglycan layer
    - ▪ Impermeable to alcohol
    - ▪ With teichoic acid
    - ▪ Sugars: n-acetylglucosamine, and n-acetyl muramic acid
  - o Gram (-) with thinner peptidoglycan layer
    - ▪ Permeable to alcohol
    - ▪ No teichoic acid
    - ▪ With CHON (proteins), phospholipids, and LPS (lipopolysaccharide)
    - ▪ Lipopolysaccharide: used for evasion of host defenses
- No cell wall:
  - o *Mycoplasma and Ureaplasma*
- M protein: major virulence factor of *S. pneumoniae* and used to prevent phagocytosis
- Mycolic acid is responsible for the acid-fastness of *Mycobacterium* and prevents digestion during phagocytosis

3. PLASMA MEMBRANE

- Surrounds the cytoplasm; site of energy synthesis
- Transport of nutrients in and out of the cell; function as golgi apparatus and lysosome

4. PILI OR FIMBRIAE

- Not common to all bacteria
- Shorter than flagella and usually uniform in length
- Common pili: for adherence or attachment to host cells
- Sex pili: for gene conjugation (transfer of genetic material)
- Usually found in gram (-) bacteria like *E. coli, N. gonorrhoeae, Pseudomonas*

5. ENDOSPORES

- Not common to all bacteria
  - o Bacteria with spores: *Bacillus and Clostridium*
    - ▪ *C. tetani*: has terminally located spores
    - ▪ *C. botulinum*: has subterminal spores
    - ▪ *B. anthracis:* has centrally located spores
- Resistant structure enabling bacteria to withstand adverse conditions; target of sterilization
- Best way to destroy spores is by autoclaving
- Resistant because of calcium dipicolinate or dipicolinic acid

6. FLAGELLA

- Not common to all bacteria
- For locomotion; longer than pili
- Many bacilli and spiral are motile while true motility is seldom observed in cocci

MOTILE	NON-MOTILE
E. coli *C. tetani* *H. pylori*	*C. diphtheria* *C. perfringens* *B. anthracis* *Staphylococci*

- Motility is best seen at room temperature
- True motility: is due to the presence of flagella
- Brownian movement: movement of non-motile organisms due to the movement of molecules surrounding them
- Axial filaments/periplasmic flagella in spiral organisms
- Microscopes used to visualize spirochetes: (1) Fluorescence microscope & (2) Dark field microscope
- Ways to demonstrate motility
  - o **Hanging drop**: place a loopful of inoculum on a cover slip; invert the coverslip on a depression slide and observe under the microscope using LPO
  - o **Flagellar stains**: Gray's & Leifson
  - o **Semi-solid media**: SIM media (dispensed as butt; inoculated by stabbing; motile if growth is found outside/beyond the line of inoculation)
- Classification of bacteria according to flagella
  - o Monotrichous: single flagellum at one end; i.e. *Vibrio and P. aeruginosa*
  - o Amphitrichous: single flagellum at both ends; i.e. *C jejuni*
  - o Atrichous: absence of flagella
  - o Lophotrichous: tuft of flagella at one or both ends; i.e. *S. maltophilia*
  - o Peritrichous: surrounded with flagella; i.e. *E. coli*
- Motility
  - o With gliding motility: *Capnocytophaga*
  - o With darting motility: *Campylobacter*
  - o With tumbling motility: *Listeria*
  - o With twitching motility: *Kingella kingae*
  - o With shooting star motility: *Vibrio cholerae*
  - o With cork-screw motility: Spiral organisms or Spirochetes

Board Exam Must-Know!
- Parts external to the cell wall: [1] Capsule, [2] Pili, & [3] Flagella

7. METACHROMATIC GRANULES/INCLUSION BODIES

- Serves as energy source; these are food reserves
- *Corynebacterium diphtheriae:* Babes Ernst Granules
- *Mycobacterium tuberculosis*: Much granules
- *Yersinia pestis*: Bipolar bodies
- *Nocardia and Actinomycetes*: Sulfur granules

8. NUCLEOID

- The DNA in the bacterial cell is generally confined to this central region

9. RIBOSOME

- Protein synthesis
- Gives the cytoplasm of bacteria a granular appearance in EM

Additional notes!
Plasmid
- Extrachromosomal DNA carries an antibiotic resistant gene
- Can replicate independently
- Types:
  - o Large plasmid: confers resistance to beta lactams
  - o Small plasmid: confers resistance to tetracycline and chloramphenicol

BACTERIAL PHYSIOLOGY/REQUIREMENTS FOR BACTERIAL GROWTH

1. NUTRITIONAL REQUIREMENTS

- Phototrophic: uses light as energy source
- Chemotrophic: uses chemical compounds as energy source; Types: Autotrophs and Heterotrophs

Autotrophs/Litotrophs	Heterotrophs/Organotrophs
Uses inorganic compounds as carbon source i.e. carbon dioxide	Uses organic compounds as carbon source i.e. glucose

2. OXYGEN REQUIREMENT

Obligate Aerobes	Organisms requiring oxygen growth i.e. *Brucella spp., Francisella, Mycobacterium*
Obligate Anaerobes	Organisms not requiring oxygen for growth i.e. *Veillonella, Actinomyces*
Facultative anaerobes	Organisms that can live with or without air i.e. family *Enterobacteriacea*
Microaerophilic	Organisms requiring small amount of air i.e. *Campylobacter spp.*
Aerotolerant	Does not grow well but survives in the presence of air i.e. *Lactobacillus* and *Propionibacterium*
Extremophiles	Extremophiles: able to survive on unusual conditions i.e. absence of oxygen and increased temperature i.e. *Bacillus infernus*

Examples	Obligate aerobes	Facultative anaerobes	Microaerophiles	Obligate anaerobes
Gram positive	*B. cereus*	*Staphylococcus, B. anthracis, Corynebacterium, Listeria, Actinomyces*	*Streptococci*	*Clostridium*
Gram negative	*Neisseria, Pseudomonas, Bordetella, Legionella, Brucella*	Most gram-negative rods	*Spirochete, Campylobacter*	*Bacteroides*
Acid fast/ slightly AF	*Mycobacterium & Nocardia*			

3. TEMPERATURE

Psychrophilic *0-20°C*	Mesophilic *20-45°C*	Thermophilic *50-150°C*	Thermoduric
Cold-loving bacteria	Organisms that grow at moderate temperature Many bacteria are mesophilic	Organisms that grow at high temperature	Bacteria which do not usually grow at high temperature but can withstand exposure to high temperature

- Optimum temperature: defined as the temperature at which an organism grows best; temperature best suited for bacterial growth
- Cold enrichment: incubation at 4°C for several weeks
- Can tolerate cold enrichment: *Yersinia enterocolitica* and *Listeria*
- Thermophilic organisms:
  - *Bacillus stearothermophilus*: biological indicator of autoclave
  - *Thermus aquaticus:* source of DNA polymerase used in PCR
- Incubation temperature for most bacteria and viruses: 37°C
- Incubation temperature for fungi: room temperature (27-30 °C)
- Incubation temperature and time for aerobes: 37°C for 18-24 hours
- Incubation temperature and time for anaerobes: 37°C for 24-48 hours

Board Exam Must-Know!
Blood bag contaminants at 4°C
- *Pseudomonas fluorescens*
- *Yersinia enterocolitica*
- *Serratia liquefaciens*

**Listeria*, although able to tolerate cold temperature (4°C), is not considered as a blood bag contaminant because it is an animal pathogen

ANAEROBIC CULTURE (Gas Pak Jar)	MICROAEROPHILIC (Candle jar)
5% CO_2 10% H_2 85% N_2	5% O_2 10% CO_2 85% N_2
Difficult to perform	Intended for *Neisseria*

4. pH REQUIREMENT

ACIDOPHILIC	BASOPHILIC	NEUTROPHILIC
- Requiring acid medium for growth - *Lactobacillus*	- Requiring alkaline medium for growth - *Vibrio* (medium: alkaline peptone water)	- Those able to grow in medium with a pH between 7.2-7.4

*pH of media for fungi: 5-6

5. MOISTURE

- 70% moisture is needed to prevent drying (drying deters survival of organisms)
- Humidophilic: those requiring increased (>70%) moisture

6. SALT CONCENTRATION

- Halophilic: those requiring high salt concentration; i.e. *Vibrio*
  - 6.5%: *Enterococci*
  - 8-10%: *Vibrio*
  - 7.5%: *Staphylococcus aureus*

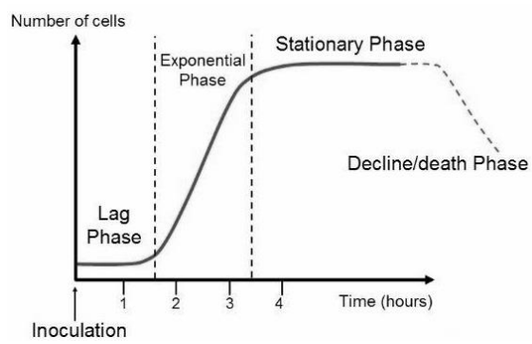
7. CARBON DIOXIDE

- Capnophilic: require increased carbon dioxide; i.e. *Neisseria* and *Haemophilus influenzae*

	Oxygen	Carbon dioxide
Aerobes	21%	0.3%
Anaerobes	0%	5-10%
Capnophilic	15%	5-10%
Microaerophilic	5-10%	8-10%

STAGES OF BACTERIAL GROWTH

LAG PHASE *Rejuvenescence phase*	LOG PHASE *Exponential phase*	PLATEAU *Stationary phase*	DEATH PHASE *Phase of decline*
- No cell division during this stage because the organism is still adapting to the environment - Increase in size but not in number - Growth rate: 0%	- Start of cell division - Growth rate starts to increase - Organism becomes susceptible to antibiotics	- Characterized by a balance in the number (equal) of dead cells and living cells - Decreased nutrients, increased toxic waste	- Increased death rate - Causes of bacterial death o Food shortage o Increased concentration of toxic products o Development of unfavorable pH



BIOLOGICAL SAFETY LEVEL/BIOLOGICAL AGENTS/HAZARDS

BSL-1	BSL-2	BSL-3	BSL-4
- Those that may pose **minimal threat** to laboratory workers - Those with no known potential of infecting healthy people - i.e. *Bacillus subtilis* and *Mycobacterium gordonae*	- Those that may pose **moderate threat** to laboratory workers - Those associated with *laboratory acquired infections* - Those acquired through ingestion, mucus membrane, and percutaneous exposure - i.e. *Bacillus anthracis* and *Yersinia pestis;* Enteric pathogens such as *Salmonella & Shigella; S. aureus;* HIV and HBV	- Those that may pose **high risk** to laboratory workers - Those with a potential for aerosol transmission (inhalation) - i.e. *Mycobacterium, Coxiella,* all agents of systemic mycoses, *Rickettsiae, F. tularensis, Brucella*	- Those that may pose **extreme risk** to lab workers - Can cause life threatening diseases - Organisms encountered in research institutions (RITM, PGH) - i.e. *MERS-CoV, Ebola virus*

BIOLOGICAL SAFETY CABINETS *A device that encloses a workspace* *Sterilizes air with infectious material; Sterilizes air through filtration using HEPA filter* *Prevents laboratory acquired infections (aerosols)*		
CLASS I	**CLASS II**	**CLASS III**
- Least effective because it has an open front - Negative pressure, ventilated cabinet - Unsterilized air enters and circulates within the cabinet, & the exhaust air from the cabinet is filtered by HEPA filter - Sterilizes air to be exhausted - Process non-pathogens, BSL-1, Low oncogenic virus, low toxic chemicals, carcinogens - With 1 HEPA filter	- Sterilizes air entering and circulating within the cabinet and the exhaust air - Sterilizes air to be exhausted and the air that flows over infectious material - A.k.a laminar flow - Process bacterial and fungal pathogens, medium risk oncogenic virus, chemicals, carcinogens - For BSL-2 & 3 organisms Types: **IIA** - With fixed opening - 70% of air is recirculated **IIB** - With variable sash opening - Exhaust air is discharged outside the building	- The most effective - The system is entirely close - The infectious material is handled using gloves attached and sealed to the cabinet - Can process/handle highly toxic chemicals and viral pathogens - For BSL-4 organisms

CATEGORIES OF POTENTIAL INFECTOUS AGENTS OF BIOTERRORISM

CATEGORY A	CATEGORY B	CATEGORY C
- These are agents that pose the greatest public health threat - Easily transmitted and highly contagious - Can cause high mortality rate	- Agents with moderate morbidity and low mortality - Not easily transmitted as category A agents	- Emerging pathogens that can be engineered for mass spread in the future
- i.e. *Bacillus anthracis* and *Francisella tularensis*	- i.e. *Rickettsia, Coxiella, Brucella, E. coli O157:H7*	- i.e. *MERS-CoV, SARS-CoV, MDR-Tuberculosis, H1N1*

*Samples for possible bioterrorism must be processed within BSC Class-II

METHODS OF MICROBIAL CONTROL

- **Sterilization:** refers to the complete destruction and removal of all forms of microbial life, including their spores
- **Disinfection:** destruction and removal of pathogens but not necessarily all microorganisms and their spores

A. *STERILIZATION*

PHYSICAL METHODS OF STERILIZATION

- Use of moist heat, use of dry heat, filtration, and exposure to ionizing radiation

1. MOIST HEAT	
AUTOCLAVE	- 121°C, 15 psi, 15-30 minutes: to sterilize used and unused media - 132°C, 15 psi, 30-60 minutes: to sterilize medical waste - Indicator: *Bacillus stearothermophilus* (New taxonomy: *Geobacillus stearothermophilus*) - Principle: steam under pressure - Not killed by autoclaving are: Prions (infectious protein particles said to cause neurologic diseases in animals and man; Mad cow disease and CJD)
TYNDALLIZATION	- Intermittent or fractional sterilization (process is not continuous; done for several days) - Uses: flowing steam - Equipment used is Arnold sterilizer - 100°C for 30 minutes, for 3 consecutive days
INSPISSATION	- Intermittent or fractional sterilization - Principle: thickening through evaporation - For sterilizing media with increased protein, like: Lowenstein-Jensen media (contains egg; for *M. tuberculosis*) - 75-80°C for 2 hours, for 3 consecutive days

Fractional sterilization will kill:
- 1st day: Vegetative cells
- 2nd day: Spores
- 3rd day: Remaining cells

2. DRY HEAT			
USE OF OVEN	**INCINERATION**	**DIRECT FLAMING**	**CREMATION**
- Often used in a microlaboratory - 160-180°C for 1-2 hours - Used to sterilize glasswares - Indicator: *Bacillus subtilis var. niger*	- Temperature to burn materials into ashes is 300-400°C - To dispose hazardous waste: 870-980°C - Not done in the Philippines because it is not environment friendly - Dry sterilization method that can eliminate prions	- For loops and needles	- Burning of the body into ashes - To control disease

3. FILTRATION - For materials that are heat-sensitive - For antibiotic solutions, toxic chemicals	
Depth filter	- Consist of granular material - i.e. Berkefield filter, asbestos
Membrane filter	- These are porous membranes 0.1mm thick to sterilize culture media, antibiotics - Composed of cellulose acetate and polycarbonate
Millipore filter	- Millipore filter with a pore diameter of 0.22µm can give 100% sterility
HEPA filter	- Can remove objects larger than 0.3µm

4. IONIZING RADIATION/COLD STERILIZATION

- Exposure to gamma rays
- Sterilize disposable gloves, catheters, and syringes
- Indicator: *Bacillus pumilus*

CHEMICAL METHODS OF STERILIZATION

a. ETHYLENE OXIDE (GAS STERILANT)
   - Most commonly used chemical sterilant
   - Used in gaseous form for sterilizing heat sensitive objects
   - Disinfectant for machine that cannot be autoclaved
   - Indicator: *Bacillus subtilis var globijii*

b. FORMALDEHYDE VAPOR AND VAPOR PHASE HYDROGEN PEROXIDE
   - Used to sterilize HEPA filters in BSCs

c. GLUTARALDEHYDE (2%)
   - Sporicidal; kills spores in 3-10 hours
   - For medical equipment

d. PERACETIC ACID

Board Exam Must-Know!
- Cold sterilization: use of glutaraldehyde or peracetic acid
- Chemical sterilants a.k.a biocides

B. *DISINFECTION*

PHYSICAL METHODS OF DISINFECTION

PASTEURIZATION	- Purpose: remove pathogens in food - 63°C, 30 minutes (low temperature holding): VAT pasteurization or batch method - 72°C, 15 seconds: High temperature short time (HTST) or Flash method - 140°C, 3 seconds: Ultra-High temperature short time
BOILING	- 100°C, 15-30 minutes - For surgical instruments
UV	- Non-ionizing radiation using UV light

CHEMICAL METHODS OF DISINFECTION

1. **ANTISEPTIC**: chemical germicide for use on the skin or tissue and not to be substituted for a disinfectant
   - Alcohols
     - Commonly used antiseptic (but not the best): 70% ethyl alcohol or isopropyl alcohol
   - Iodophors
     - Regarded as the best antiseptic
     - Iodine + detergent
   - Chlorhexidine
   - Hexachlorophene
   - 10% hydrogen peroxide is used for cleansing of wounds

Board Exam Must-Know!
- 70% ethanol is more effective than 95% EA as disinfectant
- Ethyl alcohol and isopropyl alcohol are non-sporicidal
- Iodine is prepared as tincture with alcohol or as an iodophor

2. **DISINFECTANTS**: for non-living things; for surfaces, used to remove blood spills
   - Halogens (chlorine, iodine, fluorine)
   - Heavy metals
   - Aldehydes
   - QUATS – Quaternary Ammonium Compounds
     - i.e. Benzalkonium chloride/zephiran
     - *P. aeruginosa* grown in ammonium acetate media is said to be resistant to QUATS
   - Phenolics

Additional notes!

- Best disinfectant for blood spillage ***sodium hypochlorite/ household bleach/ Clorox*** (10-30 minutes contact); the CDC recommends 1:10 dilution of bleach
- If not available, can be used as substitute is ***vinegar***
- **Standard precaution:** require that blood and body fluids from every patient be treated as potentially infectious; set of preventive measures applied to all patients designed to reduce risk of infection in health care setting based on the premise that all blood and other body fluids are infectious
- **Universal precautions:** precaution on all human blood and all other body fluids that contain visible blood; set of preventive measures designed to reduce transmission of blood-borne pathogens

Examples of bacteria which remain active in a dry environment:

- *N. gonorrhoeae:* viable for 1 hour
- *M. tuberculosis:* viable for several months
- *Bacillus and Clostridium:* viable for 10 years

COLLECTION, TRANSPORT, AND SPECIMEN PROCESSING

A. SPECIMEN COLLECTION AND TRANSPORT

- Collect before antibiotic therapy
- Ensure aseptic collection and quantity must be sufficient
- Must be placed in sterile containers
- Ideal specimens should be transported to the lab within 2 hours of collection

TRANSPORT MEDIA

- Used to maintain the viability of the organism during transport
- With nutrients to maintain the growth of organism, buffer to maintain pH, small amount of gas to maintain moisture
- Stool pathogens: Cary Blair Transport media
- Stuart's: Viral transport media
- Amies: a modification of Stuart's media; for clinical swab specimens
- Transgrow: Transport media for *Neisseria*
- JEMBEC: Transport media for *Neisseria*

COLLECTION TECHNIQUE/SAMPLE TYPE

- ✓ For **aerobic culture**: collected through swabbing or needle aspiration
- ✓ For **anaerobic culture**: needle aspiration, tissue
- ✓ Urine, CSF, Serous fluid, stool, sputum: place in a sterile container
- ✓ Use sterile cotton tip applicator for throat, nose, eyes, wound, abscess
- ✓ Sterile bottle broth with SPS: for blood culture
- ✓ For fungal culture, specimen should never be collected through swabbing

B. ANTICOAGULANTS/PRESERVATIVE

- To maintain accurate colony count, if there is delay in processing of urine, this preservative may be added: **boric acid**
- **0.025% sodium polyanethol sulfonate/SPS**: for blood specimens
- **Heparin**: anticoagulant used in viral culture, may inhibit gram positive organisms and yeast
- Viral PCR anticoagulant: **EDTA**
- Not to be used in microbiology: **EDTA and Citrate**

C. SPECIMEN STORAGE

- All specimens submitted for culture must be processed immediately, if not, adequate storage must be observed
- CSF (1st priority) should be kept at 37°C if there is delay in processing; transport temperature: room temperature
- Urine, stool, viral specimens, sputum, swab: placed inside the refrigerator (4°C)
- Serum for serology: -20°C for 7 days (1 week)
- Tissue/specimens for long term storage kept frozen at: -70°C
- Specimen suspected to contain anaerobe: should never be refrigerated; should be kept at room temperature
- Anaerobic culture is seldom performed (difficult)

Ways to facilitate anaerobic cultivation/anaerobic atmosphere: 5% CO2, 10% H2, 85% N2 (Gas Pak Jar)

- Boiling of culture medium to remove oxygen
- Use of gaspak jar with palladium catalyst (remove oxygen)
- Indicators: *Resazurin* (pink) or *Methylene blue* (blue) in the presence of air; in the absence of air they become: colorless
- Anaerobic cultures are usually incubated at 37°C
- Ambient air: organism supplied with 21-22% oxygen during incubation

Board Exam Must-Know!

- Most common failure of gaspak jar: **inactivation of catalyst** due to repeated use

D. CLINICAL SPECIMENS

Blood	- Avoid normal skin flora while collecting blood for culture
	- The collection site must be cleansed with 70-95% alcohol, followed by iodine scrub, then alcohol rinse
	- Most common anticoagulant: 0.025% SPS
	  ○ Inhibits *Neisseria, G. vaginalis, P. anaerobius*
	  ○ To counteract, add: 1% gelatin
	- Dilution factor of blood to medium: 1:10
	- Amount (adult): 10mL (5mL for aerobic culture and 5mL for anaerobic culture)
	- Amount (children): 1-5mL
	- Routine blood culture bottles are held for 5-7 days (TAT)
	  ○ Signs of growth: bubbles, hemolysis, cloudiness, pellicle
	  ○ *Brucellosis:* held for 3-4 weeks
	  ○ *Leptospirosis*: held for 8 weeks

Cerebrospinal fluid	- Usually collected using 3-4 tubes - Tube #2 is for microbiology, gram stain & culture, if with 4th tube, use the 4th for better exclusion of skin contaminants (least contaminated) - Examine immediately; priority/1st to process or held either at 37°C storage temp; transport temperature: room temperature - Process: Centrifuge, use the sediments for smear and culture - Common pathogens: *Haemophilus influenzae* and *Neisseria meningitidis*
Throat swab	- Usually for the detection of *Streptococcal* infection - Medium of choice: Sheep's blood agar (SBA) - Most abundant throat flora: *Viridans Streptococci* - Most common throat pathogen: *Streptococcus pyogenes* (Group A beta-hemolytic *Streptococci*) - Culture on Todd-Hewitt broth for fluorescence microscopy of beta *Streptococci* **Nasopharyngeal swab** - Submitted for the detection of *Bordetella pertussis* - The specimen of choice to detect carrier state of *N. meningitidis* and to detect the presence of *B. pertussis* and *H. influenzae*
Sputum	- Non-sterile specimen; may often be contaminated with normal flora so it is important to evaluate the quality of the specimen - Note the number of squamous epithelial cells/LPF and PMNs to evaluate the acceptability of the specimen - >25 PMNs, <10 SECs = sputum; specimen is suited for culture - **Bartlett's classification**: assess the quality of sputum ○ 0 or less = the specimen is saliva; >1 = sputum - A gram stain is performed on all sputum samples - Collected ideally in the morning when it is most concentrated

Urine	- Specimen of choice for bacterial culture is **clean-catch midstream;** catheterized for those who are unable to void
	- Must be preserved or refrigerated if not processed, suitable preservative: **boric acid** (to maintain accurate colony count)
	- Major cause of UTI: *E. coli*
	- Cause of UTI among young females: *S. saprophyticus*
	- Suprapubic urine: specimen preferred for anaerobic culture, infants, and children
	- MAC and BAP are suitable combination
	- Colony count should be performed on all urine sample; when performing colony count, use **calibrated loop** (1 µl or 10 µl)
	- Formula to compute for colony count per mL of urine:
	  ○ # of colonies counted x 1,000 (if 1 µl loop was used) = colony count/mL of urine
	  ○ # of colonies counted x 100 (if 10 µl loop was used) = colony count/mL of urine
	  ○ Considered significant for UTI: **>1.0 x 10^5 CFU/mL**
	  ○ 1.0 x 10^3 or 1.0 x 10^5 CFU/mL: contaminated or specimen collected during recovery following treatment

Stool	- For the detection of enteric pathogens
	- Stool specimen not processed within 2 hours of collection should be placed on a transport media like Cary Blair
	- This specimen is routinely screened for *Salmonella, Shigella,* and *Campylobacter*
	- Maybe directly plated on differential plate EMB or Mac and onto a selective media HEA or SSA and enrichment i.e. tetrathionate
	- Number of quadrants streaked: 4
	- Fecal pathogens
	  ○ Oxidase (+): *Vibrio, Aeromonas, Campylobacter, Plesiomonas*
	  ○ Oxidase (-): *Enterobacteriaceae (Salmonella and Shigella)*

Genital tract specimens	- Used to detect urogenital pathogens - To detect presence of *N. gonorrhoeae, G. vaginalis, and C. trachomatis* which is a common cause of cervicitis

Board Exam Must-Know!

- If the patient is in a ventilator (intubated), the preferred specimen is ***endotracheal aspirate***

- Centrifugation rate and duration:
  - CSF & Urine: 1000-1500 rpm for 10-15 minutes
  - Sputum: 3,000 rpm for 15 minutes

BACTERIAL IDENTIFICATION

- Manual/Traditional: Gram staining, selective media, biochemical tests
- Semi-automated: uses API (Analytical profile index)
- Automated: Vitek

MORPHOLOGY

- Consider the following: size, shape, arrangement, motility, and staining characteristics

MOTILITY

- Motile: those with flagella
- Non-motile: no flagella but may show Brownian motility

STAINING CHARACTERISTICS

- Staining: process of artificially coloring the organism with dyes/stains
- Purpose of staining:
  1. To observe and appreciate the appearance of bacteria
  2. To differentiate one organism from the other
  3. To reveal the chemical nature of bacteria

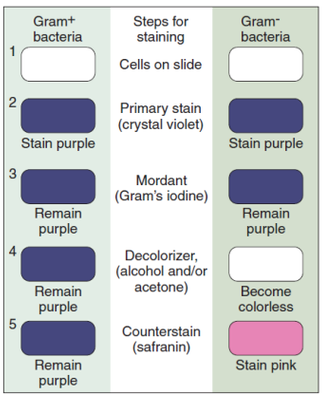
STAINING TECHNIQUES

- **Simple staining**
  - Uses only 1 dye; the color of the dye is also the resulting color
  - All organisms would retain the same color; less differentiation
  - Ex. Methylene blue
- **Indirect/Relief/Negative**
  - The background, not the organism is stained
  - The organism appears colorless
  - Ex. India ink/Borris method, Nigrosin
- **Special staining** (used to demonstrate the special features of the cell)
  - Capsular stains = Hiss, Anthony's, Tyler, Muir
  - Spore stains = Dorner's, Schaeffer and Fulton, Wirtz and Conklin
    - Schaeffer and Fulton
      - Primary dye: Malachite green
      - Counterstain: Safranin
      - Spores will take the color of the primary dye which is green
  - Flagellar stain = Gray's, Fisher and Conn, Leifson
  - Metachromatic granules = Albert's, Neisser, Ljubinsky, Ponder, Methylene blue, Lindergran, Burke's technique
  - Nucleic acid = Feulgen
  - Polar bodies = Wayson
  - Spirochetes = Levaditi, Warthin Starry, Fontana-Tribondeau
- **Differential staining**
  - Utilizes more than one dye; used to differentiate one organism from another
  - The two most common differential stains are Gram's stain and Acid-fast stain

GRAM STAIN

PURPOSE	REAGENT	GRAM (+)	GRAM (-)
Primary stain/ Initial stain	Crystal violet	Appear violet	Appear violet
Mordant	Gram's Iodine	Retain the violet color	Retain the violet color
Decolorizer	95% alcohol, Acetone, or Alcohol-acetone mixture	Remain violet; not decolorized	Become colorless; decolorized
Counterstain/ Secondary stain	Safranin red	Remain violet; Does not absorb safranin	Becomes red

- Most critical step in gram staining: **decolorization**
- **Hucker's modification**: gram staining for fungi; ammonium oxalate is incorporated to crystal violet
- **Carbolfuchsin method of gram staining** is used (counterstain = Carbolfuchsin): to improve staining of gram negative organisms; used on organisms which are poorly stained by safranin such as *Legionella, Brucella, Campylobacter, Fusobacterium, Bacteroides*



RULES

1. All **COCCI** are **gram (+)** except *Neisseria, Branhamella (Now: Moraxella)*, and *Veillonella*
2. All **BACILLI** are **gram (-)** except:
   - *Mycobacterium, Corynebacterium, Clostridium*
   - *Bacillus, Lactobacillus*
   - *Erysipelothrix*
   - *Listeria*
3. Higher forms of organisms like *Actinomyces, Nocardia, Streptomyces*, yeast and molds are gram (+)
4. All spiral organisms are reported as gram (-)

- Not gram stained:
  - Intracellular: *Rickettsiae, Chlamydia*
  - No cell wall: *Mycoplasma, Ureaplasma*
  - Cant resolved by bright field: *Spirochetes*

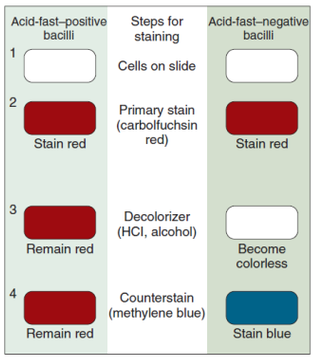
Reasons why Gram (+) becomes Gram (-)	Reasons why Gram (-) becomes Gram (+)
1. Overdecolorization 2. Use of acidic gram's iodine (alkaline gram's iodine must be used)	1. Underdecolorization 2. Use of thick bacterial smears

ACID FAST STAIN

- To distinguish acid-fast from non-acid fast organisms
- **Acid fast organisms**: are organisms that are very hard to stain but once stained, they are very hard to decolorize because of ***mycolic acid/hydroxymethoxy acid*** that envelopes the bacteria
- Rule: all bacteria are non-acid fast except:
  - *Mycobacterium and Nocardia* (slightly acid fast)

Ways to facilitate Acid-Fast staining

- Ways to remove mycolic acid temporarily:
  - Steaming process
  - Addition of wetting agents (tergitol) prior to the stain solution
- Increasing concentration of phenol (accentuator; increase the staining power of the dye) and basic fuchsin
  - Acid fast organisms are hard to identify under the microscope, even at OIO magnification, especially if they are few in number
- Prolonging contact of stain with the material



1 Fix smears on heated surface (60°C for at least 10 minutes).

2 Flood smears with carbolfuchsin (primary stain) and heat to almost boiling by performing the procedure on an electrically heated platform or by passing the flame of a Bunsen burner underneath the slides on a metal rack. The stain on the slides should steam. Allow slides to sit for 5 minutes after heating; do not allow them to dry out. Wash the slides in distilled water (note: tap water may contain acid-fast bacilli). Drain off excess liquid.

3 Flood slides with 3% HCl in 95% ethanol (decolorizer) for approximately 1 minute. Check to see that no more red color runs off the surface when the slide is tipped. Add a bit more decolorizer for very thick slides or those that continue to "bleed" red dye. Wash thoroughly with water and remove the excess.

4 Flood slides with methylene blue (counterstain) and allow to remain on surface of slides for 1 minute. Wash with distilled water and stand slides upright on paper towels to air dry. Do not blot dry.

5 Examine microscopically (see **A** and **B** below), screening at 400× magnification and confirm all suspicious (i.e., red) organisms at 1000× magnification using an oil-immersion lens.

PURPOSE	ZIEHL NEELSEN/ HOT METHOD *Considered as the best for DSSM (direct sputum smear microscopy)*	KINYOUN'S / COLD METHOD *Best method to stain AF organisms in tissues*	AURAMINE - RHODAMINE (FLUOROCHROME) *Most sensitive method*
Primary stain	Carbolfuchsin	Carbolfuchsin	Auramine-Rhodamine dye
Mordant	Steaming or heat	Tergitol	None
Decolorizer	3% acid alcohol (HCl + ethanol)	3% acid alcohol (HCl + ethanol)	0.5% acid alcohol
Counterstain	Methylene blue	Methylene blue	0.5% potassium permanganate
Result	Acid fast: red against a blue background Non-acid fast: blue		
May be used as substitute for methylene blue counterstain	Malachite green (NF = green)		

- Size of AFB smear: 2x3 cm
- AF organism in tissues is best stained using: Kinyoun's method
- AFB smear can be fixed using a slide warmer set at: 65°C for 2 hours
- Following staining, we do microscopy:

	Brightfield/ Light Microscope	Fluorescence	Darkfield	EM
Bacteria	+	+	+	+
Fungi	+	+	-	+
Viruses	+	+	-	+
Parasites	-	+/-	-	+

Other methods of Acid-Fast staining

- **Pappenheim's (urine)**: to differentiate *M. smegmatis* (blue) from *M. tuberculosis* (red)
- **Baumgarten's (tissue)**: to differentiate *M. leprae* (red) from *M. tuberculosis* (blue)
- **Fite Faraco's**: *M. leprae* (used as counterstain is hematoxylin)

STUDY OF CULTURAL/ COLONIAL CHARACTERISTICS/ PHENOTYPICAL ORGANIZATION

- Culture media: contains the cultural requirements needed for bacterial growth
- Culture: growth of microorganism in culture medium
- Types of culture
  1. Pure culture: contains only 1 species of organism
  2. Mixed culture: contains more than 1 species
  3. Stock culture: often used in research
- ATCC: American-type culture collection
- Best way to obtain pure culture is through **streak plate method**
- Other methods of obtaining pure culture: pour plate, serial dilution
- Steps to prepare media:
  - Plated: initially weigh, dissolve, sterilize, dispense
  - Tubed: weigh, dissolve, dispense, sterilize
- Water to make agar: deionized/distilled water

ACCORDING TO PHYSICAL STATE/CONSISTENCY

	Solidifying agent/agar	Examples
Liquid	None 0% agar	- Brain-heart infusion - Nutrient broth - Alkaline peptone water
Semi-solid	0.5-1% agar	- SIM
Solid	2-3% agar	**Liquefiable** - When heated again, they liquefy - EMB, Mac-Conkey, MSA, SSA **Non-liquefiable** - When heated again, they no longer liquefy - Rice medium or Rice grain agar (for fungi)
Biphasic		- Both solid and liquid - Castaneda (for *Brucella*) - Human blood bilayer tween

- Agar
  - Usual solidifying agent
  - Usually derived from red algae
  - Solidifies at 40-50°C
  - Melts at: 80-90°C
  - Cooling temperature for distribution of media to plate: 55-60°C
  - Amount dispensed: 25mL per plate

ACCORDING TO COMPOSITION

1. **SYNTHETIC/DEFINED**
   - All components are known to the user
   - For research
   - Ex. BG-11 for the isolation of Cyanobacteria
2. **NON-SYNTHETIC/COMPLEX**
   - Composed of some unknown substance (peptone, meat)
   - Useful for bacterial isolation
   - EMB, MacConkey, Blood agar, MSA
3. **TISSUE CULTURE**
   - For viral culture; for virus isolation we use living cells to allow growth of viruses
   - In tissue culture, we wait for: cytopathic effect (CPE)
     - Cytopathic effect: change in the nucleus/cytoplasm of cells due to the presence of virus
   - Types of tissue culture:
     - Primary cell line – uses normal mature adult cells (ex. Primary monkey kidney)
     - Semi-continuous/diploid – uses embryonic or fetal cells
     - Continuous/heteroploid – uses malignant, cancerous, or tumor cells
       - HeLa cells: uses cervical cancer cells
       - A549: uses cells from lung carcinoma
       - Hep2 cells: uses laryngeal cancer cells
   - Vero cells: from African green monkey kidney cells
   - McCoy cells: for Chlamydia

ACCORDING TO USE

1. **GENERAL PURPOSE/GENERAL ISOLATION MEDIA**
   - Contains only the necessary nutrition to support bacterial growth
   - Does not contain any supplement
   - For routine cultivation of bacteria
   - I.e. nutrient agar and nutrient broth
2. **ENRICHMENT MEDIA**
   - Type of media used to enhance bacterial growth
   - When we place organisms in this type of media, bacterial yield is also increased
   - Ex. Alkaline peptone water for *Vibrio*, and medias for the isolation of gram negative bacteria such as Selenite broth and Tetrathionate broth
3. **ENRICHED MEDIA**
   - Type of media used for fastidious organisms (organisms which are unable to grow in ordinary media)
   - Usually contains blood
   - Fastidious organisms: *Haemophilus* and *Neisseria*
   - Sheep's blood agar: for *Streptococci*
   - Horse blood agar: for *Haemophilus*
   - Human blood agar: for *Gardnerella vaginalis*
   - Chocolate blood agar: for *Neisseria*
4. **SELECTIVE MEDIA**
   - Promotes the growth of the desired organism while inhibiting the growth of others
   - Contains inhibitors (to make the media selective)
   - i.e. antibiotics

Examples of inhibitors for gram positive organisms	Examples of inhibitors for gram negative organisms
- Bile salts: such as sodium desoxycholate - Dyes: such as crystal violet (incorporated in MacConkey agar)	- Potassium tellurite

Examples of selective media:

Lowenstein Jensen	for *M. tuberculosis*; inhibitor is Malachite Green
Cystine Tellurite Blood Agar (CTBA)	for *C. diphtheriae;* inhibitor is Potassium tellurite
Gentamicin blood agar	For *Streptococci* Bacitracin Chocolate agar – for *Haemophilus*
Blood agar with ampicillin	Aeromonas
Chloral hydrate	Prevent swarming organisms

5. **DIFFERENTIAL MEDIA**
   - Used to differentiate organisms that are growing together
   - Ex. EMB and MacConkey (differentiation of lactose and non-lactose fermenting organisms), TCBS (differentiate *Vibrios* which are sucrose and non-sucrose fermenters)
6. **SELECTIVE AND DIFFERENTIAL**
   - EMB & MacConkey: Selective because they only allow the growth of gram negative bacilli; Differential because they can differentiate lactose and non-lactose fermenters
7. **TRANSPORT MEDIA**
   - May be required to maintain the viability of organisms during transport
   - Maintains moist environment, with nutrients to maintain growth
     - Transgrow/JEMBEC: for *Neisseria*
     - Stuart's: Viral transport media
     - Cary Blair: for stool pathogens
8. **MEDIA FOR SUSCEPTIBILITY TESTING**
   - Mueller-Hinton Agar

OTHERS

1. **TODD-HEWITT BROTH**
   - Selective enrichment media for *S. agalactiae* in female genital specimens
   - Used to detect genital carriage of Group B Streptococci during pregnancy
2. **CNA (COLUMBIA-COLISTIN-NALIDIXIC ACID)**
   - For gram positive organisms

Other commonly used media:

1. **SHEEP'S BLOOD AGAR**
   - For *Streptococci*
   - To determine hemolytic pattern
2. **PEA (PHENYL ETHYL ALCOHOL) AGAR**
   - For gram positive cocci and other anaerobic gram negative rods

Additional info!

- Bacterial cultures are held for: 48 to 72 hours
  - Anaerobes and broth culture are held for: 5 to 7 days
- In **THIOGLYCOLLATE** – enrichment broth
  - Gram negative facultative anaerobes: diffuse even growth all throughout the media
  - Strict aerobes: growth towards the surface
  - Anaerobes: growth at the bottom

SUSCEPTIBILITY TESTING

- To detect ability of antimicrobial agent to inhibit bacterial growth in vitro
- To determine susceptibility/resistance of organism against antimicrobial agents

METHODS

1. **Dilution method**
   - Standard inoculum for broth dilution: 5 x 10^5 CFU/mL
   - Standard inoculum for agar dilution: 1 x 10^4 CFU/mL
   - Total broth volume
     - Microdilution: 0.05-0.1 mL
     - Macrodilution: 1mL or greater
   - **TUBE DILUTION**
     - A quantitative technique in which measured amounts of antibiotics are prepared by serial dilution to which is added specific concentration of the suspected bacteria
     - Serial dilution of antibiotic is prepared, to each tube a uniform amount of inoculum is added
     - Process:
       - Several tubes are prepared for every antibiotic to be tested
       - Prepared different concentrations of antibiotics through serial dilution
       - Least diluted is tube 1, most diluted is tube 10
       - Add the same amount of inoculum/organism to all tubes
       - Incubate at 37°C
       - Identify tube with no growth/no turbidity = MIC
     - MIC: Minimum inhibitory concentration; lowest concentration of antibiotic that inhibited bacterial growth
     - MBC/MLC: Minimum bactericidal concentration/minimum lethal concentration; lowest concentration of antibiotic that killed the organism
   - **AGAR DILUTION METHOD**
     - Same procedure as tube dilution but must be done on petri dishes
2. **Disk diffusion method**
   - A qualitative technique
   - Kirby Bauer Method

KIRBY BAUER METHOD

1. Preparation of media
   - Media: MHA (Mueller Hinton Agar)
   - Depth of agar: 4mm
   - pH of agar: 7.2-7.4
   - Uses filter paper disks/size: 6mm

- **MHA**: standard media for Disk diffusion method
- **MHA w/ 5% sheep blood**: for susceptibility testing of *Streptococcus and Neisseria meningitidis*
- **MHA w/ 2% NaCl**: susceptibility testing of MRSA and other *Staphylococci*
- **Middlebrook 7H10**: isolates *Mycobacteria* and susceptibility testing of *Mycobacterium* only

2. Preparation of inoculum
   - Throat swab on BHI/TSB; Incubate for 24 hours at 37°C
     - Turbid = (+) growth
   - Subculture 4-5 colonies on TSB
   - Incubate at 37°C for 3-5 hours or until the turbidity of the subculture matches the turbidity of the McFarland standard
     - Too turbid = dilute
   - Compare turbidity of subculture with McFarland standard purpose: in order to standardize the inoculum
   - 0.5 McFarland Standard
     - 0.5 mL of 1.175% Barium chloride
     - 99.5 mL of 1% sulfuric acid
     - Equivalent of McFarland Standard: 1.5 x 10^8 CFU/mL
   - If turbidity is ok, inoculate on MHA (using sterile cotton swab)
   - If too turbid, dilute using NSS or distilled water (overlap streaking)
   - Wait for 3-5 minutes before applying the disk
3. Plate size & number of disks; distance of disks
   - If plate size is 150 mm place no more than 12 disks
   - If plate size is 100 mm place no more than 5 disks
   - Distance of disk from center is 24 mm, between 2 disks is 15mm
   - Invert plates and incubate 35- 37°C; incubation time: 16-18 hours
   - Aerobic incubation no CO2
   - Measure zone of inhibition using a ruler or caliper, at the underside of the plate, and interpret if S, R or intermediate
   - For media with blood: measure the ZOI from the top with cover removed
   - For Streptococci, MHA is supplemented with 5% Sheep's blood

Possible Sources of error

FALSE SENSITIVE	FALSE RESISTANT
1. Too light inoculum 2. Thin agar 3. Very dry agar 4. Acid pH affect result in Tetracycline, Novobiocin, Methicillin 5. Alkaline pH affect result in aminoglycosides/erythromycin	1. Too heavy inoculum 2. Thick agar 3. Too much moisture on surface of agar 4. Prolonged incubation

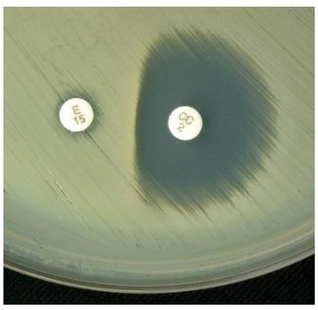
- Use of mixed culture
- Improper storage of disk
  - Indicator of improper storage: Penicillin and Methicillin
- Storage temp for antibiotics
  - WORKING SUPPLY
  - Kept inside the refrigerator (4-8°C or 2-8°C) or -20°C in a desiccator
- If there is swarming: ignore
- When using SULFONAMIDES (such as SXT) if there are 2 concentric zones measure the outer zone

Additional notes!

- Antibiotics (derived from bacteria or fungi): Streptomycin and Penicillin
- Chemotherapeutics: chemically produced (ex. SXT)
- Narrow: has limited range of action
- Broad spectrum: can act against a number of organisms
- Bacteriostatic: inhibit bacterial growth (ex. chloramphenicol, tetracycline, clindamycin)
- Bactericidal: kill the organism (ex. beta lactams)

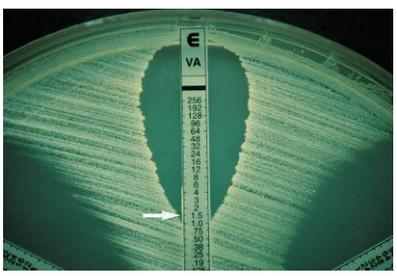
What is "D" test?

- Not usually done in the lab; susceptibility and sensitivity test
- Double disk diffusion test
- To detect inducible clindamycin resistance among strains of *S. aureus*
- Uses: 15 ug erythromycin + 2 ug clindamycin
- Positive result: appear as if it is sensitive; blunting/flattening of clindamycin zone to have "D pattern"
- Negative: erythromycin zone should appear as if it is resistant
- Report:
  - Negative = clindamycin sensitive
  - Positive = clindamycin resistant



What is "E" test?

- A susceptibility test, a quantitative method, and a dilution method
- a.k.a. MIC on a stick
- Uses a strip with single antibiotic of decreasing concentration along its length
- For fastidious organisms and anaerobes
- Positive result: ellipse of growth inhibition



BACTERIAL IDENTIFICATION	

CONVENTIONAL	- Microscopy is performed
	  - Motility
	  - Staining
	- Use of selective medium
	- Manual biochemical tests

ANALYTICAL PROFILE INDEX	- Semi-automated procedure - Biochemical tests are still performed but made smaller - Uses plastic strips and microtubes with biochemical substrates - The biochemical substrates are inoculated with pure culture suspension - Ex. API 20E (for the identification of Enterics)
AUTOMATED	- Vitek 1 & 2 (Automation program for bacterial identification and susceptibility)

ANTIBIOTICS

MODE OF ACTION		CLASS	REPRESENTATIVE ANTIBIOTICS	
Inhibit cell wall synthesis	Beta lactams	Natural penicillins	Penicillin V Penicillin G	Allergic reaction is a common side effect
		Synthetic penicillins	Methicillin Oxacillin Ampicillin Carbenicillin Piperacillin	
		Cephalosporins	Cephalexin Ceftriaxone Cefotoxin Cefepime	
		Carbapanems	Imipenem	
		Monobactams	Aztreonam	
		Glycopeptides	Vancomycin	Drug of choice for MRSA; Some strains of *Enterococcus* are now resistant
Inhibit protein synthesis	Macrolide		Erythromycin Clarithromycin Azithromycin	Bacteriostatic
	Tetracycline		Tetracycline Doxycycline	For intracellular organisms, spirochetes Bacteriostatic Can stain teeth, abnormal bone growth
	Aminoglycosides		Gentamycin Amikacin Tobramycin Kanamycin	Toxic to kidneys and ears
	Chloramphenicol			
Inhibit folic acid synthesis	Sulfonamides		Sulfamethoxazole Trimethoprim	Bacteriostatic Used primarily for UTI
Inhibit DNA synthesis	Quinolones		Ciprofloxacin Levofloxacin Ofloxacin	Ciprofloxacin used for anthrax
Cell membrane inhibitors			Bacitracin Colistin Polymixin Amphotericin B Nystatin	

RECALLS FROM PREVIOUS BOARD EXAMS

Ethylene oxide	Chemical sterilant	Critical step in gram staining	Decolorization
Increase in size	Lag phase	Bench marking is also known as	Peer comparison
0.5 McFarland	Sulfuric acid + Barium chloride	*S. viridans*	Subacute bacterial endocarditis
121°C, 15 psi, 15 minutes	Autoclaving	Species are written in	Lowercase
Incubation at 35-37°C	Bacterial and viral culture	Water used to make agar	Deionized water
Bacteria are attracted to	Moisture	To measure zone of inhibition	Use calipers or rulers
Dye retained by AFO	Carbolfuchsin	Automation program for susceptibility	VITEK
Kirby Bauer	Filter paper disk	Bacteria with the same size as virus	*Mycoplasma*
B. anthracis uses	BSC Class 2	Commonly used to test for sensitivity nowadays	VITEK
Disinfectant	10% household bleach	Alternative if there is no Clorox in the lab	Vinegar
Throat pathogen	*S. pyogenes*	Iodine + detergent	Iodophor
#1 throat flora	*Viridans Streptococci*	India ink	Negative stain Capsular stain for *C. neoformans*
Exhaust air outside the building	BSC Class 2b	7.5% salt tolerated by	*S. aureus*
Disinfectant for machine that cannot be autoclaved	Ethylene oxide	Capsular swelling test	Neufeld Quellung
Transport media for *S. agalactiae* for fluorescence	Todd Hewitt Broth	Metachromatic granules of *C. diphtheriae*	Babes Ernst granules
Lowest concentration to kill bacteria	Minimum Bactericidal/Lethal Concentration	Effect of too much moisture in disk diffusion	False resistant

Fluorescent stain will require what type of microscope	Darkfield microscope	Bacteria with darting motility	*Campylobacter*
CNA is for	Gram (+) cocci	Unique in cell wall of Mycobacteria	Mycolic acid
CSF storage temperature	37°C	Schaeffer and Fulton	Spore stain
Tumbling motility at 25°C	*Listeria*	JEMBEC	*Neisseria*
Best disinfectant for blood spillage	10% household bleach	VITEK system	Bacterial identification and sensitivity
Soap	Germicidal	Urine CS for ambulatory patients	Random urine
Treatment for MRSA	Vancomycin	Urine CS for women	Clean catch midstream
Cold AF stain	Kinyoun's	Smears of CSF are prepared from	Sediments
Possible result in antibiotic susceptibility testing	Susceptible, intermediate, resistant	Purpose of boiling thioglycollate	Drive off oxygen
Todd Hewitt broth is for	*S. agalactiae*	Chain of infection	Source – mode of transmission & susceptible host

GRAM POSITIVE COCCI

- *Staphylococci, Micrococci, Streptococci*
- *Pneumococci, Viridans Streptococci*

STAPHYLOCOCCI

- Gram positive cocci in grape-like clusters; Non-motile
- Catalase test (+)
- Able to grow in 7.5-10% NaCl
- Resistant to 0.02-0.04 units of bacitracin
  - Bacitracin = also referred to as Taxo A

As to pigment formation/growth on **Loeffler's Serum Slant (LSS)**

Staphylococcus aureus	*Staphylococcus citreus*	*Staphylococcus albus*
Golden yellow pigment	Lemon-yellow pigment	White pigment

CATALASE TEST

- A presumptive biochemical test to identify/detect the genus *Staphylococci* and *Micrococci*
- Uses 3% hydrogen peroxide (for *Staphylococci*)
  - 30% H_2O_2: used in Superoxol test; presumptive test for *Neisseria* and *Mycobacterium* (heat stable catalase test)
  - 15% H_2O_2: for anaerobes
- Can be performed on slide or in tube
- Principle:
  - Catalase expedites the breakdown of hydrogen peroxide into water and oxygen
- Procedure *(Delost)*:
  - Transfer a small portion of a well-isolated colony that is 18 to 24 hours old onto a clean microscope slide.
  - Add one to two drops of 3% hydrogen peroxide.
  - Observe for the formation of rapid and continuous bubbles.
- Positive result: **vigorous bubbling/effervescence**
  - The organism is able to produce the enzyme catalase
  - Why bubbling: Hydrogen peroxide, through the action of the enzyme catalase, will be broken down into water and oxygen; the release of oxygen will cause bubbling
- Use of colonies from BAP: False positive; colonies from BAP are not used

POSITIVE	NEGATIVE
Staphylococci *Coagulase Negative Staphylococcus* *Micrococcus*	*Streptococci*

Staphylococcus aureus

- Medically/Clinically significant
- Normal flora of anterior nares and nasopharynx
- PCR will enable detection from nasal swab
- Can cause toxin-mediated diseases and non-toxin mediated diseases
- Facultative anaerobe

GROWTH/COLONY MORPHOLOGY

- Uniform turbidity on broth
- On plates, they produce smooth circular and opaque colonies with oil-like or butyrous appearance
- Jet black colonies on Tellurite agar/Vogel Johnson medium

TESTS FOR IDENTIFICATION

- Catalase (+)
- Bound & Free Coagulase (+)
- Mannitol fermentation (+)
- DNAse (+)
- PYR (-)
- VP and Nitrate (+)
- Latex agglutination test for protein A (+)
- Beta-hemolytic on 5% sheep blood agar
- Growth and fermentation in MSA

COAGULASE (+)

- Definitive test for *S. aureus*
- Biochemical test used to differentiate the pathogenic *S. aureus* from the other *Staphylococci*
- Reagent: Rabbit's plasma obtained using EDTA
- Use of citrated plasma: false positive
- Slide coagulase = detects clumping factor (bound coagulase)
- Tube coagulase = detects free coagulase

SLIDE COAGULASE	TUBE COAGULASE
Bacteria + NSS + Rabbit plasma obtained using EDTA = fibrin clot (+) Note: if negative, perform the tube coagulase	Bacteria + rabbit plasma Incubate at 37°C for 4 hours If negative, incubate at room temperature for a total of 20 hours
Positive result: clot formation	

- Bound coagulase: produce clot formation immediately
- Free coagulase: produce clot formation after incubation
- If positive for coagulase test: you may report it as *S. aureus* already; if you are not convinced yet, you may perform mannitol fermentation test

MANNITOL FERMENTATION TEST (+)

- Media used: Mannitol Salt Agar (MSA)
- Inhibitor: 7.5-10% NaCl; inhibitory agent for gram negative and other bacteria
- Indicator: phenol red
- Colony odor: similar to an old sock
- Positive result:
  - Development of yellow halo around the colonies (fermenters)
    - Organism was able to ferment mannitol
    - Organism was able to tolerate high concentration of salt

Additional notes!

- Fermentation: organism is capable of producing acid in the presence or absence of air
- Oxidation: organism is capable of producing acid only in the presence of air

DNASE TEST/THERMONUCLEASE (+)

- A.k.a Thermonuclease test
- Detects deoxyribonuclease
- Used to determine the ability of an organism to hydrolyze DNA and utilize it as a source of carbon and energy for growth
- Not specific for *S. aureus; Moraxella and Serratia* are also positive for DNAse
- **Dye method**
  - DNAse agar with dye is used
  - Dye: methyl green or toluidine blue
    - Positive: clear zone around the colonies (if methyl green is used)
    - Positive: pink color (if toluidine blue is used)
- **HCl precipitation method**
  - Inoculate using DNAse agar
  - Add 0.1N HCl after incubation
  - Positive result: clearing of agar around the colonies

SUSCEPTIBILITY TESTS

- Sensitive to: Lysostaphin and Novobiocin
- Resistant to: Polymyxin

TEST FOR BETA LACTAMASE PRODUCTION: CEPHALOSPORINASE TEST

- Used to detect the ability of the organism to produce the enzyme beta lactamase
- Uses: Cefinase disk
- Impregnated substrate: Nitrocefin
- Procedure:
  - Inoculum is placed on disk; Wait for less than 10 minutes for a change in color
- Result:
  - Positive result: Pink to Red (change in color) within 60 minutes
  - Negative: Yellow (no change in color)

VIRULENCE FACTORS	
ENZYMES	
Lipase	- Fat splitting enzyme - Initiates skin infection
Hyaluronidase	- Also known as: Duran Raynal factor - Spreading factor; once released, it enhances the ability of *S. aureus* to invade tissues

Beta lactamase	- A.k.a penicillinase
	- Responsible for *S. aureus* resistance to penicillin and beta lactam antibiotics
	- Beta-lactam:
	  - Example: Cephalosphorins
	  - Cell wall inhibitors; kill bacteria by damaging the cell wall

Staphylokinase	- Promotes fibrinolysis - Responsible for the dissolution of clot
Coagulase	- **Major virulence factor** - Causes bacterial cell to agglutinate in plasma - Converts fibrinogen to fibrin - Considered as marker of virulence
DNAse/ Thermonuclease	- Degradation of DNA - Decreases viscosity of exudates allowing more mobility
Beta hemolysin	- Sphingomyelinase C - Hot-cold lysin - Responsible for the beta hemolysis shown by *S. aureus*; destruction of RBCs
Protein A	- Interferes with phagocytosis

TOXINS	
Enterotoxins A & B	- Food poisoning
TSST-1	- Enterotoxin F - Toxic Shock Syndrome Toxin-1; Superantigen - Causes Toxic Shock Syndrome
Exfoliatin	- Causes skin desquamation/exfoliation in scalded skin syndrome/Ritter's disease
PVL/Panton Valentine Leukocidin	- Destruction of WBCs

- Catalase (+): although it is an enzyme, it is not a virulence factor

Toxin mediated diseases	Non-toxin mediated diseases

- Food poisoning
  ○ Enterotoxin A and B
- Toxic shock syndrome
  ○ TSST-1
- Scalded skin syndrome/Ritter's disease
  ○ Exfoliatin
  ○ May occur in newborns
	- Boils, carbuncles, furuncles, cellulitis, wound infections
	- Bullous impetigo
	  ○ Skin infection affecting the epidermis
	- Sty (eye infection that develops at the edge of eyelids; *S. aureus* is an ocular pathogen)

- Drug of choice for treatment of *Staphylococcal* infections are penicillinase resistant drugs: **oxacillin, cloxacillin, methicillin**

MRSA/METHICILLIN RESISTANT *Staphylococcus aureus*

- A.k.a Oxacillin-resistant *S. aureus*
- Strain of *S. aureus* resistant to Methicillin, Nafcillin, and Oxacillin
- Resistance of *Staphylococcus* to penicillinase resistant penicillin is due to PBP2A (Penicillin binding protein 2A) in cell wall, encoded by mecA gene

To detect oxacillin resistance in *S. aureus*

1. (+) Chromogenic test – uses Chrom agar
   - MRSA: development of mauve-rose color
   - Non-MRSA: No growth; colorless or blue colonies
2. Cefoxitin Disk Diffusion test – induces expression of PBP2A
3. D-test
4. PCR
   - To detect the gene that codes for resistance
   - Specimen: nasal swab
   - Gold standard for MRSA detection

MRSA – Methicillin Resistant *Staphylococcus aureus*	MSSA – Methicillin Sensitive *Staphylococcus aureus*
Penicillin resistant Oxacillin resistant Cefoxitin resistant	Penicillin sensitive Oxacillin sensitive Cefoxitin test negative

May be mistaken for *S. aureus*

S. lugdunensis	*S. intermedius*	*S. haemolyticus*
- Slide coagulase (+) - PYR (+) - Narrow beta hemolytic - Mannitol (+) - Can cause endocarditis and bacteremia	- Slide coagulase (+) - PYR (+) - VP (-)	- Beta hemolytic - Coagulase (-)

COAGULASE NEGATIVE STAPHYLOCOCCI (CoNS)

- Catalase (+); Coagulase (-)
  - *Staphylococcus saprophyticus*
  - *Staphylococcus epidermidis*

	S. aureus	*S. epidermidis*	*S. saprophyticus*
Colony	Yellow colonies	White colonies	White colonies
Catalase test	+	+	+
Coagulase test	+	-	-
Mannitol fermentation	+	-	-
Hemolysis on BAP	Beta	Gamma	Gamma
Novobiocin susceptibility (5 ug) – *Differential test for CoNS*	Susceptible	Susceptible (>16 mm)	Resistant (<16 mm)
DNAse test	+	-	-
Phosphatase	+	+	-
Gelatinase	+	+	+
Others		- Normal flora of the skin - Causes UTI - Stitch abscess - Prosthetic heart valve infection	- Most common cause of UTI in young women - Causes pyelonephritis and cystitis in those with indwelling catheters
Virulence factors		Slime production or biofilm formation	Uncertain

- Biofilms: will enhance attachment of organism to plastic catheters

MICROCOCCI

- Gram positive cocci in tetrads; cuboidal packets
- **Strict aerobe**
- On BAP = gamma hemolysis
- Modified oxidase test (+)/Microdase (+)
  - Presumptive test for *Micrococci*
  - Use of 6% modified oxidase reagent
    - Tetramethyl p – phenylenediamine dihydrochloride in DMSO (Dimethylsulfoxide)
  - Result:
    - Positive result: development of blue color (blue or purple-blue color change within 2 minutes)
    - Negative: no change in color

	Staphylococci	*Micrococci*
Aerobic growth	+	+
Anaerobic growth	+	-
Lysostaphin susceptibility (200 mg/mL)	S	R
Modified oxidase test	-	+
Bacitracin susceptibility (0.04 U)	R	S
Furazolidone/Furoxone susceptibility	S	R
Catalase test	+	+
Benzidine test	-	+
Glucose utilization (of media)	Fermenter	Oxidizer

STOMATOCOCCUS (Now: ROTHIA)

- An emerging pathogen of immunocompromised patients causing endocarditis and bacteremia
- Modified oxidase test (-)
- Bacitracin, furazolidone, and Lysostaphin resistant
- Fermentative
- Unable to grow in 5% sodium chloride
- Encapsulated
- Weakly catalase test (+)
- Coagulase negative

STREPTOCOCCI

- Gram positive cocci in chains, spherical to ovoid
- Non-motile; Capnophilic – increased CO_2
- Catalase (-)
- Medium of choice: Sheep's blood agar
  - First step in identification:
    - Inoculate in Sheep's blood agar
    - Starting point in identification of Streptococci: determine the hemolysis pattern
- Selective medium: Phenyl ethyl alcohol agar

SMITH & BROWN CLASSIFICATION	
Alpha hemolytic	- Can cause incomplete/partial hemolysis on BAP - Can cause **greening of agar** - *Streptococcus pneumoniae* - Viridans Streptococci: *S. mitis & S. mutans*
Beta hemolytic	- Can cause complete hemolysis on BAP - Can produce **clear zones** around the colonies - *S. pyogenes and S. agalactiae*
Gamma hemolytic	- Do not cause hemolysis on BAP - *E. faecalis, E. faecium, & S. bovis*
Alpha prime	- Has 2 patterns of hemolysis - Colony is surrounded by inner alpha and outer beta - No single organism is reported as alpha prime - Usually happens because of prolonged refrigeration

ACADEMIC OR BERGEY'S CLASSIFICATION	
Pyogenic	- Unable to grow at 10°C and 45°C - i.e. *S. pyogenes*
Viridans	- Able to grow at 45°C but not at 10°C - i.e. *S. mitis, S. mutans, S. sanguis, S. salivarius, S. constellatus, and S. intermedius*
Enterococci	- Able to grow at 10 and 45°C - i.e. *E. faecalis*
Lactic	- Able to grow at 10°C but not at 45°C - i.e. *S. lactis* and *S. cremoris*

Lancefield classification

- Most significant in identifying and classifying beta hemolytic Streptococci
- Based on the extraction of C carbohydrate from the Streptococcal cell wall
- All Streptococci except Viridans and *S. pneumoniae* are placed under Lancefield

LANCEFIELD CLASSIFICATION (BETA HEMOLYTIC)	
Group A	- *S. pyogenes*
Group B	- *S. agalactiae*
Group C, F, G	- Normal flora: Skin, Nasopharynx, Gastrointestinal tract, Genital tract

GROUP A STREPTOCOCCI - *Streptococcus pyogenes*

- Flesh eating bacteria
- Major throat pathogen
- Diseases caused: pharyngitis /Strep throat; Necrotizing fasciitis, Erysipelas (a skin infection), Scarlet fever
  - Necrotizing fasciitis: a.k.a galloping gangrene; rapidly progressing skin infection
  - Erysipelas are different from Erysipeloid
    - Erysipeloid is caused by *Erysipelothrix rhusiopathiae*
  - Scarlet fever: a form of pharyngitis with rashes; manifest with "strawberry tongue"
- Repeated episodes of pharyngitis causes post-sequalae complications:
  - Acute glomerulonephritis or Bright's disease
  - Rheumatic fever
- Growth/Colony morphology: best isolated on BAP, addition of SXT can enhance its recovery from heavily contaminated specimens
- Media for Initial isolation: Sheep's blood agar

Tests for identification/Susceptibility test

- **Bacitracin (0.02-0.04 units)** – Sensitive/Susceptible
- **PYR (+)**/Pyrrolidonyl Arylamidase (PYR) Test
  - Used for the presumptive identification of group A beta-hemolytic *Streptococci* and *Enterococci*.
  - Reagent: p-dimethylaminocinnamaldehyde
  - Procedure: Get filter paper, place inoculum in the filter paper, add reagent to the colonies
  - Positive: development of red color

VIRULENCE FACTORS OF *S. pyogenes*	
Streptolysin	- Responsible for the beta hemolysis of *S. pyogenes*
Streptolysin O	- Antigenic/Immunogenic (stimulates the body to produce anti-streptolysin O or ASO) - Oxygen labile - Can cause sub-surface hemolysis on BAP (hemolysis is produced when incubated anaerobically)
Streptolysin S	- Non-antigenic - Oxygen stable - Can cause surface hemolysis (hemolysis is produced only when incubated aerobically)
Erythrogenic Toxin or Pyrogenic Toxin	- Causes rashes in scarlet fever
M protein	- **Major virulence factor** - Prevents phagocytosis - Found in the cell wall of *S. pyogenes*
Protein F	- Promotes/enhances attachment to host/epithelial cells
Others	- **Hyaluronidase**: spreading factor; enhances ability of the organism to invade tissues - **Streptokinase**: promotes fibrinolysis; responsible for the dissolution of clots

Diagnostic tests for Scarlet fever	
Dick's test	- Susceptibility test for Scarlet fever - To determine if the patient is at risk for developing scarlet fever - Positive: erythema (redness) at the site of injection
Schultz Charlton Test Or Blanche Phenomenon	- Test to determine if rashes are due to scarlet fever or not - (+) result: gradual disappearance or fading of rashes following the injection of anti-erythrogenic toxin - This means that the erythrogenic toxin was neutralized by anti-erythrogenic toxin

GROUP B STREPTOCOCCI - *Streptococcus agalactiae*

- A.k.a. Group B-Beta Hemolytic Streptococci
- Normal flora of the GIT, pharynx, and vaginal tract
- Causes septicemia
- Regarded as the # 1 cause of Neonatal meningitis
- In adults: postpartum endometriosis
- ID tests:
  - CAMP test (+)
  - Hippurate Hydrolysis (+)
  - Bacitracin & SXT RESISTANT
  - LIM broth, Granada – may be used for Group B Streptococci
- Major virulence factor: capsule (prevents phagocytosis)

CAMP TEST

- Media: BAP
- **Known organism**: *S. aureus* (beta hemolytic)
  - *S. aureus* is recommended to be used (Single streak)
  - Reverse CAMP test: for *C. perfringens*
  - *Listeria* – CAMP positive
    - *Rhodococcus equi* is used as the known organism if *Listeria* is suspected
- **Unknown organism**: Suspected *S. agalactiae* (beta)
  - Perpendicular streak
- (+) result: enhanced hemolysis as shown by arrowhead zone of beta hemolysis
- (-) result: no enhanced hemolysis; but there is still hemolysis

HIPPURATE TEST

- Detects hydrolysis of sodium hippurate to benzoic acid and glycine
- The positive result is due to the enzyme: hippuricase/hippurate hydrolase
- The reagent sodium hippurate is hydrolyzed into two end products: **benzoic acid and glycine**, in the presence of the enzyme hippuricase/hippurate hydrolase
  - Traditionally, ferric chloride is used to detect benzoic acid
  - Nowadays, (rapid test) ninhydrin reagent is used to detect glycine
  - Best answer if both ferric chloride and ninhydrin reagent are both in the choices: **Ninhydrin reagent**
- Positive result: development of purple color

GROUP C STREPTOCOCCI

- These are animal pathogens
- May cause severe pharyngitis followed by bacteremia, may also cause pneumoniae, cellulites, & abscess (*S. equisimilis, S. equis, S. dysagalactiae, S. zooepidemicus*)
- **Group F:** *S. anginosus* – causes cellulites
- **Group G:** usually found as a normal flora of human GIT, vaginal tract, and oropharynx

GROUP D - NON-ENTEROCOCCI & ENTEROCOCCI

- **Bile Esculin test**: screening test for *Group D Non-Enterococci and Enterococci*
  - 40% bile; uses Bile esculin agar (dispensed as slant)
  - Indicator: ferric ammonium citrate
  - (+) result: blackening of media
  - Positive: Both *Enterococci and Non-Enterococci*

DIFFERENTIAL TEST FOR NON-ENTEROCOCCI & ENTEROCOCCI		
	NON-ENTEROCOCCI	***ENTEROCOCCI***
Bile Esculin	+	+
Salt Tolerance Test using 6.5% NaCl	-	+
Penicillin Disk Test	S	R
PYR Test	-	+

Group D Streptococci (Group D-Non-Enterococci) *Now: S. bovis group or S. gallolyticus group*	**Enterococci** *E. faecalis* *E. faecium* *E. durans* *E. avium*
- *S. bovis and S. equines* - Growth inhibited in 6.5% NaCl - Penicillin sensitive - Vancomycin resistant - PYR test (-), Bile esculin (+) - Causes UTI, endocarditis, and septicemia - 45°C: +; can grow - 10°C: -; unable to grow - Isolation of these in the blood maybe an indication of colon cancer - Bacteria from cows causing colon cancer: *S. bovis*	- Normal GUT, GIT, and RT flora - Grows in 6.5% NaCl - Penicillin and Cephalosporin resistant - Bacitracin-R - Bile esculin and PYR tests (+) - Hippurate hydrolysis (-) - Most commonly isolated from human infection: *E. faecalis* and *E. faecium* - Specimen: blood urine, wound secretion - May cause UTI in hospitalized patients - 45°C: + - 10°C: +

PYR test (+)	Group A, Group D Enterococci, and *S. urinalis*
Hippurate (+)	Group B and Group D Enterococci

ALPHA HEMOLYTIC STREPTOCOCCI

- Viridans Streptococci *(Streptococcus mitis, S. salivarius. S. sanguis, S. mutans)*
- *Streptococcus pneumoniae*

Streptococcus pneumoniae

- *Diplococcus pneumoniae, Pneumococci*
- Morphology: Lancet-shaped or Bullet-shaped diplococci
- Normal flora of nasopharynx and oropharynx (URT)
- Regarded as # 1 cause of adult bacterial meningitis (adults who are above 29 years old)
- Also, an agent of Lobar pneumonia (rusty colored sputum), Otitis media, and pericarditis
- On plated media, young colonies are DOME SHAPED after 24 hours' incubation, but these dome shaped colonies collapse on prolonged incubation (after 48 hours) forming the so-called:
  - CRATER-LIKE COLONIES
  - NAIL HEAD COLONIES
  - Colonies with CHECKER APPEARANCE
- Capnophilic: requires 5-10% CO_2
- Tests for identification and susceptibility
  - Bile solubility test (+)
  - Susceptible to optochin
  - Capsular swelling test (+)
- Specimen: sputum, blood, CSF
- Francis test: skin test for pneumonia

Bile Solubility Test

- Media: BAP; To be added to the colonies: 10% sodium desoxycholate
  - (+) result: lysis of colonies
  - (-) result: intact colonies
- If on broth (2% sodium desoxycholate)
  - (+) result: clearing of media
  - (-) result: turbidity or partial clearing

Optochin Disk Test

- More than 14 mm zone
- Test that will differentiate *Pneumococci* from other alpha hemolytic *Streptococci*
- Chemical name: Ethyl-hydrocupreine hydrochloride; Taxo P
- 10 ug: sensitive if 14mm
- 6 ug: 10mm

Capsular swelling test

- Positive; Encapsulated
- The capsule is a major virulence factor

	Pneumococci	***Streptococci***
Bile solubility	+	-
Inulin fermentation	Ferment inulin	Does not ferment
Capsular swelling	+	-
Quinidine and Optochin	S	R

VIRIDANS STREPTOCOCCI

- Major throat flora
- Cause subacute bacterial endocarditis (SBE)
- Normal flora of the URT, GIT, and GUT
- Unique: ability to produce glucans and dextrans (enhances attachment of organisms on cell surfaces)
- Not classified under Lancefield
- Optochin-R
- Bile insoluble/Bile solubility: negative
- On BAP: Alpha hemolytic

Species

1. *Streptococcus mitis*
2. *S. salivarius*
3. *Streptococcus sanguis* causes: Subacute bacterial endocarditis due to dental procedure
4. *Streptococcus mutans* causes: dental caries and dental plaques

Tests	Group A	Group B	*S. pneumoniae*	Viridans	Group D *(previously S. bovis group)*	Enterococci
Bacitracin	S	R	S	R	R	R
SXT	R	V	S	S	R	R
Optochin	R	R	S	R	R	R
PYR	+	-	-	-	-	+
Bile Esculin	-	-	-	-	+	+
Growth in 6.5% NaCl	-	-	-	-	-	+

BETA HEMOLYTIC STREP			
	Group A	**Group B**	**Group C, F, G**
Bacitracin	S	R	R
SXT	R	R	S
CAMP	-	+	-
PYR	+	-	-

MUST REMEMBER		
S. aureus	Coagulase (+) Mannitol (+) DNAse (+) Bacitracin-R Beta hemolytic	
S. epidermidis	Novobiocin-S	In all tests where *S. aureus* is +, both are (-)
S. saprophyticus	Novobiocin-R	Gamma hemolytic
S. pyogenes	PYR (+) Bacitracin-S Beta hemolytic	
S. agalactiae	CAMP (+) Hippurate (+) Beta hemolytic	
S. pneumoniae	Bile solubility (+) Optochin-S Neufeld Quellung (+) Alpha hemolytic	
Viridans	Alpha hemolytic Bile insoluble Optochin-R	

*Colonies of *Staphylococci* are medium-sized and butyrous while those of *Streptococci* are small-pinpoint colonies

HEMOLYTIC PATTERN

Alpha	Beta	Gamma
S. pneumoniae *Viridans Streptococci*	*Group A, B, C, Some D*	*Mostly group D*

- **Milleri Streptococci Complex**, a.k.a. Microaerophilic Streptococci
  - Able to produce colonies with butterscotch or caramel odor
  - Produce liver, brain, and spleen, abscess
  - *S. constellatus, S. intermedius, S. anginosus*

STREP-LIKE ORGANISMS: NUTRITIONALLY VARIANT STREP

- *a.k.a. Pyridoxal Requiring Streptococci, Satelliting Streptococci, Thiol Requiring Streptococci, Cell wall deficient Streptococci*

Abiotrophia & Granulicatella

- These are *S. adjacens and S. defectives*
- These are Viridans Strep needing vitamin B6 for growth
- When mixed with *S. aureus* they satellite around colonies of *S. aureus*
- BAP and CAP negative

Other gram positive, catalase negative organisms isolated from clinical specimens:

- *Pediococcus, Lactococcus, Aerococcus, Leuconostoc & Rothia:* have low virulence potential and are pathogenic only in immunocompromised host
- *Pediococcus, Leuconostoc, & Enterococcus* are vancomycin: resistant

MISCELLANEOUS TEST

1. **VANCOMYCIN RESISTANCE**
   - For ID of *Pediococcus & Leuconostoc*
   - Differentiates *Pediococcus* = R from *S. viridans* = S
2. **LAP TEST (LEUCINE AMINO PEPTIDASE)**
   - Presumptive test for catalase negative, gram positive cocci
   - Involves detection of the enzyme: L
   - Substrate: Leucine-beta-naphthylamide
   - Reagent: cinnamaldehyde
   - (+) result: red; *E. faecalis* and *Pediococcus*
   - (-) no color or slight yellow; *Leuconostoc*

BIOCHEMICAL TEST	(+) CONTROL	(-) CONTROL
Catalase test	*S. aureus*	*S. pyogenes*
Coagulase test	*S. aureus*	*S. epidermidis*
Bacitracin disk test	*S. pyogenes* *M. luteus*	*S. agalactiae* *S. aureus*
CAMP test	*S. agalactiae*	*S. pyogenes*
Bile Esculin Test	*E. faecalis*	*E. coli* *S. pyogenes*
Bile Solubility Test	*S. pneumoniae*	*E. faecalis*
Butyrate disk test	*M. catarrhalis*	*N. gonorrhoeae*

- **Bacitracin disk test (Taxo A)**
  - To presumptively identify *S. pyogenes*
  - To differentiate *Staphylococci* (R) from *Micrococci* (S)
- **Butyrate disk test**
  - (+) result = blue, blue-violet, or indigo

GRAM NEGATIVE COCCI

- *Neisseria & Moraxella*

NEISSERIA

- Significant: *N. gonorrhoeae, N. meningitidis,* Pigmented *Neisseria*
- Non-motile, aerobic, capnophilic (need 5-10% CO_2)
- Cold Sensitive
- Non-hemolytic
- Most species are CHO fermenters
- Develops non-pigmented colonies
  - except: *N. flava, N. flavescens, N. subflava*
- All are catalase (+)
  - except: *N. elongata*
- All are Cytochrome Oxidase Test (+)
  - except: *N. elongata*

- Screening test: ***CYTOCHROME OXIDASE TEST***
  - Reagent: 1% oxidase reagent
  - Reagent chemical name: tetramethyl-p-phenylenediamine dihydrochoride
  - Procedure:
    - A loopful of organism is placed on a filter paper, and the reagent is added, wait for a change in color
    - (+) Result: purple
  - Modified oxidase = 6% reagent, no DMSO, positive is blue color (for *Micrococci* detection)

OXIDASE POSITIVE	*Pseudomonas* *Plesiomonas* *Aeromonas*	*Moraxella* *Vibrio* *Campylobacter*	*Helicobacter* *Neisseria*

- To speciate: ***CHO UTILIZATION TEST***
  - Media: Cystine Trypticase agar (CTA) + CHO
  - Indicator: phenol red
  - In an acid pH (+) media will turn YELLOW
- **CANDLE JAR**
  - May be used for *Neisseria* isolation because the use of this jar may provide the organism with increased CO_2
- **TRANSPORT MEDIA**
  - Ensures that the organism continuously receives increased CO2 during specimen transport
    - Amies with charcoal
    - Transgrow (traditionally used)
      - Flat bottles coated with MTM and a bottle under increased CO_2 rarely used nowadays
    - JEMBEC (commercial transport system)
      - John E. Martin Biological Environment System
      - Uses a tablet to generate CO_2

Neisseria gonorrhoeae

- Never a normal flora, always a pathogen
- #1 cause of Sexually Transmitted Disease
- Kidney, coffee bean shaped diplococci intracellular within PMN
- Major virulence factor: **Pili** (Common Pili; used for adherence or attachment)
- Superoxol test (+) — rapid test for *N. gonorrhoeae*
  - Reagent: 30% hydrogen peroxide
- Ferments glucose only
- Can cause:
  - Gonorrhea – STD
  - Opthalmia neonatorum
    - A gonococcal eye infection (gonorrheal form of conjunctivitis) acquired by newborns when discharge from infected mother accumulates in their conjunctiva
    - Prevented by placing silver nitrate on the baby's eyes
  - PID (Pelvic Inflammatory Disease)
    - Which may cause sterility, Perihepatitis or Fitz Hugh Curtis syndrome
    - Agents of PID:
      - *C. trachomatis, M. hominis, and N. gonorrhoeae*
- To collect specimen use Dacron/Rayon fibers (swabbing)
  - Never use: Calcium alginate and cotton swab; they are toxic or inhibitory to *N. gonorrhoeae*
  - Because they contain fatty acids which may be inhibitory
  - Sensitive to cold temperature
- Specimen: pus secretions from the urethra, cervix, prostate, throat, and rectal mucosa

Media used for Cultivating Neisseria gonorrhoeae

1. **Thayer Martin media**
   - Enriched chocolate agar
   - Contains VCN
     - Vancomycin: inhibits gram (+)
     - Colistin: inhibits gram (-)
     - Nystatin: prevents fungi
   - 3 antibiotics
   - Base media: chocolate agar
2. **Modified Thayer Martin media**
   - Same components as Thayer Martin but with additional antibiotic = VCN - T
   - Trimethoprim lactate – prevents swarming organisms
   - Base media: chocolate agar
3. **Martin Lewis Media**
   - Contains VCAT
   - "A" is Anisomycin - prevents fungi
   - Base media: chocolate agar
4. **New York City Media**
   - Contains VCAT
   - "A" is Amphotericin B - prevents fungi
   - Can also be used for other urogenital pathogens like *M. hominis* and *U. urealyticum*
   - Base media: yeast extract
5. **GC-LECT Medium**
   - Gonococcus-LECT medium
   - Has low concentration of Vancomycin
   - VCA-T plus lincomycin and amphotericin B
   - Lincomycin: prevent the growth of gram positive organisms and anaerobes
   - Same as New York City Media but with Lincomycin

	BAP	CAP
N. gonorrhoeae	-	+
N. meningitidis	+	+

Neisseria meningitidis

- Bean shaped diplococcic; sensitive to SPS
- Can be a normal flora; Natural habitat is oropharynx and nasopharynx
- Virulence factor: Capsule
  - Neufeld Quellung Test (+)
- Ferments maltose and glucose; Negative for lactose, sucrose, and fructose
- Specimen to detect carrier state: nasopharyngeal swab
- Can cause:
  - Bacterial meningitis (5-29 years old)
  - Meningococcemia (presence of *N. meningitidis* in the blood)
  - Waterhouse-Friderichsen syndrome
    - Severe form of meningococcemia
    - Can lead to the bleeding of adrenal glands due to severe bacterial infection
    - Positive for Disseminated Intravascular Coagulation (DIC)
- Other species:
  - *N. lactamica*: Glucose, Maltose, Lactose (+) & ONPG (+)
    - The only *Neisseria* which is lactose positive
    - ONPG = yellow (positive)
  - *N. sicca*: Glucose, maltose, Sucrose (+)
- Broth (CTA: Cystine Trypticase Agar) + Serum + Sugar + Indicator (phenol red)

SPECIE	ACID PRODUCED FROM:				
	GLUCOSE	MALTOSE	LACTOSE	SUCROSE	FRUCTOSE
N. gonorrhoeae	+	-	-	-	-
N. meningitidis	+	+	-	-	-
N. lactamica	+	+	+	-	-
N. cinerea *N. flavescens* *N. elongata*	-	-	-	-	-
N. sicca *N. mucosa*	+	+	-	+	+
N. subflava	+	+	-	V	V
Moraxella	-	-	-	-	-

N. lactamica: Glucose, Maltose, Lactose (+) and ONPG (+)
N. sicca: Glucose, Maltose, Sucrose (+)

MORAXELLA

Moraxella catarrhalis

- Produces the enzyme: Indophenol oxidase
- Morphologically and biochemically resembles Neisseria
- Encapsulated, with Pili, Non-motile
- A normal flora of oropharynx and nasopharynx but may cause otitis media
  - Agents of otitis media:
    - *M. catarrhalis*
    - *S. pneumoniae*
    - *H. influenzae*
- Catalase (-)
- Gamma hemolytic
- Does not degrade sugar — asaccharolytic/non-saccharolytic
- **BUTYRATE DISK TEST/Tributyrin test (+)**
  - Substrate: bromo-chloro-indolyl-butyrate
  - Rapid test for the identification of *M. catarrhalis*
  - (+) result: development of blue color (wait for 5 minutes)
- Develops **HOCKEY PUCK COLONIES**
  - these are colonies remaining intact when pushed across the plate using inoculating loop
- Best test to differentiate *M. catarrhalis* from other Moraxella is **DNase test**

SPECIES	OXIDASE (1%)	CHO	DNAse	Thayer Martin Agar
N. gonorrhoeae	+	Glucose only	-	+
N. meningitidis	+	Maltose & Glucose	-	+
M. catarrhalis	+	-	+	Nutrient agar +

GRAM POSITIVE, SPORE FORMING BACILLI

BACILLUS

	Bacillus	*Clostridium*
Oxygen	+/Aerobes	-/Anaerobes
Catalase (using 15% H_2O_2)	+	-
Gas	-	+

- Ethanol Survival Test: for Identification of *Bacillus spp.*

Bacillus anthracis

- Largest pathogenic bacteria; a potential bioterrorism agent
- Non-motile; forms the so-called: disjointed bamboo fishing rod appearance
- On gram staining, forms the so-called: Box car morphology
- Virulence factor:
  - D-glutamate capsule
    - Capsule is not polysaccharide in nature
    - Use Mc Fadyean's Capsular stain
  - Exotoxin with 3 components
    - edema factor, lethal factor, protective antigen
    - function of exotoxin: initiate cellular destruction
- on BAP: gamma hemolytic
- Selective medium:
  - PLET: Polymyxin Lysozyme EDTA Thallous acetate
- PEA agar in fecal specimens
- When colonies are lifted, it forms the so-called tenacious or 'BEATEN EGG WHITE CONSISTENCY"
- May also produce colonies with "comet tail" appearance
- On BAP: Non hemolytic and forms swirling projections or the so-called Medusa Head Colonies, Lion head or cut-glass appearance
- Forms the string of Pearl appearance (MHA with 10 u penicillin or .05 u penicillin on BAP
  - Inoculate on MHA + disk + coverslip incubate 3-6 hrs at 3°C, remove colonies and place on slide
  - Observe for bacilli arranged in circular manner
- In tube media (gelatin media): forms the so-called "inverted Fir tree or inverted pine tree appearance"
- Lecithinase test (+)
  - Culture media: egg yolk agar
  - (+) result: development of opaque zone around the colonies
- Causes ANTHRAX
- Mode of Transmission: contact with animals and animal products

EXTERNAL	INTERNAL	
Cutaneous anthrax	**Pulmonary anthrax**	**Intestinal anthrax**
- Most common form - Least severe - May develop black eschar on the skin	- Woolsorter's disease/ Ragpicker's disease/ Hide porter's disease - Acquired through inhalation of spores while handling wool	- Least common - Most severe - Ingestion of improperly cooked infected meat

- **Ascoli test**
  - A serologic precipitin test for the diagnosis of anthrax

Bacillus cereus *Fried Rice Bacillus*	***Bacillus subtilis*** *Hay Bacillus*
- Both motile and gamma hemolytic on BAP	

- Virulence factor: exotoxin/enterotoxin cholera like toxin
- Causes food poisoning (Diarrheal & Emetic Type)
  - Diarrheal type: diarrhea
  - Emetic type: vomiting is more frequent
- Best specimen for testing: suspected food
	- Common laboratory contaminant
	- Opportunistic pathogen
	- Source of bacitracin
	- Cause eye infection in heroin addicts
	- Blood bank contaminant @ RT
	- Penicillin: Resistant
	- Selective media: Mannitol egg yolk polymyxin B agar

Bacillus stearothermophilus

- Agent of flat sour spoilage
- Used as biological indicator of autoclave

	Bacillus anthracis	*Bacillus cereus*
Motility	NM	M
Capsule	+	-
Hemolysis	Gamma hemolytic	Beta hemolytic
Growth at 45°C	-	+
Salicin fermentation *indicator: phenol red* *(+): yellow color*	-	+
Penicillin G	S	R
Glucose, maltose, sucrose fermentation	+	+
Gelatin hydrolysis and PEA	-	+

CLOSTRIDIUM

- Strict anaerobe but some are aerotolerant
- Catalase test (-)
- All are motile
  - except: *C. perfringens, C. ramosum, and C. inoculum*
- All are with swollen spores
  - except: *C. perfringens and C. bifermentans*
- All are non-encapsulated
  - except: *C. perfringens*
- All are lactose negative
  - except: *C. perfringens and C. septicum*
- All are sucrose negative
  - except: *C. perfringens*
- All are single hemolytic (produce only 1 pattern of hemolysis)
  - except: *C. perfringens* (produce double or target hemolysis)
- All are dextrose positive
  - except: *C. histolyticum and C. tetani*
- Ethanol shock test: used to identify *Clostridium* from mixed specimens or mixed cultures

Types of Clostridium

NEUROTOXIC	HISTOTOXIC	ENTERIC
- *C. tetani* - *C. botulinum*	- *C. perfringens*	- *C. difficile*

Clostridium perfringens

- *Clostridium welchii*, Frankel's bacillus, Gas gangrene bacillus
- Encapsulated
- Aerogenic, able to produce gas On CHOPPED MEAT MEDIA, broth media, + growth plus gas
- Produces double zone hemolysis (target hemolysis)
  - Inner = beta (attributed to beta and theta toxin)
  - Outer = alpha (attributed to alpha toxin and Lecithinase)
- Reverse CAMP (+)
  - Known organism: *S. agalactiae*
  - Inoculate known organism on BAP - Single streak
  - Unknown organism is inoculated perpendicular to the known organism
  - (+) result: (enhanced) arrowhead zone of beta hemolysis
- Causes stormy fermentation of milk due to excessive gas production
- Nagler's Reaction/ Lecithinase test (+)
  - Positive reaction is due to alpha lecithinase C and phospholipase C
- Causes:
  - Gas gangrene/myonecrosis: form of necrosis specific to muscle tissues
  - Necrotic enteritis

Clostridium tetani

- Tack head Bacillus, Tetanus Bacillus, Drumstick Lollipop/Tennis racquet bacillus
- Iron loving bacteria, with terminally located swollen spores
- Asaccharolytic/Non-saccharolytic
- Produces Tetanolysin: responsible for beta hemolysis
- Virulence factor: EXOTOXIN
  - Organism will release TETANOSPASMIN
  - an exotoxin which will block release of neurotransmitters causing spastic paralysis
  - Spastic paralysis (little movement) will trigger locked jaw which causes patient to manifest Risus sardonicus or Sardonic smile
- TETANUS: diagnosis is often made based on clinical symptom (Risus sardonicus, Opisthotonus), morphologic evaluation of terminally located swollen spores
- *Clostridium ramosum:* terminally located spores but are round not swollen

Clostridium botulinum

- Not cultured
- Von ermengen's Bacillus or Canned good bacillus
- Virulence Factor:
  - Botulinum toxin
    - a neurotoxin potent in human, blocks the release of acetylcholine causing flaccid paralysis (no more movement/ muscles are totally unresponsive)
    - potent; needs only a small amount to cause flaccid paralysis
- On BAP: alpha hemolytic
- (+) lipase reaction: egg yolk agar (can detect lecithinase and lipase)
- Causes:
  - Botulism (a fatal type of food poisoning); a.k.a home canned good intoxication
  - Infant Botulism: develop because of spore ingestion via breastfeeding
    - There is no flaccid paralysis but the child may manifest with floppy baby syndrome
    - Floppy baby syndrome: child will show reduced muscle tone
  - SID/Sudden infant death syndrome/Crib Death

Clostridium difficile

- Enteric Clostridia; Major cause of diarrhea in hospitals
- Normal flora of the colon
- Causes: antibiotic associated diarrhea or pseudomembranous colitis
- Toxins: Toxin A (enterotoxin), Toxin B (cytotoxin)
- Diagnosis is usually thru: cytotoxin detection (using EIA, a serologic test)
- Specimen:
  - Freshly passed stool - liquid /unformed stools for culture & toxin assay
  - Formed stool/rectal swab – used to detect carrier state
- Media: CCFA (Cycloserine-Cefoxitin Fructose Agar)
- pH Indicator: neutral red
- On CCFA: colonies with horse stable or barnyard odor
- On BAP: fluoresce chartreuse (produce yellow fluorescence)

	Motility	Lecithinase	Lipase	Lactose	Glucose
C. perfringens	-	+	-	+	+
C. botulinum	+	-	+	-	+
C. tetani	+	-	-	-	-
C. difficile	+	-	-	-	+

- All are saccharolytic EXCEPT: *C. tetani and C. septicum*
- Swarming (+) Anaerobe: *C. tetani & C. septicum*

GRAM POSITIVE, NON-SPORE FORMING BACILLI

- *Listeria*
- *Lactobacillus*
- *Erysipelothrix*
- *Corynebacterium*

Listeria monocytogenes

- Primarily an animal pathogen
- With tumbling motility at room temperature (wet mount & hanging drop)
  - End over end motility
  - Presumptive test for *Listeria*
- In BAP, it shows a narrow band of beta hemolysis
- Forms the so-called: inverted Christmas tree pattern, umbrella like growth on agar surface (SIM media)
- Cultured on Mc Bride's media Anton's test - virulence test/ocular test of anton, organism inoculated on conjunctival sac of rabbit (+) purulent conjunctivitis
- Disease: food poisoning (commonly), meningitis, stillbirth, septicemia
- Mode of Transmission: consumption of contaminated food (usual) in particular soft cheese and coleslaw
- Virulence factor: Listeriolysin O
  - Hemolytic and cytotoxic toxin that allows survival of organism within phagocytes
- CAMP TEST (+)
  - Known organism: *Rhodococcus equi*
  - Positive result: block type hemolysis
- Can tolerate cold enrichment (4°C) along with *Y. enterocolitica*
- **REMEMBER!**
  - Neonatal meningitis = *S. agalactiae*
  - Meningitis in children less than 5 years = *H. influenzae*
  - 5 years -29 years = *N. meningitidis*
  - Meningitis in more than 20 years old = *S. pneumoniae*
  - Meningitis in elderly & immunocompromised patients = *Listeria*

Erysipelothrix rhusiopathiae

- H_2S (+) on TSI media
- Non-motile, catalase (-)
- Causes erysipeloid, a localized skin infection which is an occupational hazard for those handling meat and poultry
- On Gelatin media (stab): bottle brush-like growth or pipe cleaner pattern of growth

Lactobacillus acidophilus

- Doderlein Bacilli
- Produce large amount of lactic acid (protect women against urogenital infection)
- Non-pathogenic and has little clinical significance
- Normal flora of mouth, GIT and vaginal canal
- Presence can be noted in Pap stained smears; may also be cultivated using Tomato juice media

	L. monocytogenes	*E. rhusiopathiae*
Beta hemolysis	+	-
Growth at 4°C	+	-
Catalase test	+	-
Motility	+	-
Esculin hydrolysis	+	-
H_2S on TSI	-	+
VP test	+	-

Corynebacterium diphtheriae

- a.k.a Klebs–Löffler Bacillus
- Non-motile, highly pleomorphic
- One end typically swollen - club shaped appearance
- Palisade arrangement or side by side arrangement, picket fence arrangement, x,y,v, formation, or Chinese letter arrangement
- With Babes Ernst granules
- On BAP it produces narrow zone of beta hemolysis

Cystine Tellurite Blood Agar (CTBA): used in the initial isolation

- With 3 colony types on CTA
- *C. diphtheriae* produces gray to black colonies

	GRAVIS	MITIS	INTERMEDIUS
Colony	Large	Small	Medium sized
Hemolysis	Non-hemolytic	Beta-hemolytic	Non-hemolytic

- These can be differentiated based on colony morphology, biochemical reactions and hemolytic pattern
- Loeffler's serum and PAI's coagulated egg serum: can be used to enhance granule formation and pleomorphism
- Tinsdale medium (with tellurite): black colonies with brown halo
- Causes DIPTHERIA; caused by those able to produce toxin (toxigenic)
  - 2 types of *C. diphtheriae*: toxigenic and non-toxigenic
- VIRULENCE FACTOR: Diphtheria toxin
  - Causes necrosis and inflammation which will lead to pseudomembrane formation causing the patient to manifest Bull's neck (swollen neck) appearance
- What is the disease Diphtheria?
  - Characterized by mild sore throat, low grade fever, and body weakness

Toxigenecity tests:

- **Modified ELEK test (+)**
  - In vitro toxigenecity test
  - On agar media, place a filter paper strip with diphtheria anti-toxin, streak (+) control at right angle to the strip of anti-toxin, streak negative control in the same manner.
  - 3 streaks: unknown culture suspected of *C. diphtheriae* is streaked (single) parallel to (+) and (-) control.
  - Incubate at 35°C for 24-48 hrs.
  - Lines of Precipitation is the positive result
    - Mahon: Arc of identity (lines of precipitation of (+) and (-) control should meet)
    - Positive: toxigenic strain
- **Animal Inoculation**
  - a.k.a. Guinea pig lethal test
  - In vivo toxigenecity test
  - Cultured suspension of isolated strain of *C. diphtheriae* (0.3 ml) is injected subcutaneously into thigh of two guinea pigs, one protected intramuscularly with 500 units of diphtheria antitoxin 18-24 hours before the test.
  - POSITIVE RESULT: The unprotected animal dies within 2-3 days with evidence of hemorrhage in the adrenal glands
- **Shick's Test** – susceptibility test for diphtheria

Corynebacterium ulcerans

- Cause mastitis in cattle (inflammation of the breast)
- MOT: exposure to infected cattle or consumption of contaminated milk (unpasteurized milk)

Corynebacterium pseudotuberculosis

- formerly *C. ovix*, rare cause of lymphadenitis

Corynebacterium xerosis

- Normal flora of skin, nasopharynx and conjunctival sac
- Causes endocarditis and UTI

Corynebacterium pseudodiphtheriticum

- Normal flora of human oropharynx/oral flora
- a.k.a Hoffman's Bacillus
- Causes endocarditis

Corynebacterium minutissimum

- Causes Erythrasma - skin infection affecting axillary and pubic skin
- Produces coral red fluorescence when exposed to wood's lamp due to porphyrin

Corynebacterium urealyticum

- A urinary pathogen; agent of UTI
- Gamma hemolytic on BAP
- Rapidly urease (+) and Catalase (+)

C. jeikeium

- a.k.a JK Bacillus
- Known to cause prosthetic heart valve infection
- Gamma hemolytic

	Urease	Nitrate reduction	Starch hydrolysis/ Glycogen hydrolysis
C. diphtheriae	-	+	-
C. ulcerans	+	-	+
C. pseudotuberculosis	+	+/-	-

	Urease	CHO fermentation	Nitrate reduction
C. xerosis	-	Glucose Sucrose Maltose	+
C. pseudodiphtheriticum	+	-	+
C. jeikeium	-	Glucose	-

	Catalase test	Esculin	H_2S on TSI
Corynebacterium	+	-	-
Listeria	+	+	-
Erysipelothrix	-	-	+
Bacillus	+	V	-

	CORYNEBACTERIUM	LISTERIA
Motility	-	+
Salicin	-	+
CAMP	-	+
Esculin hydrolysis	-	+

BRANCHING/PARTIALLY ACID FAST GRAM POSITIVE BACILLI

- *Nocardia*
- *Rhodococcus*
- *Tsukamurella*

NOCARDIA	*RHODOCOCCUS*
- Gram (+), Partially AF - Strict aerobes - Beta hemolytic	- Partially AF - Tap water agar (+) ○ May be used to visualize aerial hyphae of branching bacteria - CHO fermentation (-) - Salmon pink colonies

N. asteroides	*N. brasiliensis*
- Cause pulmonary disease in immunocompromised patients	- Cause skin infection among immunocompetent individuals

	Nocardia	*Actinomyces*
Oxygen requirement	+	-
AF staining	+	-
Catalase	+	-
Urease	+	-
Sulfur granules	+	+

	Partially Acid-Fast	Appearance on Tap Water agar; Branching aerial hyphae	Lysozyme
Nocardia	+	Extensive/+	+
Rhodococcus	±	Minimal/-	±
Gordonia	±	Minimal/-	-
Tsukamurella	+	Minimal/-	+
Streptomyces	-	Extensive/+	-
Actinomadura	-	Variable/sparse	-
Dermatophilus	-	Branching	-
Nocardiopsis	-	Extensive/+	-

MUST REMEMBER

NEISSERIA Oxidase (+) Capnophilic	***MORAXELLA*** Oxidase (+) DNAse (+) Hockey puck colonies	***BACILLUS*** Aerobic Spore former	***CLOSTRIDIUM*** Anaerobic Spore former	***LISTERIA*** Tumbling motility at room temperature Cold enrichment	***CORYNEBACTERIUM***
N. gonorrhoeae With pili Glucose (+)		***B. anthracis*** Non-motile Gamma hemolytic	***C. perfringens*** Target/double hemolysis Reverse CAMP (+) Gas gangrene/Myonecrosis	***ERYSIPELOTHRIX*** Bottle brush/ Test tube brush-like growth on gelatin media	***C. diphtheriae*** Pleomorphic XYV Chinese letter Palisade-picket fence Black colonies on CTBA Modified ELEK test
N. meningitidis With capsule Glucose (+) Maltose (+)		***B. cereus*** ***B. subtilis*** Motile Beta hemolytic	***C. tetani*** Swollen terminal spore (+) Spastic paralysis		
N. lactamica Glucose (+) Maltose (+) Lactose (+)			***C. botulinum*** Flaccid paralysis Home canned good intoxication Floppy baby syndrome	***LACTOBACILLUS*** Normal vaginal flora	
			C. difficile Colonies with barn yard/horse stable odor Pseudomembranous colitis		

RECALLS

Beta lactamase substrate	Nitrocefin	Catalase (+), Coagulase (-), Non-hemolytic	CoNS
Optochin for *S. pneumonia*	Sensitive	Bacteria from Cows causing colon Cancer	*S. bovis*
Species are written in	Lowercase	DNAse test result of *S. aureus and S. epidermidis*	+, -
+ result in Bile solubility	Lysis of colonies	Otitis media (*M. catarrhalis)*	Ear infection
Duran Raynal Factor	Hyaluronidase	Oxidase (+) gram negative cocci	*Neisseria*
Coagulase test uses	Rabbit's plasma obtained using EDTA	Gas gangrene agent, double Hemolysis	*C. perfringens*
Coagulase (-) Novobiocin (R), UTI	*S. saprophyticus*	Throat pathogen	*S. pyogenes*
Coagulase (+) Novobiocin (S), PYR (-)	*S. aureus*	Differential test for *Corynebacteria and Listeria*	Motility Salicin CAMP test Esculin hydrolysis
Group D Test	Bile esculin test	Babes Ernst granules	*C. diphtheriae*

Glucose Maltose Sucrose (+)	*N. sicca*	AGN and Rheumatic fever	*S. pyogenes*
Grayish white beta hemolytic	*S. pyogenes*	(+) tube coagulase	Clot formation
Arrow head result in CAMP due to	Beta hemolysis of *S. aureus* and *S. agalactiae*	7.5% salt concentration	*S. aureus*
Kidney Bean shaped	*Neisseria*	(+) result oxidase test	Modified: blue Cytochrome: purple
CNA is for	Gram (+) cocci	Agent of floppy baby syndrome	*C. botulinum*
PYR (-) Bacitracin (R) CAMP (+)	*S. agalactiae*	Lockjaw	*C. tetani*
Optochin is also known as	Taxo P	Can cause Meningitis, still birth, food poisoning	*Listeria*
Butcher's cut, H_2S (+), catalase (-)	*Erysipelothrix*	Tumbling motility at RT	*Listeria*
#1 cause of neonatal meningitis	*S. agalactiae*	Prosthetic heart valve infection	*S. epidermidis*
Tube coagulase detects	Free coagulase	Sugar utilization test is for	*Neisseria*
Gray-black colonies	*C. diphtheriae*	Modified oxidase	*Micrococci*

GRAM POSITIVE BACILLI/ACID FAST BACILLI

MYCOBACTERIUM

- Aerobic, non-motile
- Slow growers and requires whole egg for growth
- Difficult to stain but once stained, they are difficult to decolorize due to: hydroxymethoxy acid or mycolic acid

Three Groups:

A. ***Mycobacterium tuberculosis complex*** - causes TB
   - *M. tuberculosis*: causes pulmonary TB in man
   - *M. bovis*: causes TB in cattle and intestinal TB in man
   - *M. africanum*: causes tuberculosis in man in Africa
   - *M. microti*: infect those who are immunocompromised
   - *M. canetti:* infect those who are immunocompromised

B. MOTT (Mycobacterium Other Than Tuberculosis)/Runyon's Classification
   - According to pigment production

C. *Mycobacterium leprae*

- Specimens usually processed in the laboratory for the isolation of Mycobacterium
  - Expectorated sputum
  - Secretions obtained thru bronchoscopy
  - Transtracheal aspirates
  - Laryngeal swabs
  - Gastric lavage
  - Urine
  - CSF, synovial, pericardial and peritoneal fluid

- Rules in specimen processing:
  1. Specimen from sterile body sites (ex. CSF) can be concentrated by centrifugation and inoculated
  2. Non-sterile (ex. sputum) must be decontaminated and concentrated

Decontamination and Digestion

- **Decontamination** must be carried out to remove normal flora and other contaminating organisms.
- **Digestion** must be carried out to dissolve disulfide bond in mucus which could trap MTB; to liquefy the mucus
  - NaOH (2,3, and 4%): digesting agent and decontaminant
  - NALC (N-acetyl-L-cysteine) or dithiotreitol (a.k.a Sputolysin) & 2% NaOH: NaOH acts as decontaminating agent, NALC acts as digesting or mucolytic agent
  - Zephiran (benzalkonium chloride) and Trisodium phosphate / Z-TSP: used in both digestion and decontamination
  - 1% Cetylpyridium chloride: not a decontaminant or a digesting agent; added to sputum specimens to prolong its shelf life for 8 days
  - Oxalic acid (5%): added if the specimen is likely to contain *Pseudomonas*

Culture Media for *Mycobacterium* - *M. tuberculosis* cultures are kept in the lab for 6-12 months following: identification & susceptibility			
NON SELECTIVE	**AGAR BASED MEDIA**	**SELECTIVE**	**LIQUID**
Egg based media with Malachite green (inhibits contaminating organism); growth in 6-10 weeks	Clear media, easy examination of colonies, growth in 10-12 days Can be used to detect pigment production	With antibiotics (Agar based media with added antibiotics)	These are non-conventional media
Traditional media Common disadvantage: prolonged time of growth			Growth is earlier detected
- Lowenstein Jensen Media: commonly used egg based media - Petragnani-more inhibitory: recommended for non-sterile specimens - American Thoracic Society (ATS): less inhibitory; recommended for sterile specimens - Dorset Egg Media - Wallenstein Medium: used to isolate *MAI* complex	- Middlebrook 7H10 with dextrose - Middlebrook 7H11 with casein hydrolysate - Mitchison 7H11	- Gruft Modified Lowenstein Jensen - Selective Middlebrook 7H11 - Mitchison Selective 7H11	- Bactec 12B - Bactec 13A - Septicheck - Middlebrook 7H9

Procedures to positively Identify Mycobacterium in the laboratory

1. ACID FAST STAINING

- Used as screening test for the genus *Mycobacterium* (1st step); if positive, proceed with culture
- Genus identification only; Species identification is not possible
- 2 most common methods are:
  - Ziehl neelsen - Hot Method
  - Kinyoun's - Cold Method
- AFB will stain: red against a blue background

NATIONAL STANDARD REPORTING SCALE - (RITM/DOH)

NUMBER OF AFB SEEN	REPORT
0	No AFB seen in 300 visual fields
+n	1-9 /100 fields
1+	10-99 /100 fields
2+	1-10/OIF in at least 50 visual fields
3+	More than 10 AFB/OIF in at least 20 visual fields

2. FLUOROCHROME STAINING

- Auramine rhodamine stain /TRUANT stain: primary dye
- Decolorizer: 0.5% acid alcohol
- Counterstain/quenching agent: 0.5% potassium permanganate
- (+) result: appear yellow against a black background
- Advantages: smears are easily read; the organism is easier to identify because of better contrast

NUMBER OF AFB SEEN	REPORT
0	Negative
1-2 AFB/ 70 fields	Doubtful, repeat using new specimen
1-2 AFB/100 OIF	1+
2-18 AFB /50 fields	2+
4-36 AFB /field	3+
More than 36 AFB / field	4+

3. TEST FOR PIGMENT PRODUCTION

- this is for the identification of those classified under Runyon's

4. NIACIN TEST

- useful in the rapid identification of *M. tuberculosis* (MTB +)
- Media: Lowenstein-Jensen Media
- Reagent: Cyanogen Bromide
- + result: yellow color

5. NITRATE REDUCTION TEST

- useful in the rapid identification of *M. tuberculosis* (MTB +)
- Media: sodium nitrate broth
- Reagents: n-n-dimethyl-L-naphthylamine & sulfanilic acid
- Initial (+) result: red
- Initial (-) result: no color change
  - If negative, confirm using: Zinc dust/Zinc powder (to detect unreduced nitrate)
  - + result: media colorless
  - Negative: red

6. CATALASE AND HEAT STABLE CATALASE

- Most *Mycobacterium* are catalase (+) but not all are heat stable catalase test positive.
- Media: Tween 80 media
- Reagent: 30% hydrogen peroxide
- Positive Result: vigorous bubbling
- Heat organism at 68°C for 20 minutes, cool then add 30 % hydrogen peroxide
- *M. tuberculosis* complex are heat stable catalase (-): *M. tuberculosis, M. bovis, M. africanum, M. ulcerans* and *M. leprae*

7. IRON UPTAKE TEST

- To distinguish M. *chelonae* (-) from other rapid growers (+)
- Reagent: 20% ferric ammonium citrate
- Positive result: rusty brown color

8. ARYLSULFATASE TEST

- To differentiate *M. phlei* (-) from *M. fortuitum* & *M. chelonei* (+)
- Tripotassium phenolphthalein acted upon by arylsulfatase converted to free phenolphthalein
- Media: phenolphthalein media
- Reagent: sodium bicarbonate
- (+) result: pink

9. TWEEN 80 HYDROLYSIS TEST

- in this test tween 80 is converted to oleic acid by tween 80 lipase.
- Useful in the identification of: *M. kansasii* (rapid tween +)
  - *M. tuberculosis* is (+) in 10-20 days
- + result: pink color
- Tween 80 substrate: Polyoxyethylene sorbitan mono-oleate
- Process: on a tube, 1 ml. sterile deionized water, add 2 drops of Tween 80 substrate + organism, incubate 37°C 10 days

10. PYRAZINAMIDASE TEST

- Pyrazine amidase test
- Used to identify *M. tuberculosis*
- Reagent: ferrous ammonium sulfate
- (+) result: red

11. TELLURITE REDUCTION TEST

- Useful test to identify *M. avium intracellulare complex*
- (+): black color

12. UREASE TEST

- Useful in differentiating *M. scrofulaceum* (+) from *M. gordonae*
- Media: Urea agar or Urea broth
- Media contains phenol red as indicator

INHIBITORY TESTS

13. NaCl TOLERANCE TEST

- uses an egg based media with 5% NaCl
- (+) if there is growth; (-) if no growth
- MTB (-), *M. triviale* (+)

14. TCH / T_2H SUSCEPTIBILITY TEST

- *M. bovis* is susceptible while *M. tuberculosis* is resistant
- Antibiotic used: Thiopene-2-carboxylic hydrazide

15. GROWTH ON MAC CONKEY WITHOUT CRYSTAL VIOLET

- Crystal violet: inhibits the growth of gram (+) organisms
- The absence of crystal violet allows the growth of *M. fortuitum-chelonae* complex

Mycobacterium tuberculosis

- a.k.a Koch's Bacillus or Tubercle Bacillus
- appears beaded under the microscope because of much granules
- Size: 0.2-0.6um in diameter, has a length of 1-4um
- Produces "cauliflower like growth" at 35 – 37°C
- Rough colonies exhibit cording: colonies appearing like curved strands
- Obligate aerobe; requires CO_2 for growth
- Virulence Factor: cord factor and sulfatides
- Causes Pulmonary TB in man
- Pott's Disease: skeletal TB of the spine
- Miliary TB: extrapulmonary TB that may affect the spleen or liver
- BCG vaccine: Bacillus-Calmette-Guerin; used to prevent *M. tuberculosis*
- **DOTS**: Directly Observed Treatment Strategy; a program of the DOH to prevent Tuberculosis

Types of TB:

1. **Primary**: initial exposure
2. **Secondary**: a repeat of primary TB; occurs to those who had TB in the past
3. **MDR** (multi-drug resistant TB): shows resistance to primary TB drugs
4. **XDR** (extensively drug resistant): shows resistance to primary TB drugs, quinolones, and at least one 2nd line TB drug

- **Primary TB Drugs**: Rifampin, Isoniazid, Pyrazinamide, Ethambutol, Streptomycin
- **Secondary drugs**: Ofloxacin, Kanamycin, Amikacin, Ciprofloxacin

WAYS TO DETECT EXPOSURE TO *M. tuberculosis*

- Skins tests which involves antigen introduction
- PPD: Purified Protein Derivative; a protein extracted and purified from the cell wall of *M. tuberculosis*
  - Mantoux Test: skin test that involves intradermal injection of PPD
  - Syringe used: tuberculin syringe
- Von Pirquet test: scratch test; lanolin is combined with old tuberculin to produce an ointment which is rubbed to the skin
- Killing of *M. tuberculosis* in sputum:
  - Boiling = 10 mins. (easily kills *M. tuberculosis*)
  - Direct sunlight = 20-30 hours
  - Dried sputum = 6-8 months
  - 5% phenol = 24 hours

Board Exam Must-Know!

DIRECT SPUTUM SMEAR MICROSCOPY/DSSM

- Smear size: 2 x 3 cm
- Dry prior to heat fixation: to prevent aerosol
- Examine 300 fields before reporting negative result

Mycobacterium bovis

- Causes: TB in cattle and intestinal TB in man
- Plated media: develop colonies which appear like water droplets

Mycobacterium leprae

- A.k.a. Hansen's Bacillus
- Forms the so-called: cigar packet arrangement
- Non-culturable on artificial media, but can be maintained in the lab using: mice foot pads or foot pads of armadillo
- Specimen: smears of tissue juice, AF stain for Lepra cells
  - Lepra cells are AFB within macrophage
- Causes Leprosy, it has tropism for: peripheral nerves
  - A chronic disease affecting the skin, mucus membranes, and nervous tissues
  - Those with leprosy may experience loss of sensation
- Types of leprosy
  - Tuberculoid type /Paucibacillary
    - Patient may have relatively few bacteria
  - Lepromatous type /Multibacillary
    - Patient may have large number of bacteria and the disease is widespread
- PCR has been used to identify *M. leprae* in infected tissues: molecular
  - PCR: definitive test
- Basis of diagnosis: clinical symptoms (because the organism is non-culturable
- **Lepromin** Test: skin test for leprosy; involves injection of antigen which may elicit Fernandez reaction (noted after 24-48 hours of injection) and Mitsuda reaction (delayed reaction; appear 3-4 weeks following injection)
- WADE FITE FARACO - to detect AFB in paraffinized tissues
- Phenolase test: used to separate *M. leprae* from other Mycobacteria

MOTT/RUNYOUN'S CLASSIFICATION

GROUP 1 - PHOTOCHROMGEN	*GROUP II – SCOTOCHROMOGEN*	*GROUP III – NON PHOTOCHROMOGENS*	*GROUP IV – RAPID GROWERS*

- Pigmented when exposed to constant light source
- *Mycobacterium kansasii*
  - Crossed Barred Bacillus
  - **#2 NTM to cause TB in those with AIDS**
- *Mycobacterium marinum*
  - Causes swimming pool or fish tank granuloma
- *M. simae*
- *M. asiaticum*
	- Pigmented in the dark and when exposed to light
	- *Mycobacterium scrofulaceum* - causes cervical lymphadenitis
	- *Mycobacterium szulgai*
	- *Mycobacterium gordonae* (Tap water Bacillus)
	- *Mycobacterium thermoresistible* (has ability to grow at high temperature: 55°C)
	- *Mycobacterium flavescens* - a normal flora
		- Non-pigmented
		- *M. xenopi* - on Cornmeal agar it produces branching colonies (appears like bird's nest)
		- *M. ulcerans* – known agent of Buruli ulcers
		- *Mycobacterium haemophilum* – requires hemin for growth, biochemically inert, CAP MHA with Fildes enrichment
		- *Mycobacterium gastri* - non-pathogenic, isolated from gastric lavage
		- *Mycobacterium avium* - *intracellulare complex* (MAI)
		  - Biochemically similar
		  - *M. intracellulare*: Battey Bacillus; non-pathogenic
		  - *M. avium*: #1 NTM which cause TB in those with AIDs
		- *Mycobacterium terrae complex* (*M. terrae* [radish bacillus], *M. triviale, M.* chromogenicum) - Rare agents of arthritis and osteomyelitis
		- *Mycobacterium genavense*: cause disseminated infections in patients with AIDS
		- *Mycobacterium avium subspecies paratuberculosis*: considered as an agent of Johne's disease (intestinal infection occurring as chronic diarrhea in sheep and cattle; a form of Inflammatory bowel disease which is very similar to Crohn's
			- Grows in 3-5 days
			- *M. fortuitum and M. chelonei* are: Arylsulfatase test (+); and grows in MAC without CV (crystal violet)
			  - *M. fortuitum* = (+) iron uptake and Nitrate reduction
			  - *M. chelonei* = (-) iron uptake and nitrate reduction
			- *Mycobacterium smegmatis*
			- *Mycobacterium phlei*
			- *M. mucogenicum*
			- *M. abscessus*

IDENTIFICATION of MYCOBACTERIUM

1. Biochemical tests: used for species identification
2. Pigment Production: used to detect species classified under Runyoun's
3. Photosensitivity/Photoreactivity: used to detect species classified under Runyoun's
4. Rate of Growth: used to identify the rapid growers

- GENE EXPERT: Rapid and sensitive test to detect Mycobacteria
- CULTURE: Gold standard
- MANTOUX: Skin test

BIOCHEMICAL TESTS TO SPECIATE MYCOBACTERIUM

Tests	*Mycobacterium tuberculosis*	*Mycobacterium bovis*
Niacin test	+	-
Nitrate Reduction Test	+	-
Heat Stable Catalase test	-	-
Tween 80 Hydrolysis Test	V	-
Arylsulfatase Test (3 days)	-	-
Iron Uptake Test	-	-
Pyrazinamidase Test	+	-

Tests	*M. kansasii*	*M. marinum*	*M. simiae*	*M. asiaticum*
Niacin test	-	V	+	-
Nitrate Reduction Test	+	-	-	-
Heat Stable Catalase test	+	-	+	+
Tween 80 Hydrolysis Test	+	+	Slow +	+
Arylsulfatase Test (3 days)	-	V	-	-
Iron Uptake Test	-	-	-	-
Pyrazinamidase Test	-	+	+	-

Tests	*M. scrofulaceum*	*M. szulgai*	*M. gordonae*	*M. thermoresistible*	*M. flavescens*
Niacin test	-	-	-	-	-
Nitrate Reduction Test	-	+	-	-	+
Heat Stable Catalase test	+	+	+	+	+
Tween 80 Hydrolysis Test	-	Slow +	+	+	+
Arylsulfatase Test (3 days)	-	-	-	-	-
Iron Uptake Test	-	-	-	-	-
Pyrazinamidase Test	v	-	-	NA	+

Tests	*M. avium intracellulare complex*	*M. terrae – triviale complex*	*M. malmoense*	*M. haemophilum*	*M. gastri*
Niacin test	-	-	-	-	-
Nitrate Reduction Test	-	+	-	-	-
Heat Stable Catalase test	+	+	+	-	-
Tween 80 Hydrolysis Test	-	+	+	-	+
Arylsulfatase Test (3 days)	-	-	-	-	-
Iron Uptake Test	-	-	-	-	-
Pyrazinamidase Test	+	v	+	+	-

Tests	*M. fortuitum*	*M. chelonei*	*M. phlei*	*M. smegmatis*
Niacin test	-	-	-	-
Nitrate Reduction Test	+	-	+	+
Heat Stable Catalase test	+	+	+	+
Tween 80 Hydrolysis Test	V	-	+	+
Arylsulfatase Test (3 days)	+	+	-	-
Iron Uptake Test	+	-	+	+
Pyrazinamidase Test	+	+	-	NA

GRAM NEGATIVE FERMENTATIVE BACILLI

FAMILY ENTEROBACTERIACEAE

- Facultative anaerobe or aerobe
- Fermenters
  - All ferments glucose and often with gas production
  - Aerogenic
    - except: *Shigella* (gas negative)
- Motile with peritrichous flagella
  - except: *Klebsiella and Shigella*
- Motile at room temperature but non-motile at 37 degrees
  - Genus *Yersinia*; except: *Y. pestis*
  - Non-motile at 37°C: *Klebsiella, Shigella, Yersinia*
- Some are with pili or fimbriae
- Grows well on Mac Conkey agar and EMB
- Grows well on BAP: Large dull gray colonies with variable hemolysis
- Catalase (+)
  - except *Shigella dysenteriae*
- All can reduce nitrates into nitrite
  - except *Erwinia* and *Pantoeae*
- Cytochrome oxidase (-)
  - except *Plesiomonas*
  - New member: *Plesiomonas shigelloides*
    - Previously categorized in the same group as Vibrio
- 2 groups (based on clinical infection produced):
  - **Intestinal pathogens**
    - Those that cause diarrhea
    - Never normal flora
    - *Salmonella and Shigella*
    - *Yersinia enterocolitica*
  - **Opportunistic pathogens**
    - Normal GIT flora that can produce infections outside the GIT
- **General virulence factors:**
  - Pili – for adherence/attachment/invade tissues
  - Plasmid – carries antibiotic resistant gene
  - Ability to invade tissues
- May possess 3 types of antigenic determinants which can be used in the serologic identification of a particulate organism.
- These antigens are particularly useful in the characterization of *E. coli, Klebsiella, Shigella, and Salmonella*

O antigen *Somatic antigen* *Cell wall antigen*	**K antigen** *Envelope antigen* *Capsular antigen*	**H antigen** *Flagellar antigen*

- Heat stable
- For *E. coli* there are 164 types of O antigen and specific types maybe associated with a particular disease
- Serotype 0111: causes diarrhea in infants
- Serotype 0157: associated with vero toxin production
	- Consists of capsular polysaccharide
	- Heat labile
	- With K antigens:
	  o *K. pneumoniae*
	  o *E. coli*
	  o *Salmonella*
	- Vi antigen: *S. typhi*
		- Found in motile enterics
		- Protein in nature
		- Heat labile

Board Exam Must-Know!

- Extended spectrum beta lactamase (ESBL) producing enterics: *K. pneumoniae & E. coli*
- Isolated from the urinary tract and causes bacteremia: *E. coli, P. mirabilis, K. pneumoniae*
- Antibiotic resistant: *Citrobacter, Enterobacter, Serratia*
- Associated with diarrhea: *Salmonella, Shigella, and Y. enterocolitica*

STEPS IN IDENTIFICATION (MANUAL)

- Specimen: stool
- Inoculate on plated media such as Mac Conkey Agar, EMB, or Salmonella-Shigella Agar
- Use colonies for smear preparation and do gram staining to check if it is really gram negative bacilli
- Use the colonies to do cytochrome oxidase test; result must be negative to consider it as gram negative bacilli
- Do a series of biochemical tests to identify the genera/species such as ONPG test, IMViC test, etc.
- Semi-automated: API (Analytical Profile Index)
- Automated
  - Vitek: automation program for bacterial identification and susceptibility
    - Pure isolates are used
  - MALDITOF
    - More rapid and sensitive than Vitek

MEDIA	INHIBITOR	FERMENTABLE CARBOHYDRATE	INDICATOR	CATEGORY	COLONY COLOR	
					FERMENTER	NON FERMENTER
Eosin Methylene Blue (EMB) or Levine Medium	Eosin Y and Methylene Blue	Lactose	Eosin Y and Methylene Blue	Selective and Differential	Pink-purple colonies *E. coli* = with greenish metallic sheen *K. pneumoniae* = mucoid *Enterobacter* = with black center/"fish eye appearance"	Colorless
Mac Conkey	Crystal violet and Bile salts	Lactose	Neutral red		Pink	Colorless
Xylose Lysine Deoxycholate (XLD)	Bile salts	Xylose Lactose Sucrose	Phenol red		Yellow	Red/colorless *Salmonella:* Red w/ black center *Shigella* (H_2S-): colorless colonies w/o black center

Hektoen-Enteric Agar (HEA)		Salicin Lactose Sucrose	Bromthymol blue		Yellow – w/ or w/o black	Green/colorless *Salmonella:* Green w/ black center *Shigella* (H_2S-): colorless colonies w/o black center
Salmonella Shigella Agar (SSA)		Lactose	Neutral red	Selective	Red	Colorless *Salmonella* = with black center *Shigella* = w/o black center
Bismuth Sulfite Agar (BSA)	Brilliant green	Glucose	Bismuth sulfite		*Salmonella* = black colonies	
Thiosulfate Citrate Bile Salt Sucrose (TCBS)	Bile salt pH = 8.6	Sucrose	Thymol blue and Bromthymol blue	Selective and Differential (for *Vibrio*)	Yellow	Green/colorless

OTHER MEDIA

1. **GN BROTH**
   - An enrichment media for *Salmonella*
2. **SELENITE BROTH**
   - Enrichment broth for *Salmonella and Shigella*
3. **TETRATHIONATE BROTH**
   - Enrichment broth for *Salmonella and Proteus*
4. **CIN (CEFSULODIN IRGASAN NOVOBIOCIN)**
   - A selective media for *Yersinia* and contains mannitol as the fermentable carbohydrate and neutral red as indicator
5. **BGA (BRILLIANT GREEN AGAR)**
   - For *Salmonella* except *S. typhi*
   - Inhibitory agent: Brilliant green
   - pH indicator: Phenol red

- Stool may be directly grown in plated media or it may be grown in enrichment media first prior to placing it in plated media
- 1, 2, and 3 are enrichment media; growth from any of the broth will be placed in the plated media
  - Bacterial yield is increased if grown in enrichment broth first

DIAGNOSTIC TESTS FOR GRAM NEGATIVE ENTERIC BACILLI

1. GROWTH ON MEDIA

- EMB, MAC, HEA, and SSA
- STOOL → EMB or MAC = if (+)
  1. Gram staining
  2. Oxidase test
  3. CHO fermentation test (TSI)
     - Indicators = Ferrous sulfate and sodium thiosulfate
     - (+) = Splitting of media, pulling away, cracks

2. CARBOHYDRATE FERMENTATION TEST

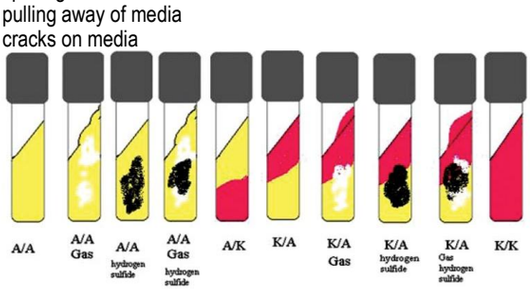
- Media: TSI/Triple Sugar Iron dispensed as butt and slant
- TSI is the conventional/traditional media
- Kligler's Iron Agar: has the same components as TSI but no sucrose
- Composition:

Fermentable CHO	pH Indicator	H2S Indicator
- 10 parts lactose - 10 parts sucrose - 1 part glucose	- Phenol red	- Sodium thiosulfate - Ferric ammonium sulfate

Steps: Inoculate → Incubate → Note for changes in color

Possible results:

- In acid pH: Yellow = reported as "A"
- In alkaline pH: Red = reported as "K"
- H_2S production: blackening
- Gas production:
  - splitting of media
  - pulling away of media
  - cracks on media



Color: Slant/Butt	Report result as	Interpretation	Classification of inoculated organism
Red/Red	K/K	No sugar was fermented	Non-fermenter
Red/Yellow	K/A	Only 1 sugar (glucose) was fermented	Non-lactose fermenter
Yellow/Yellow	A/A	2-3 sugars were fermented Glucose (+) Lactose and/or sucrose (+)	Lactose fermenter

- All members of the Enterobacteriaceae family are glucose fermenters
- No single species in the family Enterobacteriaceae are K/K; They are either K/A or A/A
- K/K: *Pseudomonas*

Rapid Lactose fermenters	Late lactose fermenters	Non-lactose fermenters
Enterobacter *Escherichia* *Klebsiella*	*Citrobacter* *Serratia* *S. arizonae* *S. sonnei* *Y. enterocolitica*	*Proteus* *Providencia* *Morganella* *Edwardsiella* *All Salmonella except S. arizonae* *All Shigella except S. sonnei* *All Yersinia except Y. enterocolitica*

TSI REACTIONS	
K/K **H2S (-)**	*Pseudomonas* *Burkholderia*
K/A **H2S (-)**	*Providencia* *Morganella* *Shigella*
K/A **H2S (+)** **GAS (+)**	*Salmonella* *Proteus* *Arizonae* *C. freundii* *E. tarda*
A/A **H2S (-)** **GAS (+)**	*Klebsiella* *Escherichia* *Enterobacter*

3. IMViC TEST (Indole-Methyl Red-Voges-Proskauer-Citrate)

- Differentiate lactose fermenters

Indole test Tryptophanase	**MR Test**	**VP Test**	**Citrate Utilization**
To detect the ability of the organism to produce indole from tryptophan and produce the enzyme tryptophanase	To detect the ability of the organism to produce acid when glucose is metabolized	To detect the ability of the organism to produce acetoin or acetylmethyl carbinol	Detects the ability of the organism to use citrate as the sole source of carbon
SIM media or Tryptophan broth	MRVP Broth or Clark-Lubs Broth	MRVP Broth or Clark-Lubs Broth	Simmon's Citrate Agar *(tubed media dispensed as slant)*
Kovac's or Ehrlich's reagent Kovac's reagent chemical name: *para-dimethyl-amino-benzaldehyde* (pDAB)	Methyl Red Indicator	**Barrit's method:** 5% Alpha naphthol 40% KOH **Coblentz method** 5% Alpha naphthol 40% KOH with creatine	No reagent will be added after incubation since the media has an incorporated indicator which is bromthymol blue
(+) Red ring at the junction (-) No color development	(+) Distinct red color pH less than 4.5 (-) No color change	(+) pink to red color (-) no color change	(+) Blue (-) Green

Additional notes!

- MRVP broth is light yellow in color
- Simmon's Citrate Agar: Green in color
- Incubation: 24-48 hours
  - After 24 hours, you can read Indole, VP, and citrate
  - MR will only be read after 48 hours
  - Reading MR results too early: false positive; all *Enterobacteriaceae* are MR positive after 24 hours
- MRVP reactions are usually opposite

Species	I	M	Vi	C	Species	I	M	Vi	C
Escherichia coli	+	+	-	-	*Citrobacter freundii*	-	+	-	+
Yersinia enterocolitica	+	+	-	-	*Serratia marcescens* *Hafnia alvei*	-	V	+	+
Proteus vulgaris	+	+	-	-	*Shigella sonnei*	-	+	-	-
Edwardsiella tarda	+	+	-	-	*Salmonella typhi*	-	+	-	-
Shigella dysenteriae	-	+	-	-	*Klebsiella oxytoca*	+	-	+	+
Morganella morganii	+	+	-	-	*Proteus mirabilis*	-	+	-	+
Yersinia pestis	-	+	-	-	*Providencia rettgeri* *Providencia stuartii*	+	+	-	+
Klebsiella pneumoniae *Klebsiella ozaenae*	-	-	+	+	*Enterobacter spp.*	-	-	+	+

BIOCHEMICAL TEST	(+) CONTROL	(-) CONTROL
Indole Test (Kovac's method)	*E. coli*	*K. pneumoniae*
Methyl Red	*E. coli*	*E. aerogenes*
Voges-Proskauer		
Citrate Utilization Test	*E. aerogenes*	*E. coli*

*Ehrlich's reagent is more sensitive for detection of small amount of indole

4. ONPG TEST (O-NITROPHENYL-BETA-D-GALACTOPYRANOSIDE)

- To detect late lactose fermenters
- To differentiate *Citrobacter* which are ONPG (+) and *Salmonella* which are ONPG (-)
  - All *Salmonella* are ONPG (-) except *S. arizonae* (+)
- May be carried out using ONPG broth/disk
- On a tube inoculate of ONPG broth, incubate at 37°C or on a tube place a loopful of organism, add 0.85% NaCl, then add ONPG disk
- (+) result: Yellow

ONPG (+)	ONPG (-)
Escherichia coli *Klebsiella spp.* *Enterobacter spp.* *H. alvei* Slow lactose fermenters: *Citrobacter spp.* *Serratia spp.* *S. arizonae* *S. sonnei* *Yersinia enterocolitica*	*Proteus spp.* *Providencia* *Morganella* *Salmonella spp.* *Shigella spp.* *Y. pestis* *Y. pseudotuberculosis*

5. UREASE TEST

- Useful in the detection of PPM – *Proteus, Providencia, Morganella*
  - Priority tests: Urease and Deaminase test
- Based on the ability of the organism to convert urea to ammonia thru the action of the enzyme urease
- PPM are regarded as the rapid urease producers
- **Rapid Urease Producers** (+):
  - *PPM, Helicobacter pylori* (Gram negative bacilli), *Ureaplasma, Mycoplasma, Nocardia* (partially fast variable), *C. neoformans, T. mentagrophytes*
- **Slow urease producers** (+); positive after 4 hours are the following:
  - KEYS-C: *Klebsiella, Enterobacter, Yersinia, Serratia, Citrobacter*
- Media: Christensen Urea Agar/Urea broth
- Indicator: Phenol Red
- Steps:
  - Inoculate in Urea agar or broth
  - Incubate; nothing is added because the media contains phenol red already
- Results:
  - (+): Pink; Red, Magenta
  - (-): Yellow

6. DEAMINASE TEST

- Useful in the identification of PPM, these are the only deaminase (+) enterics
- Media used:
  - Lysine agar
  - Tryptophan agar
  - Phenylalanine agar
- PAD + Overnight culture + 10% Ferric chloride
- 10% ferric chloride: added after incubation with tryptophan and phenylalanine agar
  - (+) result with Phenylalanine agar: Green
  - (+) result with Tryptophan agar: Brown

7. GELATIN HYDROLYSIS/LIQUEFACTION TEST

- Useful in the detection of Serratia
  - Significant: *Serratia marcescens*
- May also used to identify genus *Proteus*
- Organisms inoculated on Nutrient Gelatin Media and incubated for 14 days at 35-37°C
- Within the incubation period, tubes are removed from the incubator daily and it is placed inside the ref for 30 minutes or until the media has gelled
  - If not removed from the incubator daily: Media will liquefy
- (+) result: Partial or total liquefaction of the inoculated tube (the control tube must be completely solidified) at 4°C within 14 days
- (-) result: Complete solidification at 4°C

TESTS THAT WIL SEPARATE SALMONELLA AND SHIGELLA

- H_2S production
- Motility test
- Malonate utilization test
- Decarboxylase test

8. TEST TO DETECT H_2S PRODUCTION

- Useful in differentiating *Salmonella* from *Shigella*
- May be carried using any media provided that the media contains an indicator for H_2S detection
- Examples of media with H_2S indicator: TSI, LIA, SIM, HEA
  - SIM is the most sensitive for H_2S production
  - H_2S Indicator in HEA, SSA, and LIA: Ferric ammonium citrate
- H_2S positive:

H_2S (+)	H_2S (-)
Salmonella *Edwardsiella tarda* *Citrobacter freundii* *Proteus vulgaris* *Proteus mirabilis*	*Shigella*

9. MOTILITY TEST

- Very useful in detecting non-motile enterics: *Klebsiella and Shigella*
- May be carried out using SIM media
  - Semi-solid media, dispensed as butt
- (+): On a tube media, it is motile if growth is outside the line of streak
- (-): Non-motile if growth is at the line of streak (inoculate by stabbing)
- In SIM we use TTC or triphenyl tetrazolium chloride which is a colorless dye that turns red with organisms' growth

MOTILE	NONMOTILE
Salmonella	*Shigella* *Klebsiella*

10. MALONATE UTILIZATION TEST

- Based on the ability of the organism to use sodium malonate as the only carbon source
- Indicator: bromthymol blue
- (+) result: Blue
- (-) result: Green

POSITIVE	NEGATIVE
Salmonella *Enterobacter*	*Shigella*

11. DECARBOXYLASE TEST

- Based on the ability of the organism to remove carboxyl group from specific amino acid thru the release of the enzyme decarboxylase
- Media:
  - Moeller's broth/media
  - LIA
- **Moeller's media**
  - There are three amino acids that can be used
    - Lysine (amino acid) → Cadaverine (carboxyl)
    - Ornithine (amino acid) → Putrescine (carboxyl)
    - Arginine (amino acid) → Citrulline (carboxyl)
  - Three tubes/tests:
    - Lysine decarboxylase test
    - Ornithine decarboxylase test
    - Arginine dihydrolase test
- Indicator: Bromcresol purple
  - Moeller's broth:
    - Bromcresol purple + Amino acid + 1% glucose
- Contains glucose
- Tubes are overlaid with mineral oil
  - Mineral oil: acts as oxygen barrier/masks oxygen (makes the environment anaerobic)
- Result
  - Purple (+)
  - Yellow (-)

- Explanation:
  - If positive, cadaverine (carboxyl) was removed from lysine (amino acid)
  - The bacteria you placed will attack glucose, when glucose is attacked, the media becomes yellow and the pH becomes acid
  - But if the organism is capable of producing decarboxylase, it attacks the amino acid, and the yellow media turns purple
- Yellow: fermentation
- Purple: decarboxylation

12. LYSINE IRON AGAR (LIA)

- Detects:
  - H_2S production
  - Deamination
  - Decarboxylation
- Media which can be used to detect deamination and decarboxylation
- A media dispensed as slant and butt
  - Slant is observed for deamination
  - Butt is observed for decarboxylation
- (+) result in decarboxylase test: purple; (-) = yellow
- (+) result in deamination: burgundy red; (-) = purple

Color of slant/butt	Report it as	Interpretation	Possible organism
Purple/purple	K/K	(-) deamination (+) decarboxylation	*Salmonella*
Purple/yellow	K/A	(-) deamination (-) decarboxylation	*Shigella*
Red/yellow	R/A	(+) deamination (-) decarboxylation	*PPM*

Reported as K/K	Reported as K/A	Reported as R/A
Escherichia coli *Enterobacter aerogenes* *Enterobacter gergoviae* *Edwardsiella tarda* *Klebsiella spp.* *Serratia spp.* *Hafnia spp.* *Salmonella spp.*	*Enterobacter cloacae* *Citrobacter spp.* *Yersinia spp.* *Shigella spp.*	*Proteus* *Providencia* *Morganella*

13. KCN BROTH TEST

- Inoculate the organism to be tested and we wait for growth
- (+) = turbidity
- Useful in the identification of *Proteus*
- *Proteus* (+), *Salmonella* (-)

14. ESCULIN HYDROLYSIS

- This test is used for the presumptive identification and differentiation of *Enterobacteriaceae*
- (+) control: *E. faecalis*
- (-) control: *E. coli*
- (+) result: blackening
- Indicator: Ferric ammonium citrate

15. ACETATE UTILIZATION

- Differentiate organisms based on ability to use acetate as the sole source of carbon.
- Generally used to differentiate *Shigella* spp. from *Escherichia coli.*
- Indicator: Bromthymol blue
- Positive: Medium becomes alkalinized (blue) as a result of the growth and use of acetate.
- Negative: No growth or growth with no indicator change to blue
- Control
  - Positive: *Escherichia coli*
  - Negative: *Shigella sonnei*

A. GENUS ESCHERICHIA

- Significant species is *E. coli*

Escherichia coli

- A.k.a. Colon Bacillus
- #1 cause of Urinary Tract Infection
- #2 cause of Neonatal Meningitis
- May cause diarrhea and gastroenteritis
- Normal flora so found in normal stools
  - *E. coli* in water – indication that the water is fecally contaminated
  - Primary marker of fecal contamination
- Biochemical tests:
  - A/A in TSI
  - IMViC reaction is: ++--
- 2 serotypes:
  - **Nephropathogenic *E. coli*/Uropathogenic *E. coli*:** causes UTI
  - **Diarrheagenic/Enterovirulent *E. coli*:** known to cause diarrhea

DIARRHEAGENIC/ENTEROVIRULENT *E. coli*	
ENTEROTOXIGENIC *E. coli* **(ETEC)** 06, 08, 025	- Can cause profuse/severe watery diarrhea that is Cholera-like due to production of toxin - Severe watery diarrhea that is not bloody - Grows only on BAP - Produces heat labile and or heat stable enterotoxins
ENTEROINVASIVE *E. coli* **(EIEC)** 0124, 143, 164	- Can cause bloody diarrhea that is Shigella-like due to its ability to invade the bowel mucosa - Invades intestinal epithelium causing Shigella-like infection; bloody stool - **Sereny test (+)**: Virulence test that uses Guinea pigs as test animals; used to detect EIEC and *Shigella*; positive result is development of keratoconjunctivitis in the test animal
ENTEROPATHOGENIC *E. coli* **(EPEC)** 0111, 0114	- Non-toxigenic and non-invasive - Can cause diarrhea in infants (infantile diarrhea)
ENTEROAGGREGATIVE *E. coli* **(EAEC)**	- Can cause acute and persistent (chronic) diarrhea in children and adults - Can cause watery mucoid diarrhea without WBCs and RBCs - *E. coli* attach to Epithelial cells in a stacking manner
DIFFUSELY ADHERENT *E. coli* **(DAEC)**	- This constitutes the 6th group in the diarrheagenic *E. coli* - Can cause diarrhea in otherwise healthy patients - Can cause acute or chronic watery diarrhea - *E. coli* attach to Epithelial cells in a diffuse pattern
ENTEROHEMORRHAGIC *E. coli* **(EHEC)** VEROTOXIC *E. coli* **(VTEC)** SHIGA TOXIN PRODUCING *E. coli* **(STEC)**	- Can cause severe bloody diarrhea due to its ability to produce verotoxin - Verotoxin: a cytotoxin resembling those of *S. dysenteriae*; may induce bloody diarrhea and later cause kidney failure - Specific example is *E. coli* serotype 0157:H7 - Associated with eating poorly cooked hamburgers - Most severe manifestation: Hemorrhagic colitis or Hemolytic Uremic Syndrome

SORBITOL MacCONKEY

- All *E. coli* are lactose (+) so on Mac Conkey Agar, all can develop pink-purple colonies
- To detect EHEC/ *E. coli* O157:H7 we use Sorbitol Mac Conkey Agar/SMAC
  - SMAC: or MAC SOR media (Mac Conkey Sorbitol)
  - This is like the usual Mac Conkey Agar but instead of lactose, it contains sorbitol; only lactose was replaced
  - It has the same inhibitor and indicator as the usual Mac Conkey Agar
- All *E. coli* serotypes are Sorbitol (+) except *E. coli* O157:H7
  - *E. coli* 0157:H7 = lactose positive, SMAC negative
  - All *E. coli* serotypes can develop pink-purple colonies on SMAC except *E. coli* O157:H7
- Result:
  - (+) Result on SMAC: Pink-purple colonies
  - (-) Result on SMAC: Colorless colonies

MUG TEST

- This test is used to presumptively identify various genera of *Enterobacteriaceae* and verotoxin-producing *Escherichia coli*
- MUG disks are prepared by impregnating MUG (4-methylumbilliferyl-β-D-glucoronide)
- To detect **β-D-glucoronidase** which is produced by *E. coli* and other Enterobacteriaceae
- Result:
  - (+): Electric blue fluorescence (read under UV light)
  - (-): No electric blue fluorescence
  - (+): Yellow (in colorimetric assay; not read under UV light)
  - *E. coli* O157:H7 = MUG test (-)
- Control:
  - (+): *E. coli*
  - (-): *K. pneumoniae*

B. GENUS KLEBSIELLA

- Significant species is *Klebsiella pneumoniae*
- With large polysaccharide capsule
- Exhibits mucoid growth
- Non-motile
- (+) growth on KCN medium

Klebsiella pneumoniae

- Also known as Friedlander's Bacillus
- Normal flora of the URT and GIT
- Can cause community acquired pneumonia (produce currant jelly-like sputum)
  - *S. pneumoniae* causes lobar pneumonia and produce rusty colored sputum
- (+) encapsulated = Neufeld Quellung (+)
  - Can develop mucoid colonies in plated media (EMB)
  - (+) String test = positive result is due to mucoid colonies
    - Also done to detect the mucoid colonies of *Vibrio cholerae*
  - Reagent used in string test: KOH
- TSI: A/A + gas

Klebsiella oxytoca

- Biochemically resembles *K. pneumoniae* but can be differentiated from the latter for its being indole positive

	Indole	Urease	Growth in KCN	Lysine decarboxylase	Voges-Proskauer
K. pneumoniae	-	+	+	+	+
K. oxytoca	+	+	+	+	+

K. ozaenae

- Causes purulent sinus infection
- Agent of atrophic rhinitis

Klebsiella rhinoscleromatis

- May cause granuloma of the nose and oropharynx

Klebsiella ornithinolytica

- Biochemically indole positive
- Not biochemically similar with *K. pneumoniae*

Klebsiella granulomatis (formerly: Calymmatobacterium granulomatis)

- Intracellular; found in mononuclear cells
- Causes donovanosis or granuloma inguinale; characterized by granulomatous lesions in the skin such as in the genital and inguinal areas
- Pathogenic factors are K, Vi
- Cultured on a yolk sac or fresh egg yolk
- On Wright's Giemsa-stained smear, bacilli may show polar granules giving a safety pin appearance within cells (due to the accumulation of granules at both ends of the cell)
- Donovan bodies (Giemsa) – blue rods within mononuclear phagocytes or histiocytes
- Pleomorphic, intracellular, non-motile, not grown on routine culture media
- Media: human monocytes

C. GENUS ENTEROBACTER

- Normal flora of the GIT
- May cause opportunistic infections, UTI, RI, wound infections, and bacteremia
- Motile and ODC positive
- TSI: A/A +gas, H_2S (-)
- Most predominant isolate: *E. cloacae*
- *E. sakazakii* is now *Cronobacter sakazakii*
  - This is known to produce yellow pigment that intensifies at room temperature
  - Can cause necrotizing colitis often due to powdered infant milk formula

	Urease	Yellow pigment	Lactose fermentation	LDC	ADH	ODC
E. aerogenes	-	-	+	+	-	+
E. cloacae	V	-	+	-	+	+
E. gergoviae	+	-	V	+	-	+
E. sakazakii (C. sakazakii)	-	+	+	-	+	+
E. taylorae (E. cancerogenus)	-	-	-	-	+	+

	TSI	LIA	I	M	Vi	C	Urease
Escherichia coli	A/A +gas	K/K	+	+	-	-	-
Klebsiella pneumoniae			-	-	+	+	+
Enterobacter aerogenes			-	-	+	+	-
Enterobacter cloacae		K/A	-	-	+	+	-

PROTEUS-PROVIDENCIA-MORGANELLA (PPM)

- Tribe Proteae
  - Biochemically similar
    - Rapid urease producers
    - Deaminase (+)
    - Non-lactose fermenters
    - Motile
    - LIA: R/A
    - TSI: K/A

D. PROTEUS

- Used as a source of antigens used in Weil Felix test
  - Weil Felix test: serologic test used to detect Rickettsial infections such as Typhus fever
- Significant:
  - *P. mirabilis*
    - Most frequently isolated
    - May cause pneumonia and septicemia
    - Source of OXK
  - *P. vulgaris*
    - Causes nosocomial UTI
    - Source of OX2 and OX19
- Cause human infection only when they leave the GIT
- Shows swarming motility on BAP
- TSA: K/A +gas, H_2S+
- Clear colonies on Mac Conkey agar with burnt chocolate/burnt gun powder odor

	Indole test	ODC	Ampicillin, Cephalosporin, and Chloramphenicol
Proteus mirabilis	-	+	Sensitive
Proteus vulgaris	+	-	Resistant

E. PROVIDENCIA

- No swarming on BAP
- Agents of nosocomial outbreaks
- Normal intestinal flora but often resistant to antibiotics
- Species:
  - *P. rettgeri*: the only urease (+) Providencia
  - *P. stuartii*: urease variable

	Morganella	*Providencia*
ODC	+	-
Citrate	-	+

F. MORGANELLA

- No swarming on BAP
- Species: *Morganella morganii*
- Can cause UTI and Respiratory Tract infections
- Urease and deaminase (+)

	TSI	LIA	I	M	Vi	C	Urease
P. vulgaris	K/A +Gas +H_2S	R/A	+	+	-	-	+
P. mirabilis			-	+	-	+	+
P. rettgeri	K/A +Gas		+	+	-	+	+
M. morganii			+	+	-	-	+

G. YERSINIA

- Should be suspected if TSI is yellow over orange due to weak acid production in slant with no change in butt
- Weak acid production
- All are motile at room temperature (25-30°C) except *Y. pestis*
- Non-motile at 37 degrees Celsius
  - Except *Yersinia pestis*

Yersinia pestis "Plague Bacillus"	- Non-Lactose fermenter
	- Causative agent of plague
	  - Plague: 6th century pandemic or black death
	  - Mode of transmission: flea bites *(Xenopsylla cheopis)*
	- Bioterrorism agent
	- Assumes a safety pin appearance because of bipolar staining which can be demonstrated by Wayson stain/Methylene blue
	- On liquid media/broth it produces stalactite or flocculent growth
	  - Stalactite growth: clumps of cells adhering to one side of the tube
	- On BAP, it can develop hammered copper colonies (48 hours)
	  - Hammered copper colonies: glistening colonies (only if incubation is prolonged for 48 hours)

Yersinia enterocolitica	- A late lactose fermenter
	- Usually found in domestic animals
	- Can tolerate cold enrichment (incubation at 4°C)
	  ○ *Listeria monocytogenes* can also tolerate cold enrichment
	- Most common *Yersinia* isolated from humans
	- Can cause enterocolitis in which symptoms may be mistaken for appendicitis (pseudoappendicular syndrome)
	- Can cause Bull's eye colonies in CIN media
	  ○ Bull's eye colonies = colorless or pink colonies with red center
	- Ornithine and Urease test (+)
	- ONPG (+), ferments mannitol and sucrose

Yersinia pseudotuberculosis	- Agent of mesenteric lymphadenitis, septicemia - Usually found in rodents - Unable to ferment sucrose - ODC (-)

- Types of plague
  - **Bubonic plague**
    - Resulting from flea bites
    - Inflammation in the inguinal and axillary region
    - Bubonic plague is the most common form of plague.
    - This occurs when an infected flea bites a person or when materials contaminated with *Y. pestis* enter through a break in a person's skin.
    - Patients develop swollen, tender lymph glands (called buboes) and fever, headache, chills, and weakness.
    - Bubonic plague does not spread from person to person.
  - **Septicemic plague**
    - Occurs when plague bacteria multiply in the blood.
    - It can be a complication of pneumonic or bubonic plague or it can occur by itself.
    - Septicemic plague does not spread from person to person.
  - **Pneumonic plague**
    - Pneumonic plague occurs when *Y. pestis* infects the lungs.
    - Can spread from person to person through the air.
    - May also occur if a person with bubonic or septicemic plague is untreated and the bacteria spread to the lungs.
    - Characterized by black spots or patches found all over the body

	Y. pestis	***Y. enterocolitica***	***Y. pseudotuberculosis***
Motility at room temperature	-	+	+
Urease	-	+	+
TSI	Non lactose fermenter	Late lactose fermenter	Non lactose fermenter
Sucrose	-	+	-

SALMONELLA-SHIGELLA

- Intestinal pathogens
- Spread via fecal oral route

H. GENUS SHIGELLA

- Can cause shigellosis/bacillary dysentery
  - Bloody diarrhea; somewhat similar with amoebic dysentery
  - Spread via fecal-oral route; no bloodstream infection
- Resembles *E. coli* but are lactose negative
- Non-motile
- TSI: K/A, (-) gas
- Species can produce endotoxin
  - *S. dysenteriae* can produce endotoxin and neurotoxin
  - Invasive
- Serogroups: A, B, C, D
- *S. bongori*: animal pathogen

Serogroup	SPECIES	Lactose	Mannitol	Ornithine decarboxylase	MR	ONPG	Catalase test
A	*Shigella dysenteriae* "Shiga Bacillus"	-	-	-	+	-	-
B	*Shigella flexneri* "Strong's Bacillus"	-	+	-	+	-	+
C	*Shigella boydii* "New Castle Manchester Bacillus"	-	+	-	+	-	+
D	*Shigella sonnei* "Duval's Bacillus"	- LLF	+	+	+	+	+

I. GENUS SALMONELLA

- Acquired via fecal-oral route, may cause enteric fever/typhoid fever and AGE/food poisoning
- Species can cause a number of infections
- Virulence factor: pili
- With O and H antigen for serogrouping
- Pathogenic, Motile, except *S. gallinarum* and *S. pullorum*
- *S. typhi* causes typhoid fever
  - Most significant specie
  - Site of long term of *S. typhi* is the gall bladder
  - iMViC: -+--
- Non lactose fermenter
- KCN (-)
- Produe H_2S except *S. paratyphi A*
- LDC (+), gas (+) except *S. gallinarum* and *S. pullorum*
- TSI: K/A +gas
- LIA: K/K

Enterocolitis/Gastroenteritis	*Salmonella typhimurium* *Salmonella enteritidis*
Bacteremia	*Salmonella serotype cholerasuis* *Salmonella paratyphi C*
Typhoid fever	*Salmonella serotype typhi*
Paratyphoid	*Salmonella serotype paratyphi*

	Salmonella typhi	***Salmonella paratyphi***
Citrate	-	-
Gas	-	+
H2S	+	-
LDC	+	-
ODC	-	+

	Salmonella	***Shigella***
Resembles	*Citrobacter*	*E. coli*
Motility	Motile	Non-motile
H_2S	+	-
Malonate	+	-
Invasive	Not invasive	Invasive Bloody diarrhea
LIA	K/K	K/A

OTHER ENTEROBACTERIACEAE

J. CITROBACTER

- Resembles *Salmonella* but ONPG (+) and LDC (-)
- ONPG and LDC are the biochemical tests that will differentiate *Citrobacter* from *Salmonella*
- Late or slow lactose fermenter
- *C. freundii* & *C. diversus,* both may cause UTI and sepsis

	Indole test	**H_2S**	**ODC**
C. freundiii	-	+	V
C. diversus (now C. koseri)	+	-	+

K. EDWARDSIELLA

- Species: *Edwardsiella tarda*
- Opportunistic pathogen; Identified as a source of GI infection
- Non-lactose fermenter
- TSI: K/A, gas (+), H_2S (+)
  - Unique: produces large amount of H_2S
- Biochemicall resembles *E. coli* but NLF and H_2S+
- IMViC: ++--

L. HAFNIA

- *Hafnia alvei* (previously *E. alvei*)
- Late lactose fermenter
- Citrate (-; delayed), DNAse (-), Lipase (-)

M. SERRATIA

- DNAse, Lipase, and Gelatinase (+)
- TSI: K/A, H_2S (-)

Serratia marcescens	- Most significant specie - Produces red pigment prodigiosin when incubated at room temperature - Implicated in outbreaks of bacteremia
Serratia liquifaciens	
Serratia rubidaea	- Produces red pigment
Serratia odorifera	- Produces a rancid-like potato odor

N. PLESIOMONAS

- Previously classified under *Vibrionaceae*
- *P. shigelloides*
- Cytochrome Oxidase (+)
- Causes secretory diarrhea in HIV (+) individuals with inflammatory bowel disease; acquired following consumption of undercooked seafoods

BOARD EXAM RECALLS!

In Runyoun's, *M. fortuitum* and *M. chelonei* are	Rapid growers	Simmon citrate agar indicator	Bromthymol blue
Sulfanilic acid and alpha naphthylamine are used in	Nitrate reduction test	SIM media is used for what purpose	Motility H_2S production Indole production
MR-VP reactions are	Opposite	*Salmonella* on SSA	Colorless colonies
LOA test uses what medium	Moeller's medium	*E. coli* on XLD	Yellow colonies
Mineral oil	Oxygen barrier in Decarboxylase test	Negative result in MUG test	Absence of electric blue fluorescence
PAD (+) Urease (+)	PPM group	*M. tuberculosis* clear media	Middle brook media
Oxidase (-)	*Enterobacteriaeceae*	No color change in nitrate test following addition of zinc dust, this is interpreted as	Positive
ONPG (+) yellow	*Citrobacter*	Purpose of photoreactivity	Identify those under Runyoun's
Salmonella spp. resembles	*Citrobacter*	Purpose of nitrate reduction and niacin test	Identify *M. tuberculosis* and differentiate it with *M. bovis*
Shigella spp. resembles	*E. coli*	CIN bull's eye colonies	*Y. enterocolitica*
Sorbitol-MaConkey-colorless	*E. coli O157:H7*	Arylsulfatase test used as preseumptive test to identify	*M fortuitum* and *M. chelonei*
2-4% NaOH is used as	Decontaminating agent	(+) result in MR-VP	Oppositve
N-acetyl-L-cysteine	Digesting agent	IMViC in *E. coli* result	++--

MUST REMEMBER!

TEST	POSITIVE RESULT
Niacin	Yellow
Nitrate reduction	Initial: Red With Zing: no color change
Iron uptake test	Rusty brown color
Heat stable catalase	Vigorous bubbling
Arylsulfatase test	Pink
Tween 80 hydrolysis test	Pink
Tellurite reduction	Black
ONPG	Yellow
Indole	Red ring at the junction
Methyle Red test	Red
VP test	Pink to Red
Urease test	Red
Decarboxylase test (Moeller's)	Purple
Decarboxylase (LIA)	Purple
Deaminase	Phenylalanine – green Tryptophan – brown
Deaminase (LIA)	Burgundy – Red
Citrate Ultilization Malonate Utilization	Blue

GRAM NEGATIVE INTESTINAL PATHOGENS (NON-ENTERIC)

- *Vibrio*
- *Aeromonas*
- *Helicobacter*
- *Campylobacter*

GENUS VIBRIO

- Naturally found in seawater
- All are halophilic, requiring increased salt concentration
  - Except: *V. mimicus* & *V. cholerae*
- Motile (monotrichous), facultative anaerobes
- Oxidase (+)
  - Screening test for genus *Vibrio*
  - Except: *Vibrio metschnikovii*
- Specimen used for diagnosis: Stool
  - Must be collected and transported only in a: Cary-Blair medium
- Usual mode of transmission: consumption of raw or undercooked fish/sea foods
- (+) glucose fermentation
- Non-lactose fermenter
  - Except: *V. vulnificus*
- Media
  - TCBS (selective-differential media with sucrose)
    - pH indicator: Bromthymol blue
    - pH of the media: 8.6
  - Alkaline peptone water – enrichment broth

Vibrio cholerae

- Motile; shows shooting star motility
- (+) String Test
  - uses 0.5% sodium deoxycholate; (+) string like
- (+) Cholera red test - not specific because any organism (+) to indole and nitrate will give a (+) result
- For this test alkaline peptone water with tryptophan and nitrate + organism, incubate then add sulfuric acid, positive result: red color due to nitrosoindole
- Causes CHOLERA
  - Diarrheal disease characterized by the production of rice-water stool
  - Severe fluid loss & electrolyte loss leads to dehydration
- Choleragen
  - Virulence factor
  - Stimulate hypersecretion of water and chloride ions while preventing absorption of sodium ions
- Vibrio cholera subgroups
  - *Vibrio cholerae 01*
    - the one causing classical epidemic cholera
    - divided into 2 biotypes classical and el tor
    - toxigenic; produces choleragen
  - *Vibrio cholerae* 0139: unable to produce choleragen
  - *Vibrio cholerae non* 01: unable to produce choleragen
- *V. cholerae* 01 biogroups

	Classical	El Tor
Hemolysis on Sheep RBC	-	+
VP Test	-	+
Chicken RBC Agglutination	-	+
Polymyxin Susceptibility	S	R

- *Vibrio Cholerae* 01 SEROTYPES
  - used to follow pandemics
    - OGAWA or VARIANT F – India
    - INABA or ORIGINAL J – Philippines
    - HIKOJIMA or MIDDLE or INTERMEDIATE - Japan

	Disease	8% NaCl (Halophilic)	TCBS Yellow – sucrose fermenter Green – non-sucrose fermenter	Oxidase
Vibrio parahaemolyticus	Gastroenteritis due to contaminated seafood	+	Green colonies	+
Vibrio vulnificus	Wound infection and septicemia	+	Green colonies	+
Vibrio alginolyticus	Wound and ear infection	+	Yellow colonies	+
Vibrio mimicus	Gastroenteritis and ear infection	-	Green colonies	+
Vibrio fluvialis	Diarrheal disease	+	Yellow colonies	+
Vibrio furnissii	Diarrheal disease	+	Yellow colonies	+

GENUS AEROMONAS

- Oxidase (+); fermentative gram negative bacilli
- May cause cellulites and wound infections
- Can grow on Mac Conkey, EMB, SSA, and CIN
- Specie:
  - *Aeromonas hydrophila*
    - water loving organism associated with GIT disease
  - *A. caviae*
    - Most common isolate of *Aeromonas*
    - can grow at 4-42°C

GENUS HELICOBACTER

Helicobacter pylori

- Formerly called *Campylobacter coli*
- Habitat: human gastric mucosa (pylori end of the stomach)
- Causes: gastritis, peptic ulcer, gastric carcinoma
- With very strong urease activity
- Virulence factor: Urease
- Diagnosis
  - Endoscopy
    - Invasive; non-culture method
  - Stool antigen/Copro antigen test (Serotest)
  - Urea Breath Test
    - Involves measuring urease activity
    - Non-culture method and non-invasive
- Specimen: tissue biopsy
- Stain: Warthin-Starry Stain
- Can grow on the following media:
  - CAP
  - Brucella agar with 5% sheep blood
  - Skirrow's
  - Modified Thayer Marin

CAMPYLOBACTER

- May appear like *V. cholerae*
- Appears as S-shaped bacilli (appear like wings of sea gulls)
- Motile with single polar flagellum with darting motility
- Requires low oxygen 5% (microaerophilic) and increased CO_2 10% (capnophilic)
- Optimum temperature for growth 42-43°C
- Antecedent cause of Guillain-Barre Syndrome
- Media for Primary isolation
  - Modified Skirrow's Media (blood agar base, 7% horse lysed blood, vancomycin, trimethoprim and polymixin B)
  - CAMPY-BAP (Brucella agar base with trimethoprim, polymyxin B, cephalotin, vancomycin and amphotericin B and 10% sheep blood)
  - Modified Charcoal Cefoperazone Deoxycholate agar
  - CAMPY - CVA (Brucella base agar with cefoperazone, vancomycin and amphotericin B and 5% sheep blood)

	Campylobacter jejuni	*Campylobacter coli*	*Helicobacter pylori*
Growth at 25°C	-	-	-
Growth at 42°C	+	+	+
Hippurate Hydrolysis	+	-	-
Catalase test	+	+	+
H_2S in TSI	-	-	-
Reduction of Nitrate to Nitrite	+	+	V

- Species:
  - *Campylobacter coli* and *Campylobacter jejuni*
    - Associated with gastritis and diarrhea
    - Thermophilic

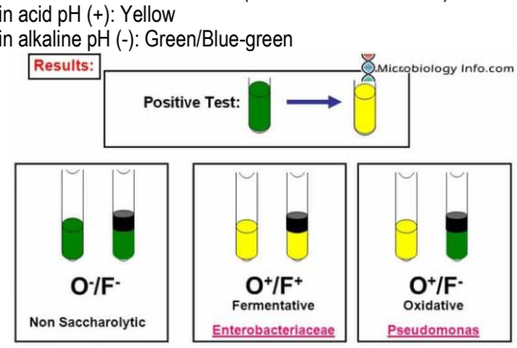
	VIBRIO	AEROMONAS	PLESIOMONAS
NaCl requirement 8-10%	+	-	-
Motility	+	+	+
Oxidase test	+	+	+
Hemolysis, DNAse, and Esculin hydrolysis	+	-	+
Mannitol	+	+	-
Growth on TCBS	+	+	-
O129 Susceptibility	S	R	V
Lysine-Ornithine-Arginine	++-	+-+	+++
Gelatin liquefaction	+	+	-

O129 = 2,4 diamino 6,7 diisopropylpteridine

GRAM NEGATIVE NON-FERMENTATIVE BACILLI

- Pseudomonas	- Acinetobacter
- Burkholderia	- Alcaligenes, Moraxella
- Stenotrophomonas	- Chromobacterium

- **OF MEDIA** (Oxidative -Fermentative Media of Hugh and Leifson)
  - Detects ability of organism for acid production (same with TSI)
  - Contains 17% glucose, 1% agar, peptone
  - pH indicator: Bromthymol blue
    - If CDC method: the indicator is phenol red
  - High carbohydrate and low peptone content
  - 2 tubes are prepared:
    - Open tube = aerobic
    - Closed tube = anaerobic (overlaid with mineral oil)
  - in acid pH (+): Yellow
  - in alkaline pH (-): Green/Blue-green



POSSIBLE RESULTS	INOCULATED ORGANISM MAY BE CLASSIFIED AS:	POSSIBLE ORGANISM/S
Positive in both open & closed tubes	Fermentative	*Enterobacteriaceae*
Positive in open tube Negative in closed tube	Oxidative/Oxidizer Non-fermenter	*Pseudomonas*
Negative in both open and closed tubes	Non-oxidizer Non-saccharolytic/ asaccharolytic	*Moraxella catarrhalis*

Characteristics of non-fermenters:

1. Never normal but are transient colonizing flora of hospitalized patients; implicated in cases of nosocomial infections

FAMILY PSEUDOMONADACEAE

- All are OXIDASE (+) except: *Stenotrophomonas maltophilia*
  - *B. gladioli*: oxidase (-) and a plant pathogen
- TSI: K/K
- All are MOTILE except:
  - *Burkholderia mallei*

PSEUDOMONAS

Pseudomonas aeruginosa

- #1 ICU isolate
- Causes pneumonia in those with Cystic fibrosis
- Strict aerobe, motile
- Most frequently isolated non-fermenter, does not ferment carbohydrate
- TSI = K/K
- Grows well at 37-42°C
- Oxidase test (+); Catalase (+)
- Beta hemolytic on BAP
- Colonies tend to spread to give off a characteristic odor resembling that of an overripe grape "Corn Tortilla Like Odor" or Corn Taco Odor
- Produces:
  - Pyocyanin: water soluble blue pigment
  - Pyoverdin/Fluorescein: green pigment
- May cause
  - (1) Wound infections in burn patients = blue pus
  - (2) UTI
  - (3) Pneumonia & sepsis
  - (4) Mild otitis externa /Swimmers ear or Shanghai fever/ Diver's ear infection
  - (5) Jacuzzi or hot tub syndrome (manifest as skin rash)
- Virulence factors
  - Exotoxin A = once released, it interferes with protein synthesis
  - Endotoxin
  - Pili = for adherence
  - Alginate = promote the ability of the organism to produce biofilms
- On BAP = gray spreading beta hemolytic colonies
- MAC = colonies with green pigment

- **Cetrimide Agar**: for initial isolation and purify *P. aeruginosa*; selective media that is toxic to many bacteria but not to *P. aeruginosa*
  - (+) control: *P. aeruginosa* (growth and color change – yellow green to blue green)
  - (-) control: *E. coli* (no growth and color change)
- Acetamide Utilization Test (+)
  - Uses Bromthymol blue as indicator
  - (+) = blue color

Other Pseudomonas:

Pseudomonas fluorescens* and *Pseudomonas putida	***Pseudomonas stutzeri* & *Pseudomonas mendocina***
Motile, oxidase (+) Pyocyanin (-) Pyoverdin (+)	Soil denitrifiers (organisms that use ammonium as sole nitrogen source; use acetate as carbon source)

- *P. fluorescens* = common blood bag contaminant
- *P. fluorescens and P. putida* = have been implicated in transfusion-associated septicemia
  - Differential test = Gelatin hydrolysis

	P. aeruginosa	***P. fluorescens***	***P. putida***
Pyocyanin	+	-	-
Pyoverdin	+	+	+
Growth at 42°C	+	-	-
Acetamide	+	-	-
Gelatin hydrolysis	+	+	-
Nitrate reduction	+	-	-

	37°C	**42°C**	**Pyocyanin**	**Pyoverdin**
P. aeruginosa				
P. fluorescens				
P. putida				

BURKHOLDERIA

Burkholderia cepacia	- Previous name: *P. kingie or P. multivorans* - Found near water sources - Causes onion bulb rot in plants and foot rot in man
Burkholderia mallei a.k.a. "Glander's Bacillus"	- Causative agent of Glander's disease – a disease of animals like horses - Transmitted to man through direct contact, through skin abrasions, inhalation of organism - Unable to grow at 42°C
Burkholderia pseudomallei a.k.a. "Whitmore's Bacillus"	- Causative agent of Melioidosis or Vietnam Time Bomb – Glander's like disease characterized by severe pneumonia; takes several years for symptoms to develop - Colonies with earthy odor on ashdown medium

STENOTROPHOMONAS

Stenotrophomonas maltophilia

- May cause nosocomial infections like pneumonia and UTI
- Produce large colonies with yellow pigmentation on TSA and lavender green pigment
- Able to grow at 42°C, colonies with "ammonia-like" odor
- For rapid detection, we use: Oxidase (-) & DNAse test (+)

	B. cepacia	***B. pseudomallei***	***S. maltophilia***
Oxidase	Weak +	+	- DNAse (+)
Nitrate to Nitrite	V	+	V
Arginine	-	+	-
Lysine	+	-	Slow +
Glucose	+	+	Weak +
Lactose	+	+	Strong +
Mannitol	+	+	-

	Lysine	**Ornithine**	**Arginine**
Pseudomonas	-	-	+
B. cepacia	+	-	-
S. maltophilia	+	-	-

ACINETOBACTER

- Colonize moist areas of skin
- Oxidase (-), Catalase (+)
- Non-motile
- Grows well on BAP (produce gummy colonies) and MAC

Acinetobacter baumannii	***Acinetobacter lwoffi***
- A.k.a. *Iraqibacter* - Regarded as the gram negative MRSA - More virulent - Glucose oxidizer	- Less virulent - Non-glucose oxidizer/non-saccharolytic

	A. baumannii	***B. cepacia***	***Chryseobacterium meningosepticum***
Motility	-	+	-
Oxidase	-	+	+
MAC	+	+	-

Alcaligenes faecalis

- Catalase, oxidase (+)
- Apple-like fruity odor
- Motile with peritrichous flagella
- Alpha hemolytic
- Mac Conkey (+)
- UTI, wound, diarrhea, obligate aerobe

	A. faecalis	***Acinetobacter***
Motility	+	-
Oxidase	+	-

Moraxella lacunata

- Causes Blepharoconjunctivitis or Angular conjunctivitis
- Oxidase and catalase (+)
- (-) Mac Conkey
- Mistaken as *Neisseria*

Chromobacterium violaceum

- Not part of the normal flora, found in the environment; found in patients with neutrophil deficit including patients with chronic granulomatous disease (CGD)
- On DNAse agar, produces violet pigment: Violacein
- On BAP, beta hemolytic and with ammonium cyanide odor

GRAM NEGATIVE BACILLI (FASTIDIOUS)

HAEMOPHILUS

- Fastidious organism
- Use either horse or rabbit's blood for isolation (horse blood is preferred); Sheep's blood agar is not used for isolation
- Most are normal flora of the upper respiratory tract and oral cavity
- Non-motile, non-spore former, facultative anaerobes
- Most species are catalase (+) and oxidase (+)
- Prefer incubation at: 35-37°C with 5-10% CO_2
- Chocolate agar (heated blood agar) can be used for isolation since it can provide X and V factors
  - Blood loving; requiring blood factors (X and V factors)
  - **X Factor** (Hemin or Protoporphyrin IX)**:** heat-stable product released upon the degradation of hemoglobin
  - **NAD/Co-enzyme 1/V Factor**: Heat-labile; Produced by certain bacteria (i.e. *Neisseria & S. aureus)* and yeast; can be obtained from potato or yeast extract.

Haemophilus are identified through:

- Hemolytic reactions on horse blood agar – if hemolytic or non-hemolytic
  - 2 species which are beta-hemolytic: *H. haemolyticus & H. parahaemolyticus*
  - The rest are gamma hemolytic
- Carbohydrate fermentation tests
  - Using glucose, sucrose, lactose, or fructose
- Growth requirement of X and V factors
  - Require X & V factors: *H. haemolyticus, H. aegypticus, H. influenzae*
  - Require V factor only: *Haemophilus* species with "para" in their name (i.e. *H. parainfluenzae, H. parahaemolyticus, H. paraphrophilus*)
  - Require X factor only: *H. ducreyi*
  - Does not require X and V factors: *H. aphrophilus* (now classified under *Aggregatibacter*)

Haemophilus influenzae

- A.k.a Pfeiffer's Bacillus
- Regarded as normal flora of the respiratory tract; associated with respiratory conditions including epiglottis and meningitis in children less than 5 years' old
- Encapsulated strains are pathogenic
- Colonies with mousy (mouse nest) odor on plated media; colonies are: dew drops
- Exhibits satellitism: luxuriant growth near the source of the incorporated V factor; demonstrated by inoculating unknown organism on BAP, inoculate *S. aureus* in a single streak, incubate. If the growth is found surrounding *S. aureus,* it is called satellitism
- Hemolytic pattern: Gamma
- There are 6 serotypes (a, b, c, d, e, f); the most common serotype is B (implicated in cases of meningitis)
- Major virulence factor: capsule

Haemophilus aegypticus	*Haemophilus ducreyi*
- A.k.a Koch Week's Bacillus - Causes pink eye; a contagious form of conjunctivitis - Causes Brazilian Purpuric fever – affects children	- Highly fastidious (culture media must be supplemented) - An agent of STD - Causes: Soft chancre - Forms the so called: school of fish appearance (coccobacilli lined up); railroad track or finger print appearance

Haemophilus haemolyticus	*Haemophilus parainfluenzae*	*Haemophilus aphrophilus & Haemophilus paraphrophilus*
- An occasional normal flora of the upper respiratory tract - Beta hemolytic	- Rarely infectious - Normal flora of the upper respiratory tract	- Normal flora of the oral cavity - Found in dental plaque - Rarely associated with endocarditis and pneumonia

SPECIES	X FACTOR	V FACTOR	BETA HEMOLYSIS	D-ALA
Haemophilus influenzae	+	+	-	-
Haemophilus parainfluenzae	-	+	-	+
Haemophilus haemolyticus	+	+	+	-
Haemophilus parahaemolyticus	-	+	+	+
Haemophilus aegypticus	+	+	-	-
Haemophilus aphrophilus	-	-	-	+
Haemophilus paraphrophilus	-	+	-	+
Haemophilus ducreyi	+	-	-	-

D-aminolevulinic acid or Porphyrin test

- Detects the ability of the organism to produce the enzyme needed in converting the substrate D-ALA into porphyrin and porphobilinogen
- Porphobilinogen is detected using Kovac's; (+) = red
- Porphyrin is detected when it shows red fluorescence under wood's lamp

SPECIES	GLUCOSE	SUCROSE	LACTOSE	FRUCTOSE
Haemophilus influenzae	+	-	-	-
Haemophilus aegypticus	+	-	-	-
Haemophilus ducreyi	+	-	-	-
Haemophilus haemolyticus	+	-	-	WEAK+
Haemophilus parainfluenzae	+	+	-	+
Haemophilus parahaemolyticus	+	+	-	+
Haemophilus aphrophilus	+	+	+	+
Haemophilus paraphrophilus	+	+	+	+

MISCELLANEOUS GRAM NEGATIVE BACILLI

- *Brucella*	- *Legionella*	- *Streptobacillus*
- *Bordetella*	- *Capnocytophaga*	- *Spirillum*
- *Francisella*		

A. BRUCELLA

- Agents of bioterrorism (except: *B. canis*)
- Normal flora of the UT and GI of animals; Zoonotic
- In man, it causes: Undulating Fever/Malta Fever/Gibraltar Fever/Mediterranean Fever
- Mode of transmission: ingestion of unpasteurized milk, inhalation of organism and direct inoculation into the bloodstream, abortion in animals
- Organism can survive in extended periods in milk/ice cream = 3 weeks, cheese = several months
- Non-motile, strict aerobe, intracellular, localized in animal tissues rich in erythritol (such as placental tissues)
- Catalase and oxidase (+)
- Hemolytic pattern: gamma
- Should be incubated in an atmosphere containing 5-10% CO_2
- Growth is inhibited by dyes; sensitive to heat: killed by pasteurization
- Optimal specimen for Brucellosis: blood
- Rapid test for Brucella: Serum or particle agglutination test
  - Significant titer: 1:160
- Definitive test: Culture
- Gold standard for detection: Culture and Rapid Test for Brucellosis
- Media for isolation: Castaneda's media
- Other media used:
  - Trypticase soy agar
  - Tryptose agar
  - Brucella broth agar or Wisconsin media

	Brucella abortus *Bang's Bacillus*	*Brucella melitensis*	*Brucella suis*	*Brucella canis*
Animal host	Cattle and buffalos	Sheep, goats, or camels	Swine/Pigs	Dogs
H_2S production	+	-	V	-
Urease (+) in	2-24 hours		0-15 mins.	
Growth on media with basic fuchsin	+	+	-	-
Growth on media with thionine	-	+	+	+
Need for CO_2	+ (Capnophilic)	-	-	-

- More virulent in man: *B. abortus and B. melitensis*

B. BORDETELLA

- Strict aerobe, replicates in ciliated respiratory ET cells
- Non-motile except: *B. bronchiseptica*
- Media:
  - Bordet Gengou media/Potato Blood Glycerol: commonly used
  - Modified Jones Kendrick Charcoal
  - Regan Lowe: preferred media; with horse blood, charcoal, cephalexin and amphotericin B; enrichment and selective media

Bordetella pertussis

- Bordet Gengou Bacillus; resembles *H. influenzae*
- Non-motile; major virulence factor: pertussis toxin (increase secretion of mucus)
- Produce mercury like droplet colonies
  - Causes Whooping Cough/Pertussis
    - Catarrhal phase: characterized by flu-like symptoms; recovery of organism in nasopharyngeal swab is highest in this phase; coughing is not yet severe
    - Paroxysmal: coughing is severe
    - Convalescence: period of recovery
  - Response to pertussis infection: lymphocytosis
- Ideal specimen: nasopharyngeal swab
- Best time to collect sample: catarrhal phase
- For collection, cotton swabs are not used; use Dacron or Rayon fibers instead
- Diagnosis PCR (polymerase chain reaction) - most sensitive for the diagnosis of pertussis

	Growth in Regan-Lowe	Urease	Nitrate	Motility	Oxidase	Citrate	BAP
Bordetella pertussis	3-6 days	-	-	-	+	-	-
Bordetella parapertussis	2-3 days	+	-	-	-	+	+
Bordetella bronchiseptica	1-2 days	+	+	+	+	+	+

Bordetella bronchiseptica	*Bordetella parapertussis*
Motile, most active and can grow on conventional media A rapid urease producer Rare cause of respiratory disease Causes Kennel cough in dogs	Causes pertussis-like symptoms

C. FRANCISCELLA

- Species: *Francisella tularensis*
- Potential agent of bioterrorism
- Causes Tularemia
- A.k.a Deerfly fever, Rabbit fever, Water rat trapper disease, Market Men's Disease
  - May be acquired thru direct contact with blood or thru animal bite or scratch
- Exhibit "faint bipolar staining", can multiply with some amoeba (*Naegleria and Acanthamoeba)*
- Alpha hemolytic
- Catalase: (+), Oxidase: (-), Beta lactamase: (+)
- Classified under BSL: 3
- Forshay test: susceptibility test for Tularemia
- Requires cystine, cysteine, and thiosulfate
- Media: Blood Cystine Glucose Agar (BCGA)

D. LEGIONELLA

- Ubiquitous, naturally found in both natural and artificial water
- Poor staining characteristics
- Media: Buffered Charcoal Yeast Extract (BCYE) - charcoal detoxifies media to allow organism to proliferate easily
- Optimal specimen: bronchial washings
- ***Legionella pneumophila***
  - with 15 serotypes
  - Serotype 1: causes Legionnaire's disease or Broad street pneumonia or Pontiac fever
  - Serotype 6: causes pneumonia
- ***Legionella micdadei***
  - causes Pittsburgh pneumonia
- ***Legionella bozemanir***
  - Wiga's agent of pneumonia

E. *Pasteurella multocida*

- Causes Pasteurellosis
- MOT: contact with domestic animals harboring the organism
- Agent of Shipping fever in cattle
- Wound infection is the most common type of pasteurella infection in man (animal bite wound); a form of hemorrhagic septicemia
- Produces colonies with musty odor (smelling like mushrooms)

F. *Gardnerella vaginalis*

- Previously called *Corynebacterium /Haemophilus vaginalis*
- Causes: Bacterial Vaginosis characterized by foul smelling grayish excessive vaginal discharge
- Diagnosis:
  - Cytology/Pap's look for clue cells: squamous cells studded with bacilli
  - Whiff test/Sniff test
    - Addition of KOH to vaginal discharge
    - Fishy amine-like odor is noted
  - Isolation on HBT (Human blood bilayer Tween)
  - Pattern of hemolysis: beta
- Nugent scoring- system used for diagnosis of vaginosis

G. *Capnocytophaga gingivalis*

- Capnophilic
- Does not have flagella but exhibits: gliding or sliding motility
- Normal flora of the human oral cavity

H. *Streptobacillus moniliformis*

- Forms string of beads appearance
- When inoculated on liquid media: forms puff balls or fluff balls growth in broth (cotton balls at the bottom of the tube)
- Causes rat bite fever and Haver Hill fever (acquired by consumption of contaminated milk)

I. *Spirillum minus*

- Causes rat bite fever called as Sudoku

HACEK

- Gram negative coccobacilli
- Causes Subacute Bacterial Endocarditis (SBE)

Haemophilus parainfluenzae

- Normal flora of the oral cavity

Aggregatibacter (Actinobacillus)

- *Aggregatibacter aphrophilus*: foam-loving bacteria; does not require X or V factors
- *Aggregatibacter actinomycetemcomitans*: committed to grow with *A. israelii*; produce star-shaped colonies

Cardiobacterium hominis

- Normal flora; pleomorphic
- Usually found in aortic valves
- Forms rosette or flower-like arrangement in gram staining

Eikenella corrodens

- Fastidious gram negative capnophilic rod
- Part of the gingival and bowel flora
- Non-saccharolytic
- With typical bleach-like odor
- Causes the so-called: human bite wound
- When allowed to grow on plated media, it pits and corrodes the agar

Kingella kingae

- Most common *Kingella* causing SBE
- Shows twitching motility
- Has pronounced tropism for bones and heart

	Oxidase	Catalase	Indole	LDC
A. aphrophilus	+	-	-	-
A. actinomycetemcomitans	+	+	-	-
Cardiobacterium hominis	+	-	+	-
Eikenella corrodens	+	-	-	+
Kingella kingae	+	-	-	-

ANAEROBES

- Organisms that die in the presence of oxygen

GRAM POSITIVE ANAEROBIC RODS

ACTINOMYCES

- *Actinomyces israelii*
  - Produces white colonies with molar tooth appearance (old/aged colonies)
  - Causes Actinomycosis: disease affecting soft tissues
  - Actinomycosis of the jaw: lumpy jaw
  - Produce sulfur granules

PROPIONIBACTERIUM

- *Propionibacterium acnes*
  - Often isolated as a blood culture contaminant
  - Agent of acne in adolescence
  - Appears as clubbed, pleomorphic rod in palisades that appear like aerobic diphtheroids; often regarded as anaerobic diphtheroids
  - Forms log jam arrangement under the microscope

BIFIDOBACTERIUM

- Highly pleomorphic rod with bifid or bifurcated ends
- May appear like bone-shaped dog biscuit

EUBACTERIUM

- Pleomorphic rod; usually in pairs and chains
- Normal flora of the mouth and intestine
- Metabolic product: butyric acid

Organism	Vancomycin (5 ug)	Kanamycin (100 ug)	Colistin (10 ug)	Catalase (15% H_2O_2)	Indole	Nitrate reduction	Oxygen tolerance
Actinomyces israelii	S	S	R	-	-	+	MA
Propionibacterium acnes	S	S	R	+	+	+	MA
Bifidobacterium species	S	S	R	-	-	-	OA
Eubacterium species	S	S	R	-	-	-	OA

***MA:** Minimally aerobe; **OA:** Obligate anaerobe

GRAM NEGATIVE ANAEROBIC RODS

Bacteroides fragilis group

- **Species: *Bacteroides fragilis***
  - Most common isolate; causes bacteremia
  - On KVLB: appear as large, wet, gray colonies
  - On BBE (Bacteroides Bile Esculin): black colonies with dark halo
- **Species: *Bacteroides urealyticus***
  - Pit/corrodes the agar

FUSOBACTERIUM

- **Species: *Fusobacterium nucleatum***
  - Predominant in the oral cavity; may cause aspiration pneumonia
  - Toothpick (because of the tapered ends)
  - Colonies with rancid butter odor due to production of butyric acid
  - With 3 different colony morphology
    1. Irregularly shaped colonies (bread crumbs or salt grains)
    2. Large, smooth with a ground glass or speckled appearance
    3. Small, smooth, gray-white

BILE SENSITIVE – PIGMENTED: Genus *Prevotella* and *Porphyromonas*

- Asaccharolytic
- Show brick-red fluorescence when exposed to UV
- Unable to tolerate bile
- Differential test is Vancomycin susceptibility test

Organism	Vancomycin (5 ug)	Kanamycin (100 ug)	Colistin (10 ug)	Catalase	Indole	Stimulated by bile	Esculin
Bacteroides fragilis	R	R	R	+	-	+	+
Fusobacterium nucleatum	R	S	S	-	+	-	-
Prevotella	R	R	V	-	-	-	-
Porphyromonas	S	R	R	-	+	-	V

GRAM POSITIVE ANAEROBIC COCCI

Peptostreptococcus anaerobius

- Indole (-), PYR (-); SPS: Sensitive
- Non-hemolytic and with sweet odor

GRAM NEGATIVE ANAEROBIC COCCI

VEILONELLA

- Common isolate by rarely involved in disease
- Produces red fluorescence under UV light

DO'S & DON'T'S IN ANAEROBIC CULTURE

1. Ideal specimen is whole tissue or aspiration via needle and syringe, remove the needle following aspiration
2. Swab – least favorable method for collection.
   Throat swab for anaerobic culture = reject.
3. Specimens on swabs can dry out during transportation and expose anaerobes to room air, not to be accepted unless transported in proper anaerobic environment
4. Have special nutritional requirements for hemin, Vitamin K, and yeast extract

Specimens ACCEPTABLE for ANAEROBIC CULTURE	UNACCEPTABLE for ANAEROBIC CULTURE
Blood, body fluids, bone marrow aspirate, percutaneous lung aspirate or biopsy, suprapubic bladder aspirates, transtracheal aspirate, wound	Expectorated sputum, feces, gastric juice, swabs, voided or catheterized urine, bronchial washings

Media:

1. **BBE/BACTEROIDES BILE ESCULIN**
   - Selective media for *Bacteroides fragilis group*
   - Gentamicin: inhibits aerobes
   - 20% bile: inhibitory to most anaerobes
   - Laked – blood hemolyzed by freezing and thawing
2. **KVLB – KANAMYCIN, VANCOMYCIN, LAKED BLOOD**
   - Selective
   - Kanamycin: inhibitory to most facultative gram negative bacilli
   - Vancomycin: inhibitory to gram (+) organisms and some porphyromonas
3. **CCFA**
   - Cycloserine-Cefoxitin Fructose Agar
   - For isolation of *Clostridium difficile*
4. **EGG YOLK AGAR**
   - Detection of lecithinase and lipase production by *Clostridium* and *Fusobacterium*
5. **THIOGLYCOLLATE BROTH**
   - Enrichment media
   - Supports growth of aerobes and anaerobes
   - It acts as reducing agent, aerobes grow at TOP, strict anaerobes at BOTTOM, and facultative anaerobes throughout
   - Store at room temperature; Boil and cool before use

SPIROCHETES

- Gram negative organisms appearing like corkscrews; moves in a unique spinning fashion via: axial filaments (periplasmic flagella)
- Replicate by transverse fission
- Darkfield microscope, immunofluorescence, silver stains are required to view these organisms
- Silver stains: Levaditi's, Fontana Tribondeau

TREPONEMA

Treponema pallidum

- Causative agent of: Syphilis
- Transmitted through sexual contact, blood transfusion, transplacental transfer
- Diagnosis is often through serologic tests
- Non-culturable in-vitro; to maintain its viability in the lab, we use testicular chancre of rabbits

STAGES OF SYPHILIS	
PRIMARY STAGE	- Genital lesions: Hard chancre or Hunterian chancre
SECONDARY STAGE	- Genital lesions spreads to other parts (extra-genital lesions): Condyloma latum
LATENT STAGE	- Absence of symptoms (no lesions) - When the blood is tested, it is positive for Syphilis
TERTIARY STAGE	- Lesions in deep organs of the body: Gummas

METHOD OF DIAGNOSIS

1. Direct Visualization: using darkfield microscope or silver stains
2. Serologic Tests: most common (non-treponemal & treponemal tests)

3 subspecies of *Treponema pallidum* that causes non-venereal diseases Mode of transmission: Acquired through direct contact with infected lesions	
Treponema pallidum subspecies pertenue	Yaws or Frambesia
Treponema pallidum subspecies carateum	Pinta
Treponema pallidum subspecies endemicum	Bejel or Endemicum Syphilis

LEPTOSPIRA

- Tightly coiled spirochetes with one or both ends bent to form a hook
- Culture media: Fletcher's media; Ellinghausen Mc Cullough Johnson

Leptospira biflexa

- A non-pathogenic specie found in water and soil

Leptospira interrogans

- Cause of human and animal leptospirosis; a zoonosis
  - Anicteric phase: septicemia and high fever; may complain of headache
  - Icteric leptospirosis or Weil's disease: there is involvement of the kidney or liver
  - Hallmark of immune phase: aseptic meningitis
- The organism is shed in the urine of animals (rats, dogs, etc.). Man acquires infection through direct contact with urine of animals carrying the organism
- Blood, CSF: specimens on the 1st week
- Urine: during the immune phase or the 2nd week

L. interrogans servovar icterohemorrhagia	Weil's disease
L. interrogans servovar canicola	Infectious jaundice
L. interrogans servovar mitis	Swineheard disease
L. interrogans servovar hebdomadis	7-day fever
L. interrogans servovar grippotyphosa	March fever
L. interrogans servovar autumnalis	Fort Bragg/Pretibial fever

Leptospirosis diagnosis

1. Darkfield microscopy: to observe the corkscrew motility
2. Microscopic Agglutination Test: confirmatory test
3. Serology

BORRELIA

- Loosely coiled spirochetes, arthropod-borne, blood spirochetes
- Transmitted through arthropod vectors: lice and ticks
- Diagnosis: culture & examination of Giemsa stained smears
- Culture of blood drawn on febrile state, culture media: Kelly's media, Barber Stoenner Kelly media, 33°C for 6 weeks

Borrelia burgdorferi

- Agent of: Lyme disease or tickborne relapsing fever
- Best diagnostic method: serologic tests
- Tick-borne relapsing fever; transmitted by ticks/Ixodes (deer ticks, black-legged ticks)
- Stages:
  1. Stage 1 – ECM (Erythema chronicum migrans): Bull's eye rash
  2. Stage 2- Multiple smaller ECM disseminated through blood; may affect other parts of the body
  3. Stage 3 – May develop neurologic abnormalities
- Best Diagnostic test: Sero tests such as (1) EIA/Immunofluorescence & (2) Western Immunoblot

Borrelia reccurentis *Louse-borne relapsing fever* *Famine fever*	- Vector: *Pediculus (P. humanus capitis/corporis)* - Best diagnostic method: Examining a Giemsa-stained smear
Borrelia duttoni ***Borrelia parkeri*** ***Borrelia hermsii*** *Tick-borne relapsing fever*	- Vector: Soft ticks

CHLAMYDIAE, MYCOPLASMA, & RICKETTSIAE

CHLAMYDIA/*Chlamydophila*

- Viewed as gram negative organism; obligate intracellular
- With 2 distinct forms: elementary body (infectious body; enters the cell) and reticulated body (reproductive body); when it goes out of the cell, it becomes elementary body again
- Cultivated in tissue culture i.e. McCoy cells, HeLa cell line, monkey kidney cell

Chlamydia psittaci	- Causes: Parrot fever/Psittacosis/Ornithosis - MOT: contact with infected birds and their droppings (aerosol transmission) - Immunoassay and PCR are the preferred means of detection
Chlamydia pneumoniae	- Formerly TWAR strain (Taiwan Acute Respiratory Strain) - Humans are the only known host - Associated with mild respiratory infections - Grows poorly in cell culture
Chlamydia trachomatis	- Causes: Trachoma - May cause LGV (lymphogranuloma venereum): STD - TRIC: Trachoma inclusion conjunctivitis (a form of preventable blindness) - Frei test: susceptibility test for LGV - McCoy cells – best medium; gold standard - Iodine or Giemsa stain of glycogen containing inclusion bodies: "Halberstaedter-Prowazek" - PCR: definitive test

	C. trachomatis	***C. psittaci***	***C. pneumoniae***
Host	Man	Animals	Man
Elementary body	Round	Round	Pear-shaped
Sulfonamide	S	S	R
#of Serovars	20	10	1
Diseases	Trachoma, LGV	Psittacosis	Pneumonia, pharyngitis

MYCOPLASMA

- Smallest free living organism found in several animals and plants; requires sterol for growth
- Pleomorphic and lacks cell wall; resistant to penicillin, since it lacks a cell wall where penicillin acts
- Dienes stain not gram stain
- Culturable in artificial media producing tiny colonies which may have a fried egg appearance

Mycoplasma pneumoniae

- Formerly known as EATON agent
- Causes primary atypical pneumonia or walking pneumonia
- Originally called PPLO – Pleuro pneumonia-like organism

GENITAL MYCOPLASMA

- Causes non-gonococcal urethritis in males and post partal fever in females
- Acquired through sexual contact
- Agents of non-gonococcal urethritis:
  - *C. trachomatis*
  - *M. hominis*
  - *U. urealyticum*

Mycoplasma hominis

- Post abortal fever, post-partum fever, pelvic inflammatory disease

Ureaplasma urealyticum

- Produces tiny fried egg colonies, causes NGU, can cross the placenta

	Glucose	Arginine	Urease
M. hominis	-	+	-
M. pneumoniae	+	-	-
U. urealyticum	-	-	+

RICKETTSIAECEAE

- Genera: Rickettsia, Ehrlichia (Excluded from *Rickettsiaceae*: Coxiella and Bartonella)
- Fastidious, obligate intracellular organism, arthropod-borne
- Stains poorly with gram's; gram negative, pleomorphic, NM, with both RNA and DNA
- Rickettsial infections except Q fever and Ehrlichiosis typically manifested by triad of fever, headache, and skin rashes
- Rickettsial infections can be diagnosed by means of serologic tests like Weil-Felix reactions
- Appears blue with Giemsa, red with Machiavello stain, reddish black with Jimenez stain

A. SPOTTED FEVER GROUP

SPECIES	DISEASE	TRANSMISSION/VECTOR
Rickettsia rickettsi	Rocky mountain spotted fever	Wood, dog, brown ticks
Rickettsia conorii	Boutounneuse fever Mediterranean spotted fever Indian Tick Typhus	Tick bites
Rickettsia sibirica	North Asian Tick Typhus Fever	Tick bites
Rickettsia akari	Rickettsial pox	Mouse mite
Cat Flea Rickettsia	Murine Typhus-like Illness	Presumably flea bites or feces
Rickettsia australis	Queensland Tick Typhus Fever/ Australian	Tick bite
Rickettsia japonica	Oriental Spotted fever	Presumably tick bites

B. TYPHUS FEVER GROUP

SPECIES	DISEASE	TRANSMISSION/VECTOR
Rickettsia prowazekii	Epidemic typhus	Body Louse Squirrel flea and louse
	Brill Zinsser Disease	No vector/ Recrudescence of latent infection / relapse of past infection
	Flying Squirrel Typhus Fever	Presumably feces or flea or louse of flying squirrel
Rickettsia typhi	Murine Typhus/ Endemic Typhus	Rat flea

C. SCRUB TYPHUS GROUP

SPECIES	DISEASE	TRANSMISSION/VECTOR
Rickettsia tsutsugamushi	Scrub typhus	Chigger or mite bite

D. COXIELLOSIS

SPECIES	DISEASE	TRANSMISSION/VECTOR
Coxiella burnetti	Q fever	Inhalation of aerosols from infected animals, possibly ingestion of animal products or tick bites

E. BARTONELLOSES

SPECIES	DISEASE	TRANSMISSION/VECTOR
Bartonella bacilliformis Destroys RBCs	Orroya or verruga peruana Carrion's disease	Sandfly bite
Bartonella henselae	Peliosis hepatitis Cat scratch disease	Kitten scratch or bite
Bartonella clarridgeiae	Minor Cat Scratch Disease No endocarditis	No vector
Bartonella elizabethae	Endocarditis, bacteremia	Bite of rat fleas
Bartonella quintana (Rochalimaea quintana)	Trench fever	Feces or pediculus louse

F. EHRLICHIOSES

SPECIES	DISEASE	TRANSMISSION/VECTOR
Ehrlichia chaffeensis	Human Monocytic Ehrlichiosis	Lone star tick
Anaplasma phagocytophilia/ Ehrlichia equi	Human Granulocytic Ehrlichiosis	Presumably deer tick
Ehrlichia sennetsu	Sennetsu Rickettsiosis	Tick bite

*All can infect WBCs

- Species under genus Ehrlichia are obligate intracellular organism
- Infects either monocytes or granulocytes, they are gram negative
- Clusters of Ehrlichia are called morulae
- Diagnosis of Ehrlichiosis is through observation of typical morulae in WBCs (monocytes and granulocytes)
- May also be detected through serologic tests: IFAT and PCR

Pneumocystis jirovecii

- #1 cause of pneumonia in AIDS patients
- #1 opportunistic infection in AIDS
- Specimen: bronchoalveolar lavage

QUALITY CONTROL FREQUENCY			
Autoclave	Spore test weekly	**Gaspak**	Check for anaerobiosis with methylene blue strip
Balances	Check for accuracy annually	**Gram stain**	Control slides weekly (*E. coli/S. aureus*)
BSCs	Check airflow annually	**Temperatures**	Daily checks for incubators, water baths, refrigerators, and freezers
Centrifuge	Check rpm every 6 months	**Microscopes**	Clean and adjust 4x per year
CO_2 incubator	Check CO_2 and temperature daily		

	Bacteria	*Chlamydia*	*Mycoplasma*	*Rickettsiae*	Viruses
Size	0.4-2µm	0.3-0.5 µm	300 nm	0.3-0.5 µm	25-300 nm
DNA & RNA	Both	Both	Both	Both	Either
Reproduce by binary fission	Yes	Yes	Yes	Yes	No
Cell wall	Yes	Yes	No	Yes	No
Growth on non-living medium	Yes	No	Yes	No	No
Gram stain	Yes	No	No	No	No

RECALLS

Gold standard SERO TEST for Rickettsiae	Indirect immuno-fluorescence	Step by step procedure	Disinfection of area, remove gloves, remove gown & wash hands
Amniotic, Serous fluid and CSF must be delivered to the laboratory in	Less than 15 minutes	Bacteria grows on	Optimal pH
Serpentine pattern, Niacin & Nitrate (+)	*M. tuberculosis*	PPE used for Enteric isolation	Gowns and gloves
Organism which can be found on Enteric Isolation	Salmonella	H_2S Positive; Urease (-); LDC (-)	Citrobacter
Rapid Urease (+)	PPM	Rickettsiae infects	Endothelial cells
Temperature sensitive organism	*N. gonorrhoeae*	Positive control for Indole test	*E. coli*
pH sensitive organism	Shigella	Relationship of MIC & Zone of Inhibition	Inversely proportional
(-) Growth on routine media	*M. leprae and Chlamydia*	Component of BAP	Nutrient agar and 5% sheep's blood
BSC with 30% recirculated air	Class 2-B1	(-) control for Acetate Utilization	Shigella
Urine Preservative	Boric acid	Bacteria implicated in animal/ commonly involved in zoonotic infection	Capnocytophaga
Reagent for String test	Sodium desoxycholate	Oxidase and H_2S positive	*Shewanella putrefaciens*
Gram (-); K/K; (-) fluorescence; LDC (+), cause of cystic fibrosis	*Burkholderia cepacia*	Specimen transport must be within	2 hours following collection

ADDITIONAL NOTES IN MICROBIOLOGY (Source: Henry's latest edition)

- *Gemella spp.* and *Aerococcus* are gram positive catalase negative cocci resembling Viridans Strep but with smaller colonies
- *In E. rhusiopathiae*, biopsy and tissue aspirates from erysipeloid lesions are the BEST specimens for culture
- *Nocardia* are partially AF gram positive bacilli, partial acid fastness can be enhanced by growing organism for 4 days on Middlebrook 7H10 or Litmus Milk broth. Pulmonary disease is the Most common manifestation of Nocardiosis
- *Rhodococcus equi*, classified as an aerobic actinomycetes is an opportunistic pathogen of immunocompromised host causing slowly progressive granulomatous pneumonia
- *M. catarrhalis* produce beta lactamase which can be detected using nitrofecin
- *Acinetobacter* spp. can be distinguished readily from *Pseudomonads* on the basis of their lack of motility, inability to reduce nitrates and negative oxidase
- Virulence factors of *Aggregatibacter actinomycetemcomitans* are LEUKOTOXIN and COLLAGENASE
- Growth of *Eikenella* enhanced by 5-10% CO2 and presence of hemin on medium
- The finding of gram negative bacilli on BAP only and is oxidase and indole +, ONPG negative provides STRONG presumptive evidence for isolation of *Pasteurella multocida*
- *G. vaginalis* is PRESUMPTIVELY identified based on hemolysis on HBT following 48 hrs incubation
- The appearance of AFB on smears does not provide species identification, however certain species have features that may be useful diagnostically like (1)" formation of CORDS in smears of broth is typical of *M. tuberculosis* (2) *M. kansasii* often appears as CROSS BARRED bacilli resembling a Shepherd's Crook

CULTURE MEDIA

Enrichment media for Vibrio	Enrichment media for *Vibrio*
Amie's Media	Transport media, to maintain viability of the organisms from the source to the laboratory
Alkaline Peptone water	Enrichment media for *Vibrio*
Bacteroides Bile Esculin (BBE)	Selective and differential media. Selective for *Bacteroides*. Used to differentiate bile resistant from bile-sensitive *Bacteroides*
Bile Esculin Agar	Nutrient agar base media with ferric citrate. Differential isolation & presumptive identification of *Group D Streptococci and Enterococci*
Bile Esculin Azide agar with Vancomycin	Selective & differential for cultivation of Vancomycin resistant enterococci from clinical and surveillance specimens
Brain Heart Infusion (BHI)	High protein media for growing a wide variety of bacteria
Bismuth Sulfite Agar	Used for isolation of *Salmonella typhi*. Produces black colonies. Not recommended for isolation of all Salmonella
Blood Agar (BA)	Cultivation of fastidious microorganisms, used in determining hemolytic reactions
Blood Cystine Dextrose Agar	For growing *Francisella tularensis*
Bordet-Gengou Media	Isolation of *Bordetella pertussis*
Brilliant Green Agar (BGA)	Selective media for isolation of *Salmonella* other than *S. typhi*. Not recommended for *Salmonella*
Buffered Charcoal Yeast Extract Agar with Antibiotics	Enrichment and selective for *Legionella spp.*
Chapman Stone agar	Selective media for the isolation of *Staphylococcus*
Campy-Blood Agar	Selective for *Campylobacter spp.*
Campylobacter Thioglycollate Broth	Selective holding medium for recovery of *Campylobacter spp.*
Cary and Blair Media	Transport media
Casein Media	Used for separating *Nocardia* & *Streptomyces*. Made from skimmed milk. Used as a media to test for hydrolysis
Cefoperazone, Vancomycin, Amphotericin (CVA)	Selective media for isolation of *Campylobacter spp.*
Cefsulodin-Irgasan-Novobiocin (CIN)	Selective for *Yersinia spp.* Maybe useful for isolation of *Aeromonas spp.*
Chocolate Agar (CAP)	Cultivation of *Haemophilus spp.* and pathogenic *Neisseria*
Chopped Meat Glucose	Used for primary isolation of anaerobes. Used for *Clostridium spp.*
Columbia Colistin Nalidixic Acid Agar (CNA)	Selective isolation of gram positive cocci
Corn Meal Agar with 1% Tween 80	Used to stimulate production of Chlamydospores of *Candida albicans*
Cystine Tellurite Blood Agar	Isolation of *Corynebacterium diphtheria*
Cystine Trypticase Agar (CTA)	CTA with carbohydrates added is excellent for determining the fermentation reactions of groups such as *Neisseria*
Cycloserine cefoxitin Fructose agar (CCFA)	Selective media for *C. difficile*
Deoxycholate Agar	Used for gram negative enteric bacilli differentiating lactose fermenters from non-lactose fermenting organisms
Deoxycholate Citrate	For isolation of *Salmonella and Shigella*
Dieudonnes media	Selective media for *Vibrios*
"E" agar and Edward Hayflick agar	Selective media for the isolation of *Mycoplasma and Ureaplasma*
Egg Yolk Agar	Used for the isolation and identification of *Clostridium*
Eosin Methylene Blue (EMB)	Isolation & differentiation of lactose fermenting & non-lactose fermenting enteric bacilli

Ellinghausen, McCullough Johnson and Harris Media (EMJH)	Selective media for *Leptospira Interrogans*
Fetal Bovine serum with Vancomycin	Selective media for *H. ducreyi*
Fildes Enrichment	Used for isolation of *H. influenza*
Gram Negative Broth (GN)	Selective enrichment (liquid) medium for enteric pathogens
Hektoen-Enteric (HEA)	Differential, selective media for the isolation and differentiation of *Salmonella and Shigella* spp. from other enteric bacilli
Human Blood Bilayer tween	Selective media for the isolation of *Gardnerella vaginalis*
Leoffler's Coagulated Serum	Used for *C. Diphtheria*
Lowenstein Jensen Media	Used for the isolation of *Mycobacterium tuberculosis* and other *Mycobacterium*. Malachite green is used to inhibit all organisms except *Mycobacterium*
Mac Conkey Agar	Peptone base media with lactose. Used for the isolation and differentiation of lactose and non-lactose fermenting enteric bacilli
Mac Conkey Sorbitol Agar	A modified Mac Conkey that contains D-sorbitol instead of lactose as primary carbohydrate. For the selection and differentiation of *E. coli O157:H7* in stool specimens
Mannitol Salt Agar	Selective isolation of *Staphylococci*
Mc Bride Media	For the isolation of *Listeria monocytogenes*
Middelbrook 7H10 or 7H11	For the Isolation of *Mycobacterium tuberculosis*
Modified Kelly's medium or Barbour Sloenner Kelly	Selective media for the isolation of *Borrelia species*
MRVP broth	Used in Methyl Red and Vogues Proskauer tests. These are biochemical tests that help in the identification of enteric bacilli
New York City Agar	Selective of *Neisseria gonorrhea*
Petragnani's Media	Same use as Lowenstein Jensen with same ingredients except that this contains more malachite green
Phenylethyl Alcohol Agar (PEA)	Selective isolation of gram positive cocci and anaerobic gram negative bacilli
Regan Lowe	Enrichment and selective media for *Bordetella pertussis*
Sabouraud Dextrose or Maltose Agar	Used for fungus culture
Salmonella-Shigella Agar (SSA)	Selective for *Salmonella and Shigella*
Schaedler Agar	Non-selective media for the recovery of anaerobes and aerobes
Selenite Broth	Enrichment for the isolation of Salmonella spp.
Simmon Citrate Agar	Green colored media used to test the organism's ability to use citrate as source of carbon
Skirrow Agar	Selective for *Campylobacter spp*.
Streptococcal Selective Agar	Selective for *Streptococcus agalactiae* in female genital specimens
Tetrathionate Broth	Selective for *Salmonella and Shigella spp*.
Thayer Martin Agar	Selective for *Neisseria gonorrhoeae and Neisseria meningitidis*
Thioglycollate Broth	Supports growth of anaerobes, aerobes, microaerophilic, and fastidious microorganisms
Thiosulfate Citrate Bile Salts (TCBS)	Selective and differential for *Vibrios*
Todd Hewitt Broth supplemented with antibiotics	Selective and enrichment for *Streptococcus agalactiae* in female genital specimens
Trypticase Soy Broth (TSB)	Enrichment broth used for subculturing various bacteria from primary agar plates
Triple Sugar Iron Agar (TSI)	Media used in biochemical tests to differentiate gram negative bacilli especially enteric bacilli
Urea Agar	Used for Urease test to detect urease production which is ideal for identification of *Proteus*
Xylose Lysine Desoxycholate Agar (XLD)	Isolation and differentiation of *Salmonella* and *Shigella spp.* from other enteric bacilli

ADDITONAL NOTES IN BACTERIOLOGY

SPECIMENS - SPECIMEN COLLECTION - PROCESSING & STORAGE

- Specimen collected to detect cases of bacteremia: blood
- Where to collect blood if patient has an existing IV line: drawn below the IV line (if drawn above the IV line, specimen will be diluted with fluid being infused)
- Skin preparation prior to drawing blood for cultures: phlebotomy site must be cleansed with 70-95% alcohol, followed by iodine scrub or iodophor, then alcohol rinse
  - Iodophor = iodine + detergent
  - Iodine tincture = iodine + alcohol
  - If iodine is not available, use chlorhexidine as substitute
- Blood culture set should include:
  - 1 tube for aerobic culture and 1 tube for anaerobic culture
- Amount of blood per culture
  - In adults: 10-20 mL
  - Children/infants: 1-5 mL
- INSTRUMENT BASED SYSTEMS: are used to improve identification of pathogens in blood specimens; advantages: more rapid and accurate determination
  - BACTEC system: fully automated with incubator, shaker & detector; Uses FLUORESCENCE to measure CO_2 released by the organism (uses fluorescent sensor).
  - BacT ALERT system: uses colorimetric sensor
- Considerations during sputum collection for detection of LRTI
  - There should be no food intake, 1-2 hours before expectoration
  - The mouth should be rinsed with saline or water prior to collection
- Specimen type that will give high diagnostic yield for *Pneumocystis jirovecii*: bronchoalveolar lavage; if sputum: aerosol-induced specimen
- Gastric aspirate: specimen type that when collected is exclusive for isolation of AFB and collected from patients who are unable to produce sputum i.e., children
- Tracheostomy aspirate: specimen type that must be treated as sputum in the laboratory
- When to reject endotracheal aspirate from adult patient: if more than 10 squamous epithelial cells were noted per LPO or no organism seen under OIO
- Cells that will indicate that samples collected thru bronchoscopy or BAL are obtained from the lower GIT: Ciliated columnar bronchial epithelial cells, Pulmonary macrophage
- Presence of colonies inside a Zone of Inhibition may indicate mixed-culture
- Identification of bacterial strains (a subtype of a species) can be made thru:
  - SEROTYPING
  - BIOTYPING: identified based on differences in biochemical test results (biovar/biotype)
  - PHAGE TYPING: to detect outbreak source and single strains of bacteria or different strains of bacteria within single species
- Manipulation of autoclave will require: Lab coat; Goggles, Closed toe shoes & Heat resistant gloves
- Collection of catheterized urine: aspirate aseptically from catheter tubing (not on bags or tips)
- Processing of CSF
  - If amount is MORE THAN 1 mL: centrifuge, remove supernatant, sediment is inoculated to media and used for smear preparation.
  - If amount is LESS THAN 1 mL: no centrifugation required, directly inoculate sample to plated media, and use directly for smear preparation
- Alcohol & iodine preparation (betadine) must be in contact with the object for 1-2 minutes for them to kill microorganisms
- Disinfectants are generally used at 20-22°C (room temperature)
- Sterilization method for heat stable objects: moist heat (autoclaving)
- Procedures that will not eliminate spores:
  - Boiling
  - Pasteurization
- Sterilization method for heat sensitive solutions: filtration
- Sterilization of disposable supplies can be thru: exposure to gamma rays
- Tincture of iodine is a skin disinfectant prepared as: 20% iodine in 70% alcohol
- Not sporicidal: alcohol
- Often used to disinfect biosafety hoods: formaldehyde gas
- Forms of IODINE:
  - Tincture (iodine + alcohol)
  - Iodophor (iodine + detergent)
- Phenolics are used to disinfect hospitals, institutions, household environment
- Main goal of handwashing: to eliminate transient flora

BIOTERRORISM AGENTS:

CATEGORY A	CATEGORY B	CATEGORY C
Small pox, anthrax, tularemia Easily transmitted and can cause very high death rate	Q fever, typhus fever Not as easily transmitted Cause lower death rate	Nipah & Hanta Virus Emerging pathogens Can be reproduced or engineered for mass spread

- Quality control: measures designed to ensure reliability of results
- Quality assurance: involves taking measures to ensure high quality of patient care; Involves monitoring of all components of a procedure

FREQUENCY OF EQUIPMENT MONITORING

- Incubator, water bath, refrigerator, freezer, heating block: daily
- Autoclave (efficiency/spore testing): weekly
- Autoclave temperature: every load (every use)
- GASPAK jar: every use
- Centrifuge function/rpm: every 6 months
- Microscopes: cleaning and adjustment (quarterly or as needed)
- Weighing balance: accuracy is checked annually

- Lim Broth a.k.a. Todd Hewitt Broth with CNA: used for isolation of Group B Streptococci (*S. agalactiae)*

PRIORITIZATION OF SPECIMENS

1. Level 1 = CSF, Amniotic fluid, blood, pericardial fluid, heart valves
2. Level 2 = Unpreserved feces, sputum, wound drainage
3. Level 3 = Catheter tip urine, tissues for quantitation
4. Level 4 = Preserved urine, feces, swab in holding media

- Proper technique to visualize hemolysis is by passing of bright light thru the bottom of the plate = transillumination technique
- Antibiotic disk storage
  - Long Term Storage: -20°C in a non-frost freezer
  - Storage of working supply: 2-8°C for at least 1 week
- MALDI-TOF (Matrix Assisted Laser Desorption ionization - Time of Flight)
  - More rapid and efficient than Vitek
- PCR: molecular method, allow detection of bacteria even following antibiotic therapy
  - Steps:
    - Denaturation: extraction of nucleic acid (done at 92°C)
    - Annealing: carried out between 50-58°C
    - Elongation: a.k.a. extension; carried out at 72°C
- Amniotic fluid, pericardial and pleural fluid must be transported to the lab: in less than 15 minutes

BIOCHEMICAL TESTS, CULTURE MEDIA

- Media that can provide best recovery for *Brucella*: Brucella broth or Trypticase soy broth
- UREASE TEST: used for the presumptive identification of PPM
  - Inoculate in Urea agar or broth
  - Phenol red is indicator
  - (+) result in 2-4 hours: rapid urease producers; development of pink to red color
    - *Proteus*
    - *Providencia*
    - *Morganella*
    - *Helicobacter pylori*
    - *Ureaplasma*
    - *Nocardia*
    - **C. neoformans*
    - **T. mentagrophytes*
  - (+) result after 4 hours: slow/late urease producers; development of orange color

- Enzyme detected in Indole test: Tryptophanase
- Indicator in Oxidation-Fermentation medium (CDC Method): Phenol Red
- Indicator in Oxidation-Fermentation medium (Hugh & Leifson): Bromthymol blue (best answer if phenol red and Bromthymol blue are both in the choices)
- Quality Control for Indole test: (+) control
  - KOVAC's method: *E. coli*
  - Ehrlich's: *H. influenzae*
  - Ehrlich's (anaerobes; Spot indole test): *Porphyromonas asaccharolytica*
- Confirmatory test for Leptospirosis: Microscopic Agglutination Test

TSI REACTIONS

K/K, H_2S-	*Pseudomonas, Burkholderia*
K/A, H_2S-	*Providencia, Morganella, Shigella*
K/A, H_2S+, Gas+	*Salmonella, Proteus, Arizonae, C. freundii, E. tarda*
A/A, H_2S-, Gas+	*Klebsiella, Escherichia, Enterobacter*

LIA REACTIONS

K/K, H_2S+	*Salmonella*
K/A, H_2S-	*Shigella*
R/A, H_2S-	*PPM;* Although Proteus is H_2S+, it is H_2S- in LIA

- *Burkholderia gladioli* and *Stenotrophomonas maltophilia* are both Catalase (+) and Oxidase (+)
  - DIFFERENTIAL TEST: LDC (Lysine decarboxylase test)
    - *S. maltophilia* = (+)
    - *B. gladioli* = (-)
- Biochemical test which is more specific to *S. pyogenes*: PYR test
- Starting point of *Streptococcus* identification: Inoculation on BAP (hemolytic pattern)
- All Brucella are Serum Agglutination Test (+) except *B. canis*
  - Particle Agglutination Test; a rapid test to detect the genus *Brucella*
- Carbon dioxide requiring Brucella: *B. abortus*
- Buffered Glycerol Saline: toxic to Vibrio
- LAP Test (Leucine Aminopeptidase)
  - Purpose: for the presumptive identification of catalase (-), gram (+) cocci
  - Substrate: leucine-β-naphthylamide
  - Reagent: cinnamaldehyde
  - (+) result: development of red color
  - (-) result: no color change or slight yellow
- Microdase Disk: used to rapidly differentiate *Staphylococci* (-) and *Micrococci* (+)

GRAM (+) and GRAM (-) SPECIES

- Bull's eye colonies, semi-opaque with organism concentrated at the center: *Streptococcus agalactiae*
- Nocardia: produces crumbly, wrinkled resembling bread crumbs (like *Fusobacterium)*
- Colonies with an odor similar to an old sock on MSA: *S. aureus*
  - *P. mirabilis*: putrid odor
  - *Haemophilus*: mousy odor
  - *Nocardia:* freshly plowed field
  - *S. anginosus*: honey suckle or butter scotch odor
  - *Eikenella corrodens & H. influenzae*: bleach-like odor
  - *H. influenzae & Pasteurella*: mousy or pungent odor
  - *Proteus*: burnt chocolate/burnt sugar/ burnt gun powder
- "Vine or Streamer effect" on thioglycollate: attributed to some *Streptococci*
- Important marker of *S. aureus*: clumping factor or bound coagulase
- *S. aureus* is PYR (-); *S. lugdunensis, S. schleiferi, S. intermedius* are PYR (+)
- Alpha prime is also known as wide zone, characterized by a very small zone of "no zone of hemolysis & a wider zone of beta hemolysis". This may be exhibited by some Viridans Streptococci.
- PRIMARY CONTRIBUTOR of DENTAL CARIES: *S. mutans*
- As to hemolysis, Group D Non-Enterococci and Enterococci are: Non-Hemolytic
- Abiotrophia and Granulicatella: Nutritionally Variant Streptococci
- EAEC and DAEC causes: pediatric diarrhea
- Legionella - slow growth, colonies with "ground glass appearance"; does not degrade sugar (asaccharolytic); requires L-cysteine
- Growth of *P. aeruginosa* on Cetrimide agar appear serrated and confluent
- Salmonella colonies growing on BG agar may appear like: snow flakes
- Tellurite reduction - best for *M. avium* identification (+) result: smooth fine black precipitate; smoke-like action
- *C. psittaci* and *C. pneumoniae* has no glycogen inclusion
- *C. pneumoniae* is resistant to sulfonamides
- *C. trachomatis* is sensitive to sulfonamide and has glycogen inclusion

EXAMPLES OF CRITICAL VALUES IN MICROBIOLOGY

- Critical values: if any of this is noted with the lab, results must be communicated immediately to the attending physician; results may be indicative of a life-threatening situation
  - Positive blood culture
  - Positive CSF gram stain or culture
  - Positive Cryptococcal antigen test or culture
  - Positive blood smear for Malaria
  - *S. pyogenes* from a sterile site
  - Positive acid-fast smears or positive *Mycobacterium* culture
  - *S. agalactiae* or HSV from genital site of a pregnant woman at term
  - Detection of significant pathogen (i.e. *Bordetella pertussis, Brucella spp., Legionella spp.)*